DSAA 4011 · LECTURE 6 · THURSDAY 15 OCTOBER 2026

Too many stories to check

Guessing what we never see, four ways: the evidence lower bound and approximate inference

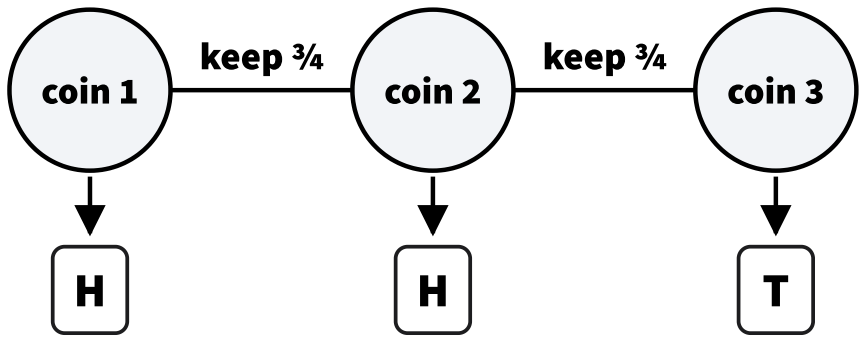
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THE STORY

A friend secretly switches coins, and we see only the flips

A friend secretly flips **coin A** (often heads) or **coin B** (rarely heads). The friend usually **keeps** the same coin but sometimes **switches**.

We see only **heads, heads, tails**. Which coin made each flip, and what are the friend's **habits**?

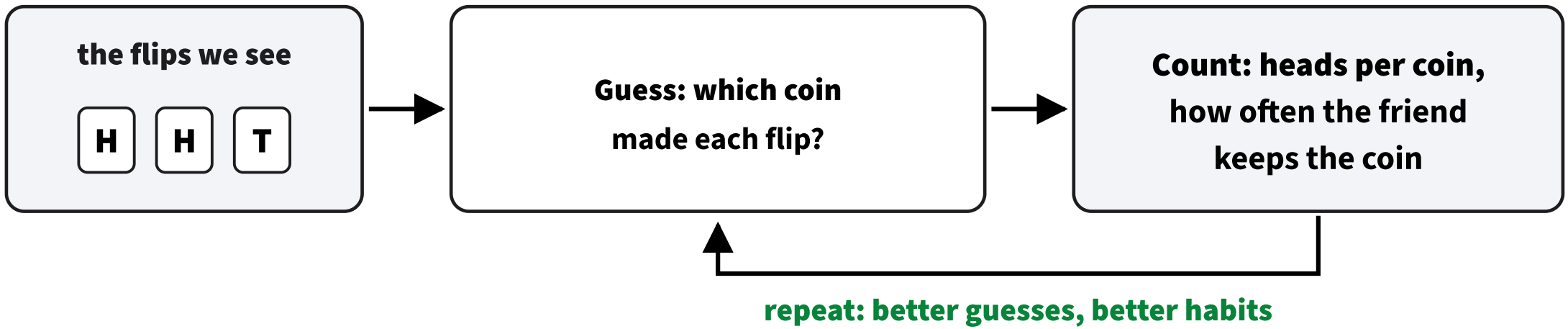


coin A: heads ¾ **coin B: heads ¼**

> **The coins are hidden; we must learn from what we see.**

Last week, we guessed the coins and then counted

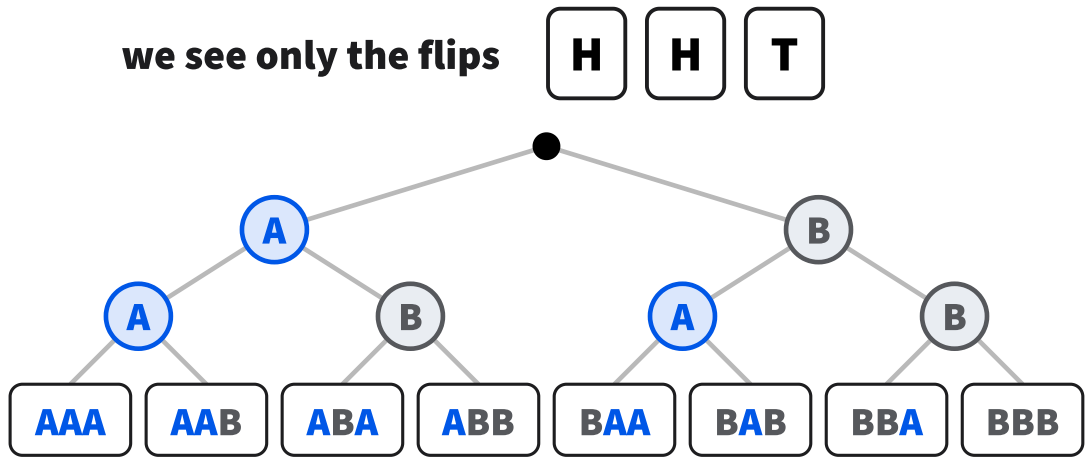
Guess: find the chance that coin A made each flip. **Count:** with these guesses, count each coin's **heads** and the friend's **switches**. Then repeat.



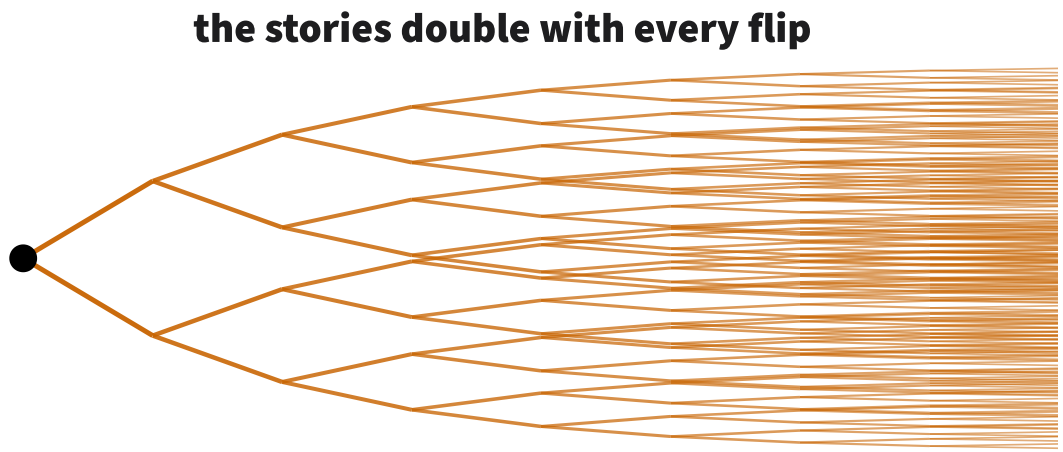
This loop is EM, expectation–maximization: each round, the guesses and the habits get better.

To guess well, we must weigh every secret story

To guess the coin of flip 2 exactly, we must weigh **every secret story**: every choice of coins.



3 flips: 8 possible secret stories



300 flips: more stories than atoms in the universe

Today's question: how can we guess well when there are **too many stories** to check?

> **With few flips we can check every story; with many flips we cannot.**

WARM-UP

Two ideas from earlier lectures that come back today

1. In Lecture 4, a sum was too large to compute exactly. What did we do instead?

nothing | drew random examples and averaged them | used only the first term

Answer. **We drew random examples** and averaged them: today's Way 2, act it out.

2. In Week 2, messages were exact on a tree. What goes wrong on a loop?

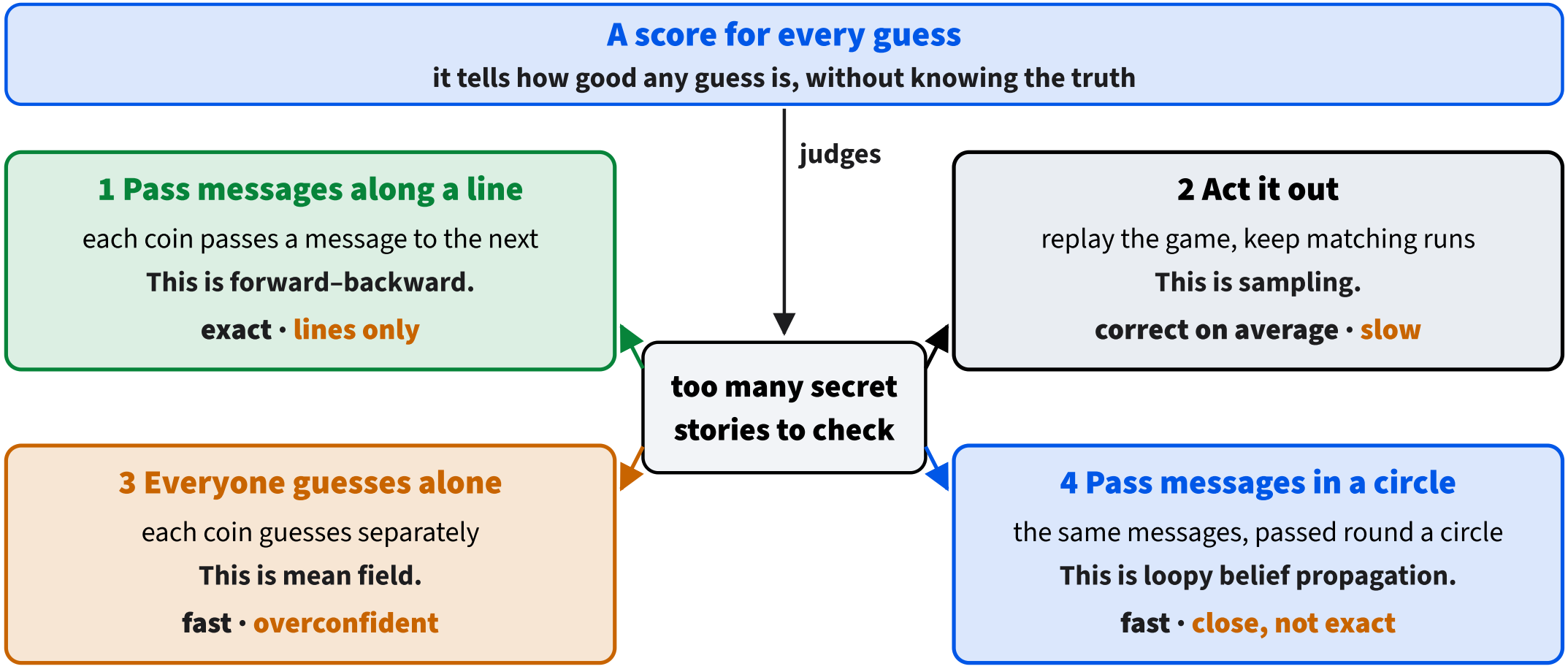
nothing | messages cannot be sent | a message travels around the loop and returns to its sender

Answer. **It returns**, so its news is **counted twice**: today's Way 4, messages in a circle.

Both ideas return today, as Way 2 and Way 4.

TODAY'S MAP

One score and four ways to guess when stories are too many



Each way gives up some exactness to gain speed.

OBJECTIVES

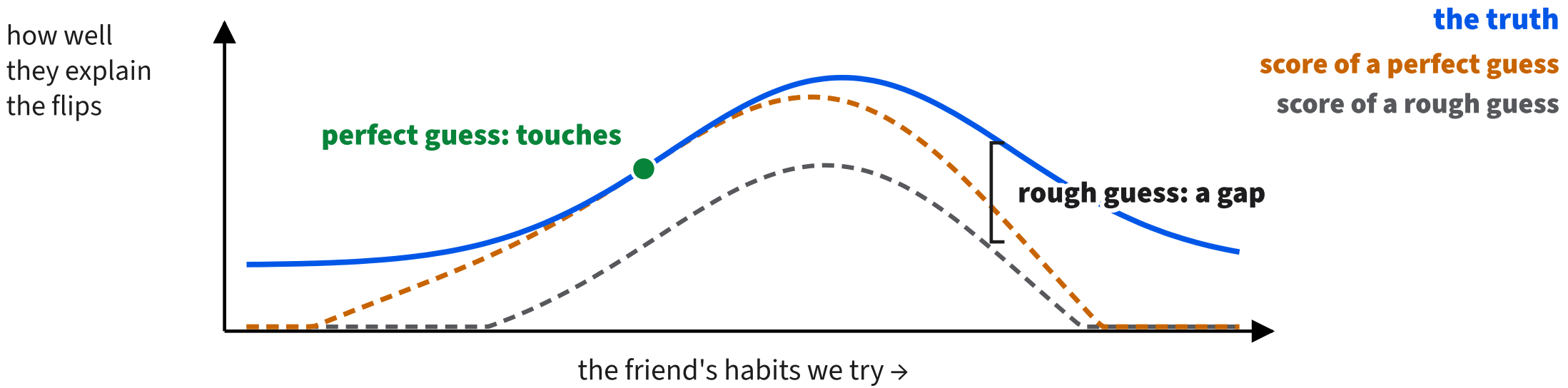
By the end of today, you can explain five things

1. why a **score** judges a guess without the truth — **the score**
2. how **messages along a line** give the exact answer cheaply — **Way 1**
3. why **acting it out** is right on average but slow — **Way 2**
4. why **everyone guessing alone** is fast but overconfident — **Way 3**
5. why **messages in a circle** are close but not exact — **Way 4**

One friend, two coins, four ways to guess.

We cannot compute the truth, but we can score any guess

For any guess of the hidden coins, we can compute a **score**. It is always **below** the true answer and **equals it only for a perfect guess**.



A better guess has a higher score, so we can improve a guess without knowing the truth.

The score is the evidence lower bound (ELBO)

Four ingredients for our three flips:

1. **the flips**, x: H, H, T;
2. **the hidden coins**, z: one secret story, such as A, A, B;
3. **the habits**, θ: heads $\frac{3}{4}$ for A, $\frac{1}{4}$ for B, keep $\frac{3}{4}$;
4. **our guess**, q: a chance for each of the 8 stories.

The score: $B(q, \theta) = \sum_z q(z) \cdot \ln \frac{p_\theta(x, z)}{q(z)}$, summed over the 8 stories

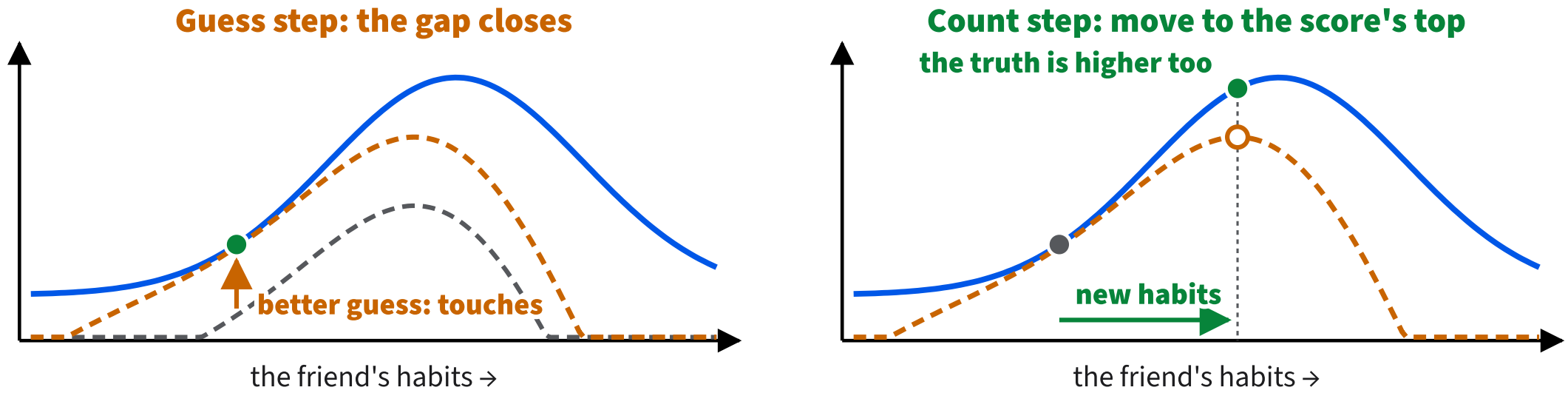
Truth = score + gap: $\ln p_\theta(x) = B(q, \theta) + \text{gap}$; the **gap** is never negative, and zero only for a perfect guess

So the score is a **lower bound** on the evidence, the log-chance of the flips: the **evidence lower bound (ELBO)** (Lecture 5, Lemma 1).

Raising the score closes the gap or improves the habits.

Expectation–maximization raises this score in two steps

Guess step: make the guess q as good as possible. **Count step:** with q fixed, choose the habits θ with the best score.



blue: the truth · orange dashes: the score of our guess

But the best guess weighs **every** secret story, which is impossible with many flips.

Today: four ways to make the guess step cheap enough to compute.

Way 1: if the coins form a line, they can pass messages



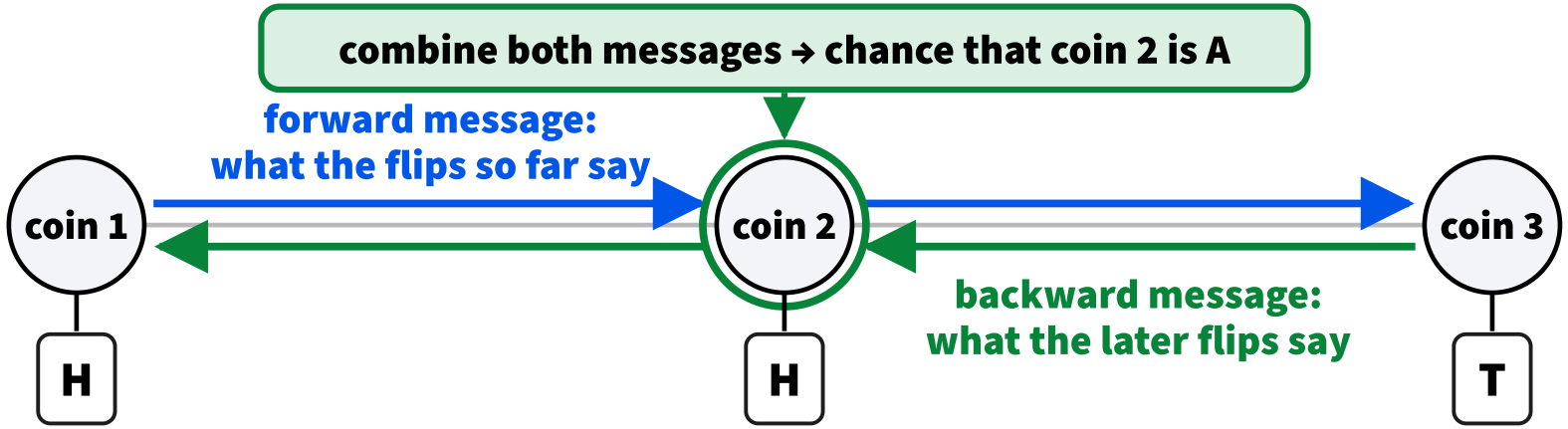
Each coin depends only on the coin before it (a **Markov chain**). This assumption lets us **avoid listing the stories**.

> **Way 1 gives the exact guess at low cost.**

Coins in a line can pass messages instead of listing stories

The one assumption: the friend chooses each coin from **the previous coin only**. This is a **Markov chain**: $p(z_t \mid z_1, \ldots, z_{t-1}) = p(z_t \mid z_{t-1})$.

So each coin sends its neighbour a short **message** that summarizes the flips on its side. We need **one pass forward and one pass back**.



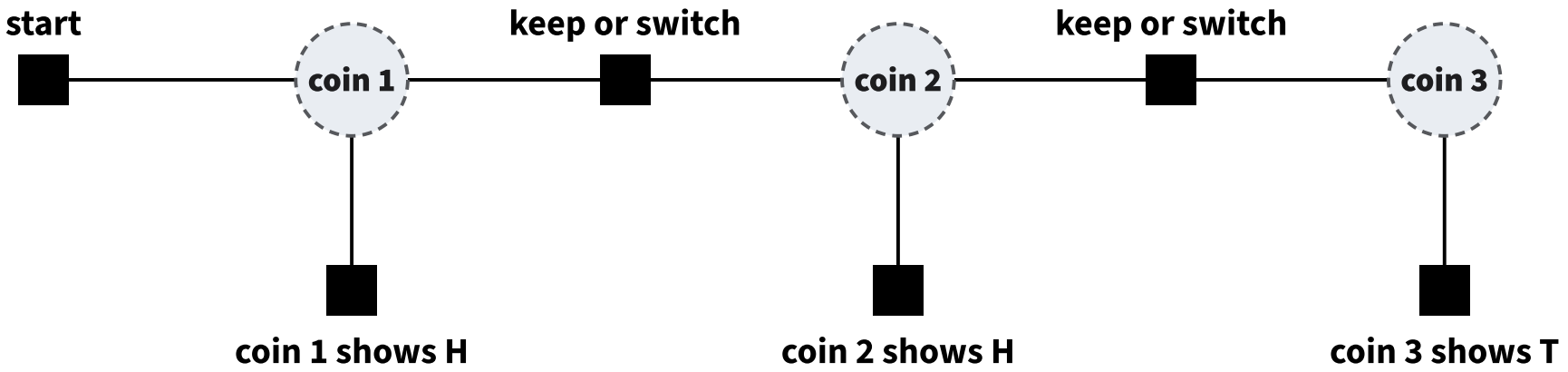
everyone learns the whole story without listing every story

Because the coins form a Markov chain, two passes replace checking every story.

The coin game is a chain factor graph

Take our three coins. Each **circle** is a hidden coin z_t, and each **square** is one habit:

1. the **start**: which coin the friend picks first;
2. **keep or switch**: how each coin follows the one before;
3. **shows H** or **shows T**: how each coin made its flip x_t.



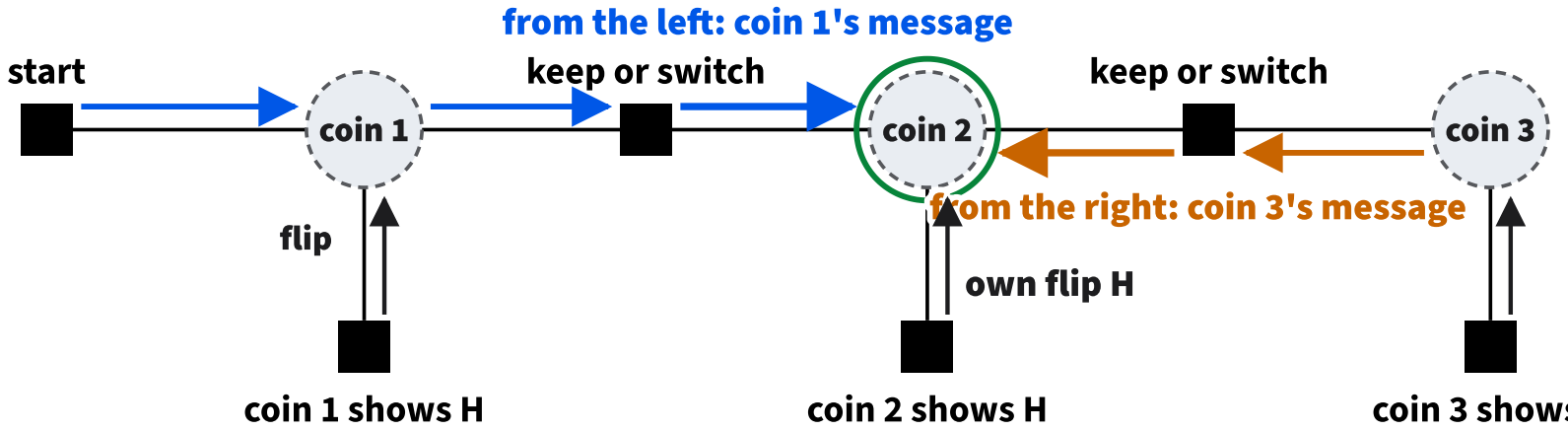
Chance of a story = product of the squares: $p(z, x) = p(z_1)\,p(x_1 \mid z_1) \cdot p(z_2 \mid z_1)\,p(x_2 \mid z_2) \cdot p(z_3 \mid z_2)\,p(x_3 \mid z_3)$

Each coin links only to its neighbours: no loop, so the graph is a tree.

Forward–backward is the sum-product algorithm on this chain

Take the **chance that coin 2 is A**. Three things reach coin 2; each message **sums out the far coin**, Lecture 2's rule:

1. the **message from the left**: coin 1's news, carried over **keep or switch**;
2. coin 2's **own flip**, H;
3. the **message from the right**: coin 3's news, explaining the last flip T.



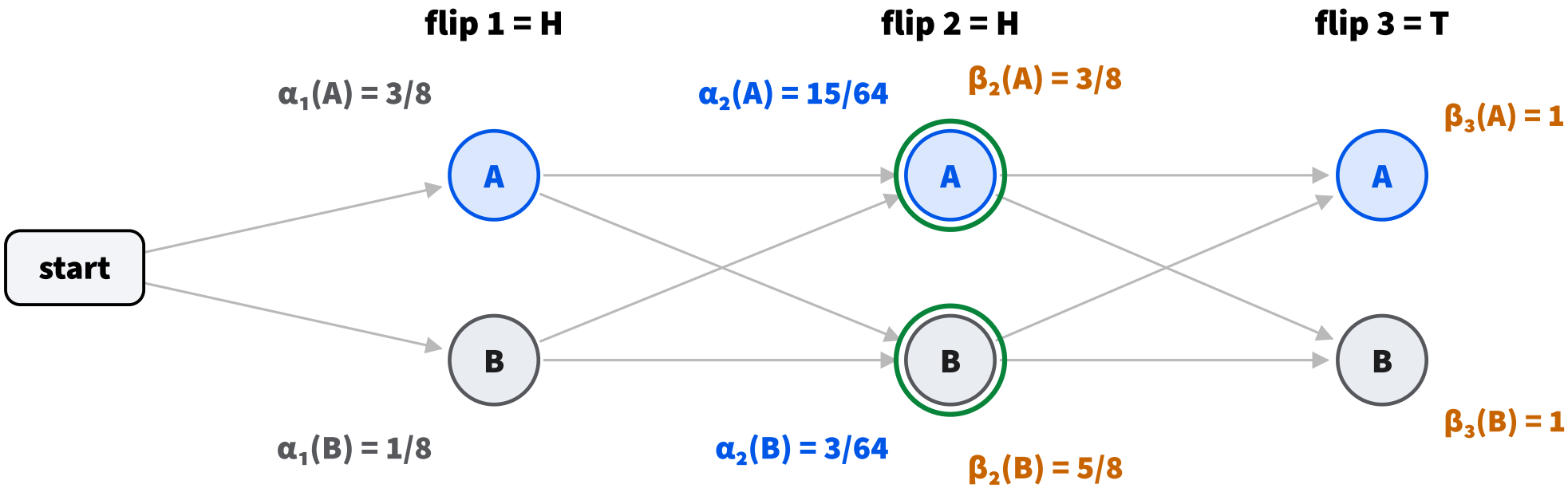
Messages: $\alpha_2(z_2) = p(x_2{=}\mathrm{H} \mid z_2) \sum_{z_1} p(z_2 \mid z_1)\, \alpha_1(z_1)$ $\qquad$ $\beta_2(z_2) = \sum_{z_3} p(z_3 \mid z_2)\, p(x_3{=}\mathrm{T} \mid z_3)$

Chance coin 2 is A = left × flip × right, rescaled: $p(z_2{=}A \mid x) = \alpha_2(A)\, \beta_2(A) \,/\, p(x)$

Forward–backward is Lecture 2's sum-product algorithm on a chain.

On H, H, T, the two forward–backward messages meet at coin 2

The **forward message** α_2 is the chance of the first flips H, H, ending at coin A or B. The **backward message** β_2 is the chance of the last flip T, starting from A or B.



Next: where each of these four numbers comes from, one step at a time.

The starting numbers are our first guess of the habits

We never see the habits, so expectation–maximization (EM) starts from a **first guess** θ and improves it each round:

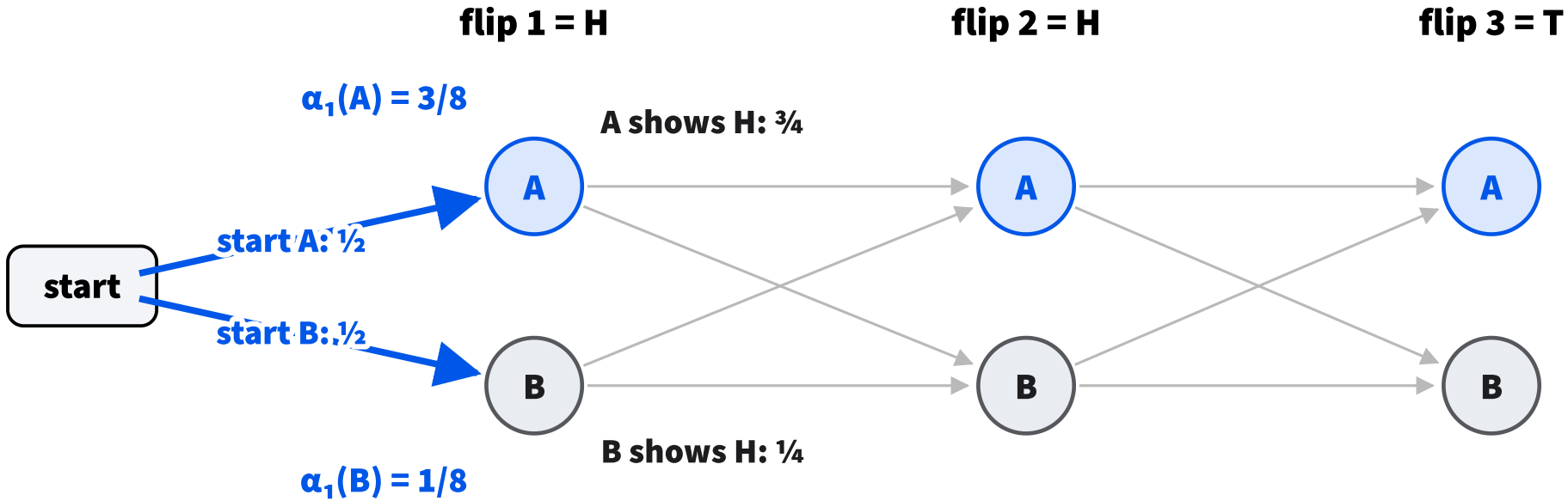
habit	first guess	why this number
first coin	**A:** $\frac{1}{2}$, **B:** $\frac{1}{2}$	**before any flip, no reason to prefer either coin**
between flips	**keep** $\frac{3}{4}$, **switch** $\frac{1}{4}$	**the friend usually keeps the coin**
heads chance	**A:** $\frac{3}{4}$, **B:** $\frac{1}{4}$	**the coins must differ: both at** $\frac{1}{2}$**, no flip could separate them, and EM would keep them equal forever** **(Lecture 6, p. 111)**

Tails takes the rest: A shows T with $1 - \frac{3}{4} = \frac{1}{4}$, B with $1 - \frac{1}{4} = \frac{3}{4}$.

Only a first guess: the count step will replace it.

Step 1: the first flip is H, starting from coin A or coin B

To be at coin A after an H, the friend must **start with A** *and* A must **show H**. For "and", we multiply the chances.

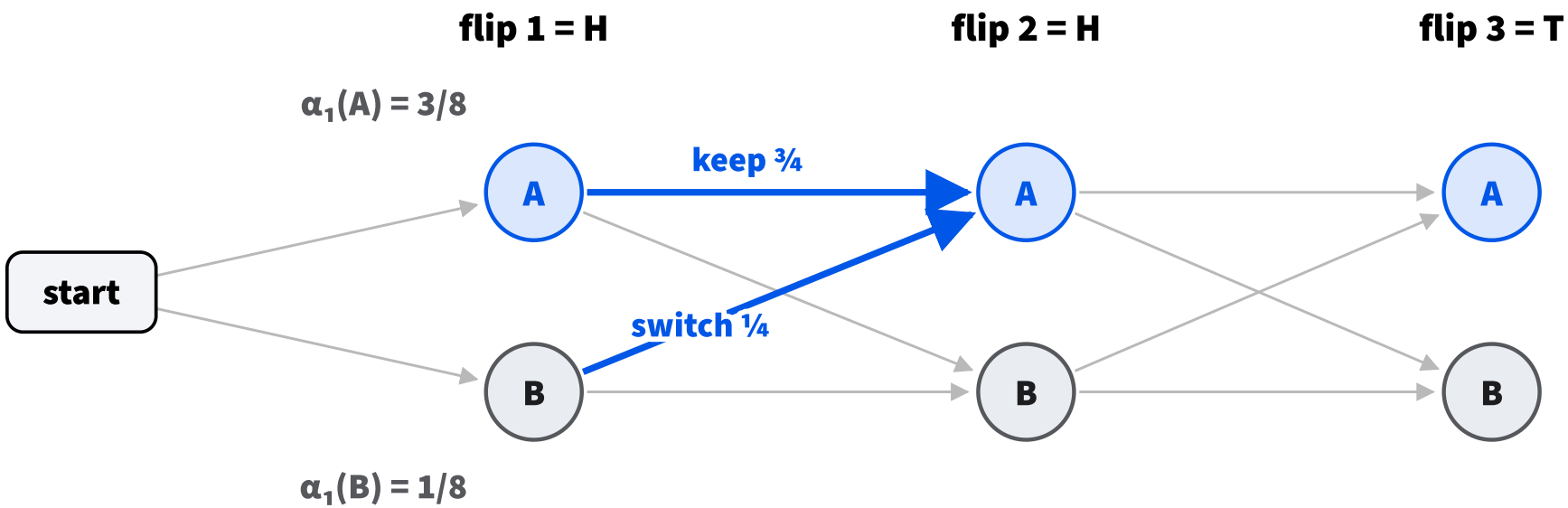


$$\alpha_1(A) = \underbrace{\tfrac{1}{2}}_{\text{start A}} \cdot \underbrace{\tfrac{3}{4}}_{\text{A shows H}} = \tfrac{3}{8} \qquad \alpha_1(B) = \underbrace{\tfrac{1}{2}}_{\text{start B}} \cdot \underbrace{\tfrac{1}{4}}_{\text{B shows H}} = \tfrac{1}{8}$$

After one H, coin A is already three times as likely as B.

Step 2: two paths lead to coin A at flip 2, so we add them

Coin 2 is A if the friend **was at A and kept**, *or* **was at B and switched**. For "or", we add the chances. Then A must show H.

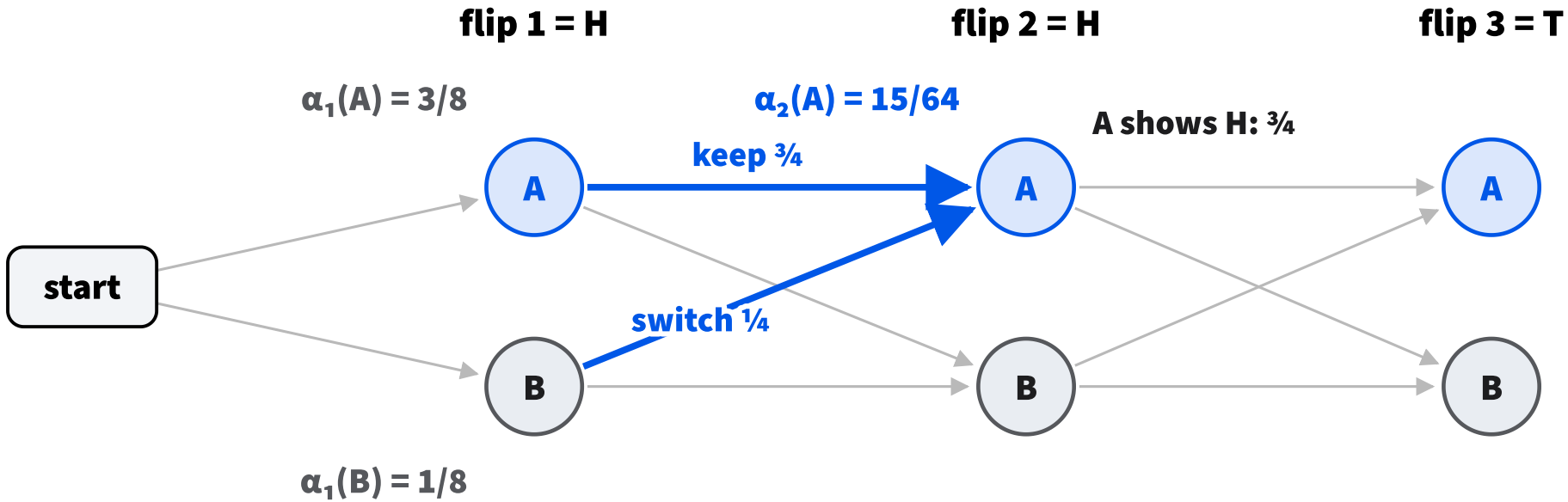


$$\text{arrive at A} = \underbrace{\frac{3}{8}\cdot\frac{3}{4}}_{\text{was A, keep}} + \underbrace{\frac{1}{8}\cdot\frac{1}{4}}_{\text{was B, switch}} = \frac{9}{32} + \frac{1}{32} = \frac{10}{32}$$

Adding the two ways in gives the chance of arriving at coin A.

Step 3: coin A must show H, so we multiply by its heads chance

After arriving at coin A, the friend flips it, and A must show H. For "and", we multiply again.

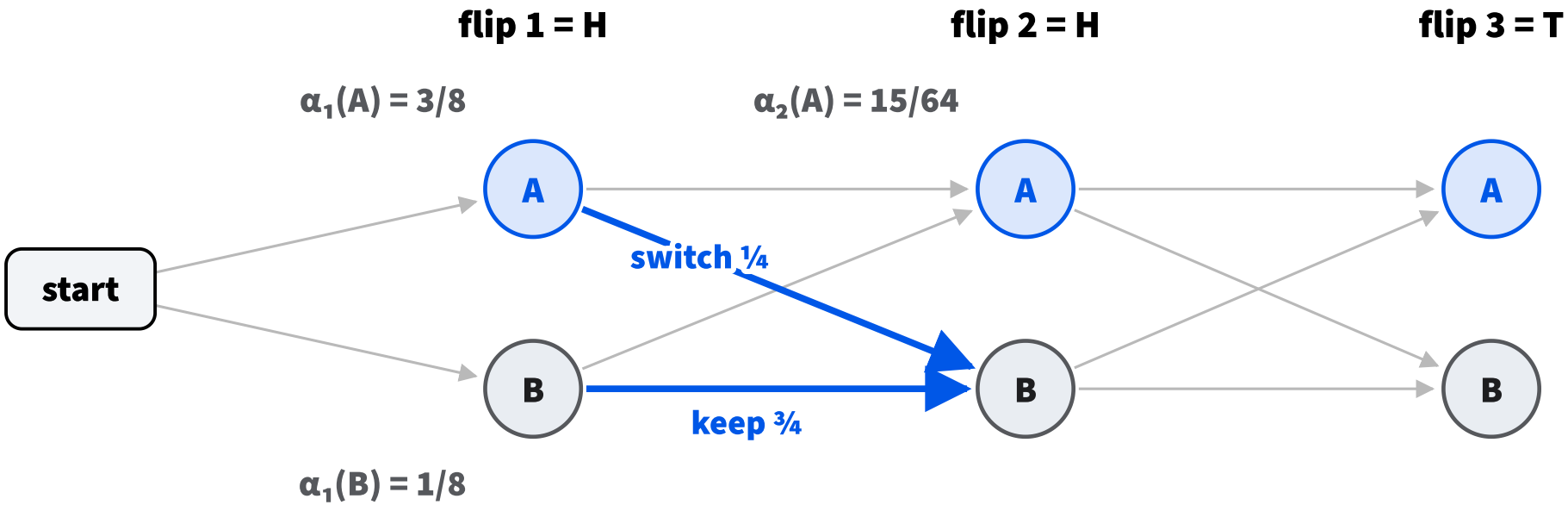


$$\alpha_2(A) = \frac{10}{32} \cdot \underbrace{\frac{3}{4}}_{\text{A shows H}} = \frac{30}{128} = \frac{15}{64}$$

We reuse α_1, so we never return to the start.

Step 4: the same two paths now lead into coin B

Coin 2 is B if the friend **was at A and switched**, *or* **was at B and kept**. Then B must show H.

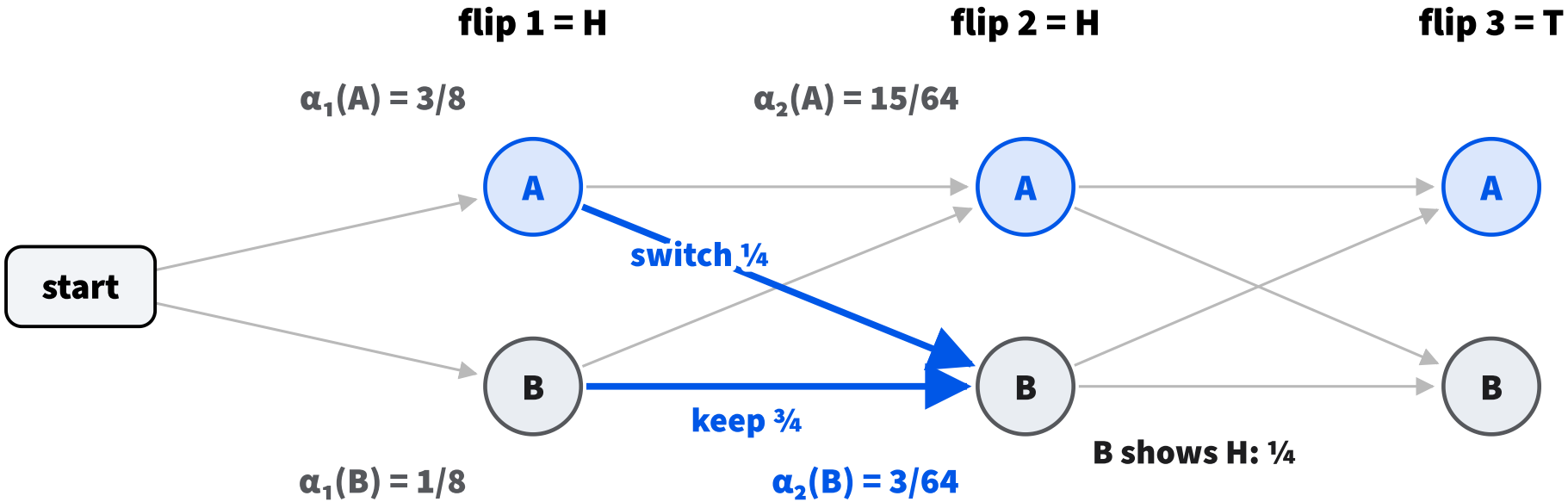


$$\text{arrive at B} = \underbrace{\tfrac{3}{8}\cdot\tfrac{1}{4}}_{\text{was A, switch}} + \underbrace{\tfrac{1}{8}\cdot\tfrac{3}{4}}_{\text{was B, keep}} = \tfrac{3}{32}+\tfrac{3}{32} = \tfrac{6}{32}$$

Adding the two ways in gives the chance of arriving at coin B.

Step 5: coin B must show H, so we multiply by its heads chance

After arriving at coin B, the friend flips it, and B must show H. For "and", we multiply again.

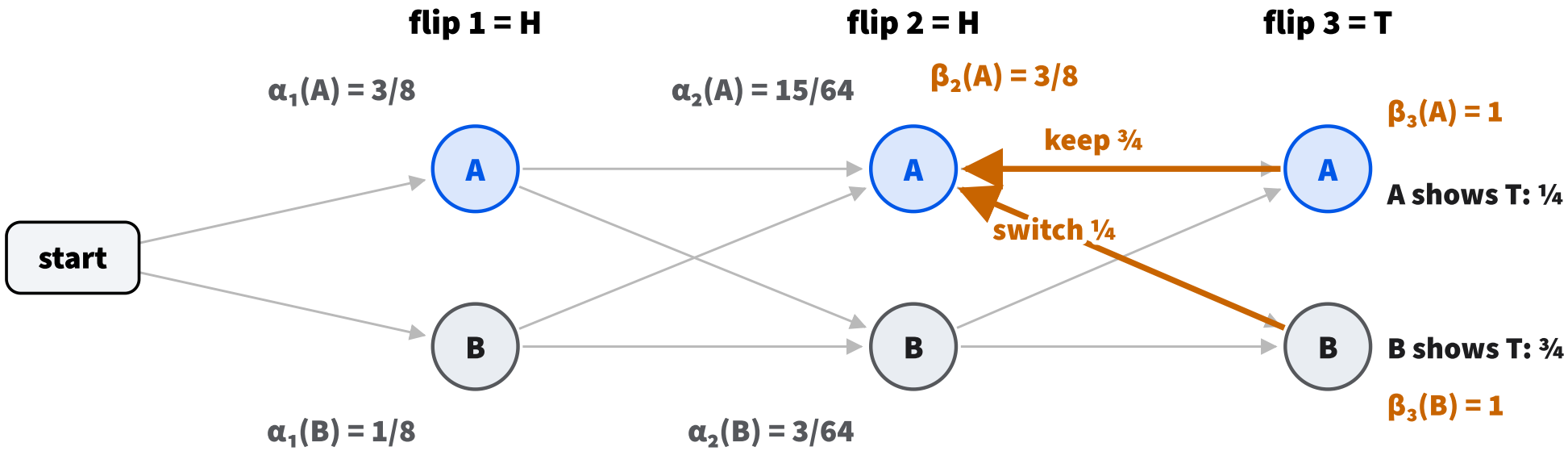


$$\alpha_2(B) = \frac{6}{32} \cdot \underbrace{\frac{1}{4}}_{\text{B shows H}} = \frac{6}{128} = \frac{3}{64}$$

After two heads, coin A is five times as likely as coin B.

Step 6: from coin A, we find the chance of the last flip T

The backward message starts at the end. After flip 3, nothing is left to explain, so $\beta_3(A) = \beta_3(B) = 1$. From A at flip 2, the friend **keeps** (A shows T) *or* **switches** (B shows T).

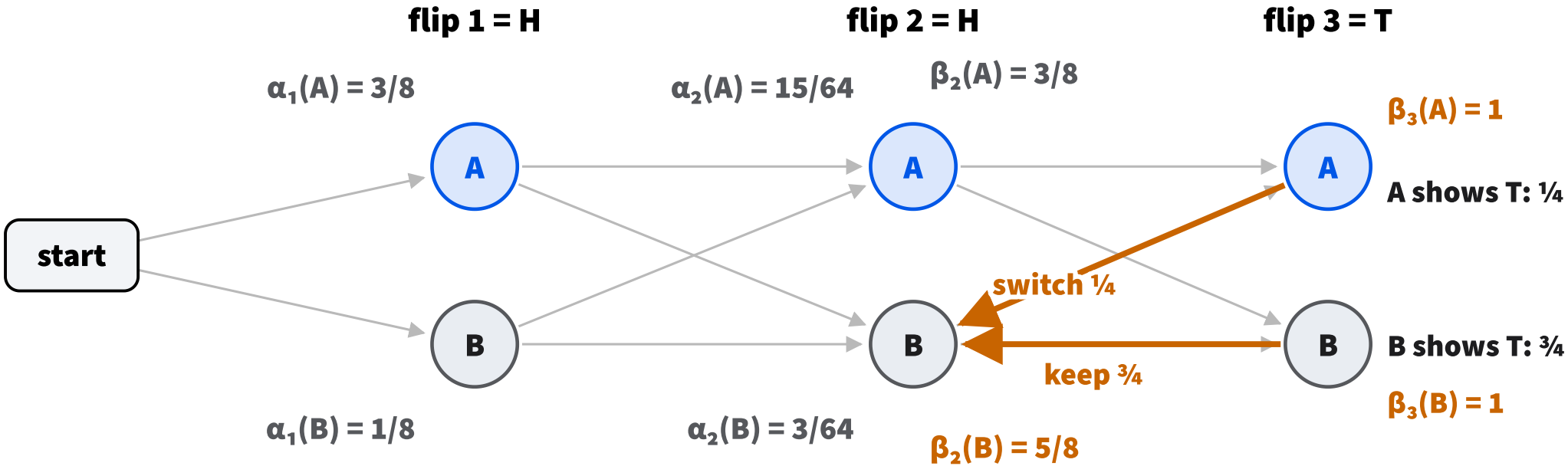


$$\beta_2(A) = \underbrace{\tfrac{3}{4} \cdot \tfrac{1}{4}}_{\text{keep, A shows T}} \cdot 1 + \underbrace{\tfrac{1}{4} \cdot \tfrac{3}{4}}_{\text{switch, B shows T}} \cdot 1 = \tfrac{3}{16} + \tfrac{3}{16} = \tfrac{6}{16} = \tfrac{3}{8}$$

The backward message runs the same rule, from the end towards the start.

Step 7: from coin B, the last flip T is more likely

From B at flip 2, the friend **switches** (A shows T) *or* **keeps** (B shows T).



$$\beta_2(B) = \underbrace{\tfrac{1}{4}\cdot\tfrac{1}{4}}_{\text{switch, A shows T}} + \underbrace{\tfrac{3}{4}\cdot\tfrac{3}{4}}_{\text{keep, B shows T}} = \tfrac{1}{16} + \tfrac{9}{16} = \tfrac{10}{16} = \tfrac{5}{8}$$

Tails favours coin B, so $\beta_2(B)$ is larger than $\beta_2(A)$.

Multiplying the two messages gives chance 0.75 that coin 2 is A

To combine the past *and* the future, multiply each coin's **forward message** by its **backward message**.

coin 2	forward × backward	= weight
A	$\frac{15}{64} \cdot \frac{3}{8}$	$\frac{45}{512}$
B	$\frac{3}{64} \cdot \frac{5}{8}$	$\frac{15}{512}$
sum	**chance of all the flips,** $p(\mathrm{HHT})$	$\frac{60}{512} = \frac{15}{128}$

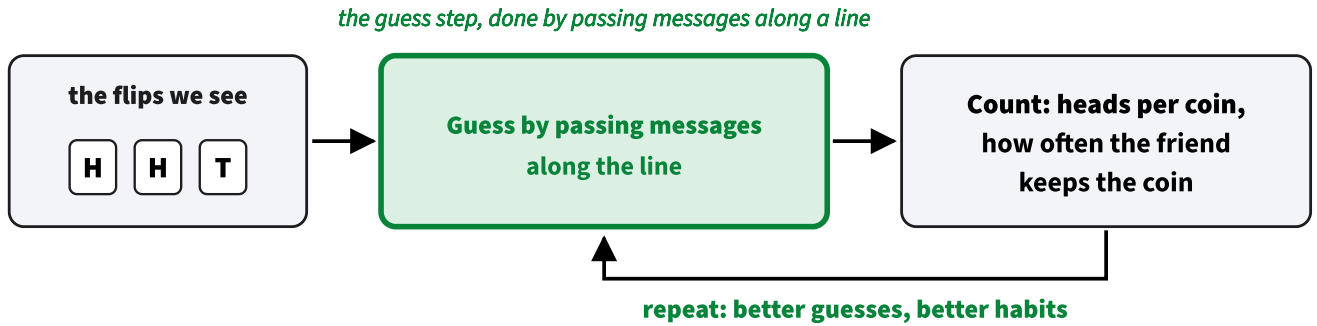
$$\text{chance coin 2 is A} = \frac{45}{45+15} = \mathbf{0.75} \qquad \text{(the same at flips 1 and 3 gives 0.8 and 0.4)}$$

Seven small sum-product steps replace listing all 8 stories.

We count the habits with the messages' guesses, then repeat

Take **coin A's heads chance**. The messages say coin A made flips 1, 2, 3 with chances 0.8, 0.75, 0.4 **(Lecture 6, p. 24)**. Count:

1. **heads made by A** = 0.8 + 0.75 = 1.55, because only flips 1 and 2 are H;
2. **flips made by A** = 0.8 + 0.75 + 0.4 = 1.95;
3. **new heads chance of A** = 1.55 ÷ 1.95 ≈ 0.795.



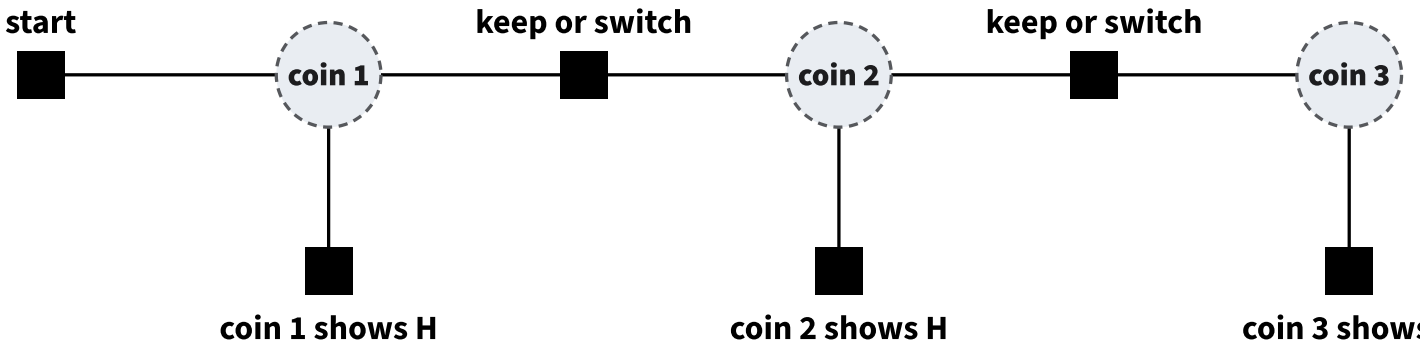
New heads chance of A $= \dfrac{\sum_{t:\, x_t = \mathrm{H}} p(z_t = A \mid x)}{\sum_t p(z_t = A \mid x)}$; coin B and the keep chance: the same way.

Count with the messages, pass new messages, repeat: this is Baum–Welch, expectation–maximization (EM) for coins in a line.

Without a cycle, these messages give the exact posterior

The messages gave 0.8, 0.75, 0.4: **exactly** what listing all 8 stories gives, because

1. the coins form a **line**, so the factor graph has **no cycle**;
2. so each flip's news reaches a coin by **one path**: **nothing is counted twice**.



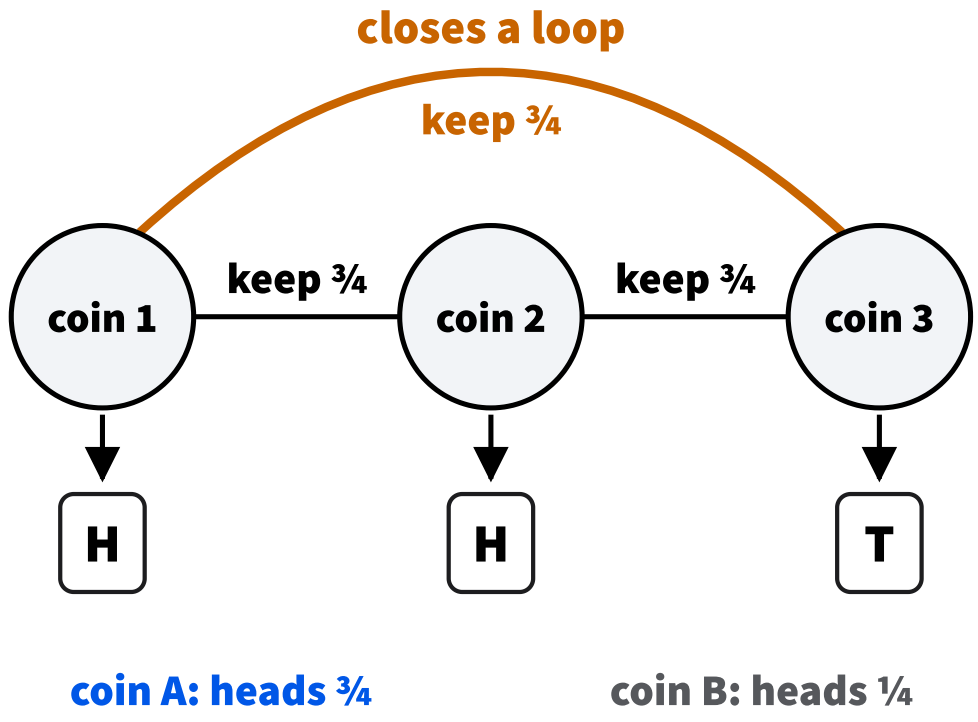
Forward–backward theorem (Lecture 5, p. 71). On any chain, at every flip t, $p(x) = \sum_z \alpha_t(z)\,\beta_t(z)$ and $p(z_t{=}z \mid x) = \dfrac{\alpha_t(z)\,\beta_t(z)}{p(x)}$, exactly.

This is Lecture 2's Theorem 1, **sum-product on a tree**: no cycle, so **every marginal is exact**.

With no cycle, each flip is counted once, so the posterior is exact.

If the line becomes a circle, the messages have no starting point

If coin 3 also depends on coin 1, the coins form a **circle**, and **no first coin** exists to start the forward message.



Messages along a line are exact only when the coins really form a Markov chain.

Way 2: when we cannot add up the stories, act them out



As in Lecture 4, instead of adding up every case, we **try many cases at random** and count.

Way 2 is correct on average but needs many runs.

Replay the game many times and keep the runs that match

1. **Play the friend's game:** choose coin 1 at random, keep or switch with the friend's chances, and flip each coin.
2. **Compare:** if your flips are H, H, T, **keep** the run; otherwise, **discard it**.
3. **Count:** among the kept runs, find how often coin 2 was A.

run	coins we picked	flips they showed	same as H, H, T?
1	A A B	H H T	✓ yes: keep
2	A A A	H T H	✗ no: throw away
3	B B A	T H T	✗ no: throw away
4	B A B	H H T	✓ yes: keep
5	A B B	H H T	✓ yes: keep
6	B B B	T T T	✗ no: throw away

kept runs 1, 4, 5: coin 2 (thick ring) was A in 2 of these 3

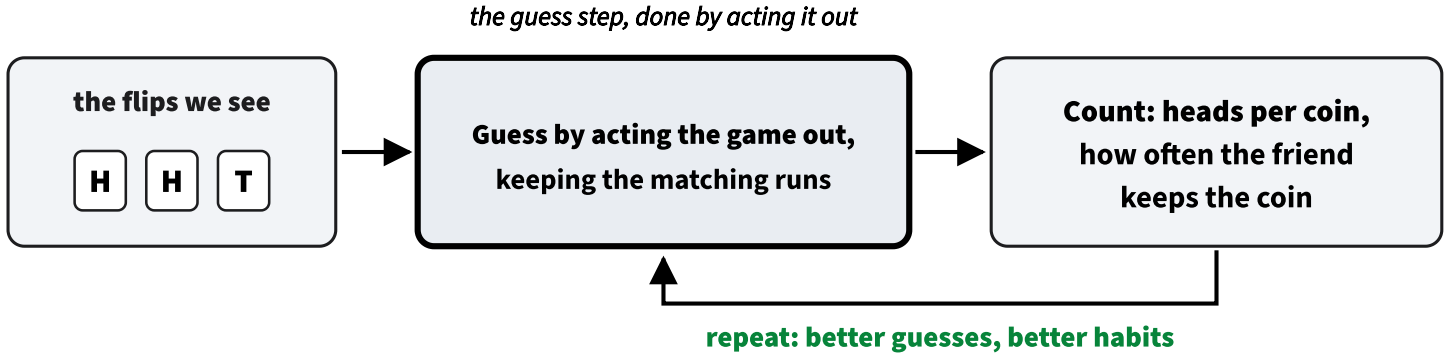
with many more runs, this share converges to the true chance

The kept runs appear in the right proportions, so no sum over all stories is needed.

In a kept run we see the coins, so we can simply count

The count step needs each coin's **flips and heads** (Lecture 5, p. 11). In a kept run **we chose the coins ourselves**, so we can count them.

1. **Kept run 1**, A, A, B over H, H, T (Lecture 6, p. 29): coin A **2 flips, 2 heads**; coin B **1 flip, 0 heads**; **kept once, switched once**.
2. **Add the counts over all kept runs**: the new habits, as in Baum–Welch's count step (Lecture 6, p. 118), e.g. coin A heads **0.795** (Lecture 6, p. 122).



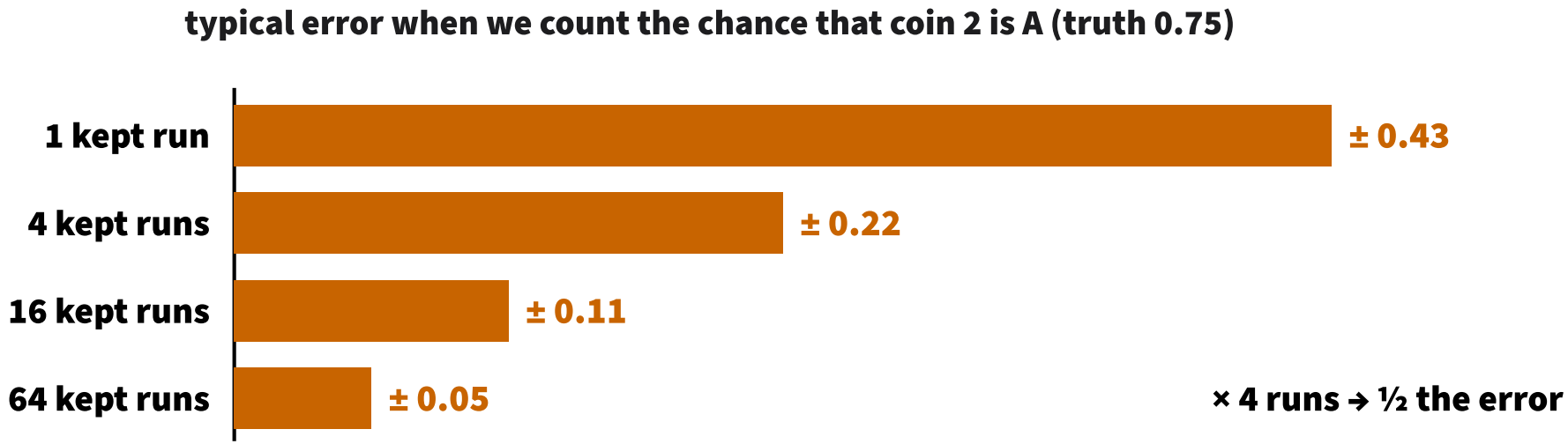
They agree because kept runs come in the **right proportions**. This is **Monte Carlo expectation–maximization (EM)**: acting out replaces the guess step.

> **Simulate, keep, count, and repeat.**

Random runs add error; four times as many runs halve it

Take the **chance that coin 2 is A**, truly 0.75, counted from the kept runs:

1. **16 kept runs**: typically off by about 0.11;
2. **64 kept runs**, four times as many: off by about 0.05, **half as much**;
3. halving it again needs **four times as many again**.



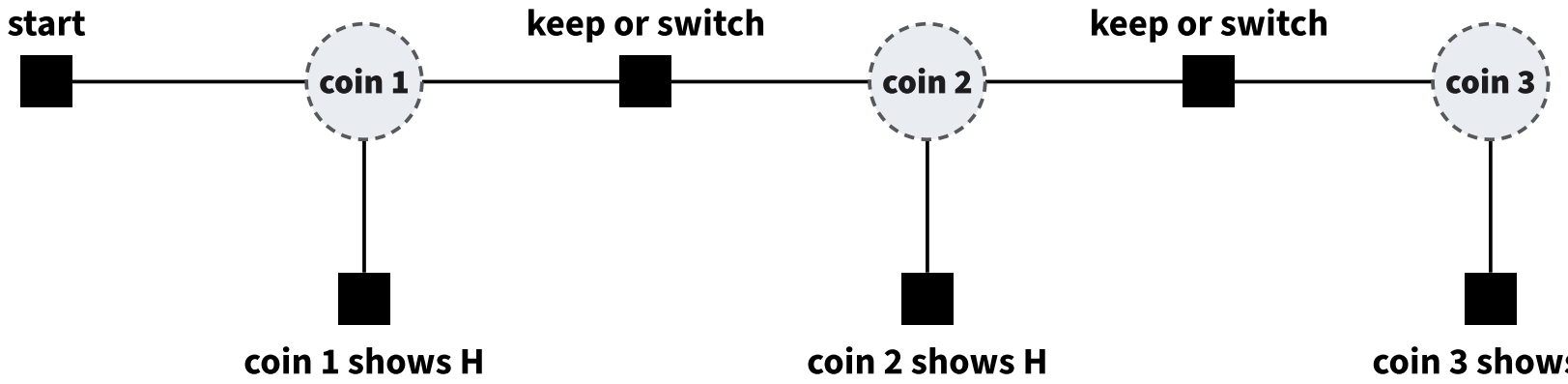
Typical error, the standard deviation: $\sqrt{q\,(1-q)/m}$, with q the true chance and m the kept runs

The error falls like one over the square root of the runs: correct on average, but slow.

No first coin on a circle, so we redraw one coin at a time

Replaying (Lecture 6, p. 29) went **in order**, coin 1, 2, 3. On a circle coin 3 ties back to coin 1: **no first coin**. Instead:

1. Keep a **current story**, say A, A, B. Pick one coin, say coin 2.
2. **Redraw** coin 2 from the **three things that touch it**: coin 1, coin 3 and its own flip H (worked: (Lecture 6, p. 121)).
3. Repeat round the circle **many times**, and **count** how often each coin is A.



In symbols: $p(z_2 \mid z_1, z_3, x_2) \propto \underbrace{p(z_2 \mid z_1)}_{\text{move } 1\to 2}\,\underbrace{p(z_3 \mid z_2)}_{\text{move } 2\to 3}\,\underbrace{p(x_2 \mid z_2)}_{\text{its flip}}$: only the squares touching coin 2.

This is Gibbs sampling from Week 4: in the long run the counts are exact (Lecture 6, p. 66).

Way 3: let each coin guess alone, which is fast



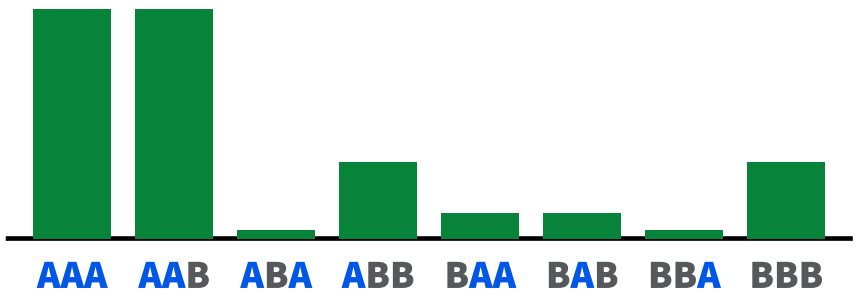
Acting out is slow. What if each coin made **its own guess**, using its neighbours' guesses?

> **Way 3 gives up a little accuracy for much more speed.**

Mean field approximates the joint by a product

1. **The truth** $p(z \mid x)$ is one table over all **8 stories**: 7 free numbers.
2. **Mean field** replaces it by a **product of three one-coin tables**: $q(z) = q_1(z_1)\, q_2(z_2)\, q_3(z_3)$, 3 numbers.
3. **The price:** a product cannot say "coins 1 and 2 agree". Each coin's table is chosen **using the other two** **(Lecture 6, p. 37)**.

the truth p(z | x): one table over the 8 stories



7 free numbers

≈

mean field q: three one-coin tables, multiplied

coin 1: q_1	
A	0.866
B	0.134

×

coin 2: q_2	
A	0.848
B	0.152

×

coin 3: q_3	
A	0.417
B	0.583

3 numbers

one story: $q(A, A, B) = q_1(A) \cdot q_2(A) \cdot q_3(B)$
$= 0.866 \cdot 0.848 \cdot 0.583 = 0.428$

Joint table → product of small tables: cheap, but no links between coins.

Among all products, choose the one with the best score

Mean field: $\max_{q_1,q_2,q_3} B(q_1 q_2 q_3, \theta)$: the product with the **largest score (Lecture 6, p. 9)**.



The simplest **variational inference**: guessing becomes **maximizing a score over a family**, here the products.

Three numbers instead of a table over every story.

Theorem A: guessing the coins is climbing the score

Theorem A (variational principle). For every guess q: $\ln p(x) = B(q) + \mathrm{KL}\big(q \,\|\, p(z \mid x)\big)$, with $\mathrm{KL} \geq 0$ and zero only at the posterior (Lecture 5, p. 47). So the **best guess is the posterior**, and no score passes $\ln p(x)$.



Insight. Too many stories? **Climb inside a smaller family.** The climb is cheap, the height is a **safe lower bound**, and the gap is the **price of the family**.

Remember: best guess = posterior; smaller family = cheaper guess + a gap.

 KL = Kullback–Leibler

Each update uses the neighbours' current tables as weights

To update coin 2, **fix** q_1 and q_3 and use them as **weights**:

$$q_2(z_2) \;\propto\; \underbrace{p(x_2 \mid z_2)}_{\text{own flip}} \cdot \exp\Big(\underbrace{\sum_{z_1} q_1(z_1)\,\ln p(z_2 \mid z_1)}_{\text{coin 1's table as weights}}\Big) \cdot \exp\Big(\underbrace{\sum_{z_3} q_3(z_3)\,\ln p(z_3 \mid z_2)}_{\text{coin 3's table as weights}}\Big)$$

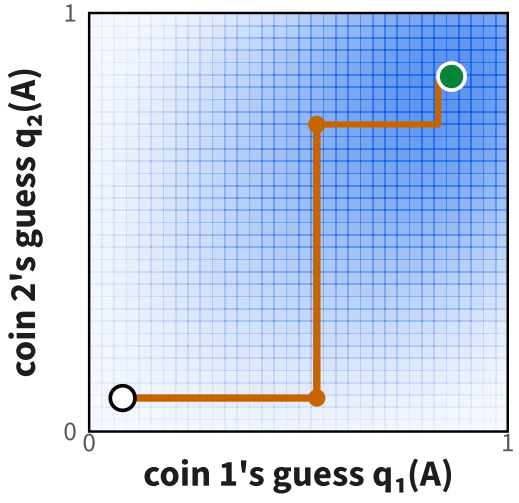
The weights are **fixed numbers**, so q stays a **product**.

what coin 1 (flip H, table 0.866) sends coin 2	combines	uses	says A
Way 1: a sum-product message	**a sum of tables**	**all but what coin 2 sent back**	**0.625**
Way 3: its whole current table	**average of logs**	**all, including coin 2's push**	**0.691**

Message passing, but not Way 1's: each update raises the score, so the updates converge (stop changing).

Theorem B: each mean-field update can only raise the score

Theorem B (coordinate ascent). With the other coins fixed, B is largest at $q_t(z_t) \propto \exp\left(\mathbb{E}_{\text{others}}[\ln p(x, z)]\right)$ (Lecture 6, p. 37). Updating the coins in turn **never lowers** B.



darker = higher score B (coin 3 fixed at 0.417)

○ **start in the corner: coins 1, 2 at 0.08**

→ **coin 1: left–right only, to the best point**

↑ **coin 2: up–down only, to the best point**

each move goes uphill: B never falls

● **stop at (0.866, 0.848):**
no single move helps, a local top

Proof in one line: with the others fixed, $B = -\mathrm{KL}(q_t \,\|\, \tilde{q}_t) + \text{const}$, and a Kullback–Leibler (KL) is ≥ 0 (Lecture 5, p. 47).

Insight. **One dial at a time**, each to its best position: never downhill. It stops where no single dial helps: a **local top**.

Remember: one coin at a time, always uphill, to a local top.

Try it: coin 1 leans to A, so does coin 2's guess go up or down?

Coin 2 showed H: alone, it says A with 0.75. Its neighbours' tables: coin 1 is A with 0.75; coin 3 with 0.5, no opinion. After one update, coin 2's guess of A is …

below 0.75 | still 0.75 | above 0.75

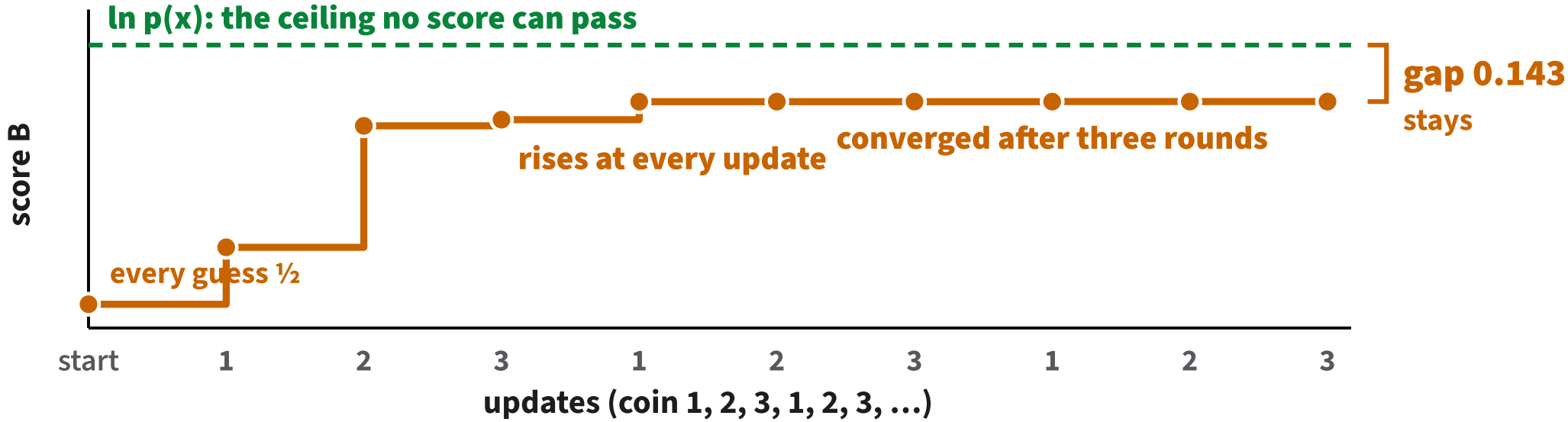
Answer. Above: 0.839.

1. **Coin 3 adds nothing:** a table at 0.5 has no opinion.
2. **Coin 1 adds evidence for A:** the friend usually keeps, so coin 2 probably follows coin 1.
3. The rule **adds that evidence** to coin 2's own flip: 0.75 rises to 0.839 **(Lecture 6, p. 119)**.

A neighbour's lean counts as extra evidence, so the guesses push each other up.

On H, H, T, every update raises the score B

Start from **every table at ½** and update coin 1, 2, 3 in turn with the rule (Lecture 6, p. 37): the score **rises at every update** and **converges after three rounds**.



It converges **below the ceiling** $\ln p(x)$, and the converged chances are **too sure**. All guesses and scores: (Lecture 6, p. 120).

The score never falls and converges fast, but it stops short of $\ln p(x)$.

Separate guesses cannot express that two coins tend to agree

Take **coins 1 and 2** on the line and compare their four pairs:

1. the truth gives **both B** a real chance, because the friend usually keeps;
2. separate guesses **multiply** q_1 by q_2, so cannot say "coin 2 follows coin 1";
3. to make **both A** likely, they push both towards A and **almost rule out both B**.

the truth

	coin 2 is A	coin 2 is B
coin 1 is A	0.675	0.125
coin 1 is B	0.075	0.125

both B keeps a real chance

separate guesses $q_1 \times q_2$

	coin 2 is A	coin 2 is B
coin 1 is A	0.734	0.131
coin 1 is B	0.114	0.020

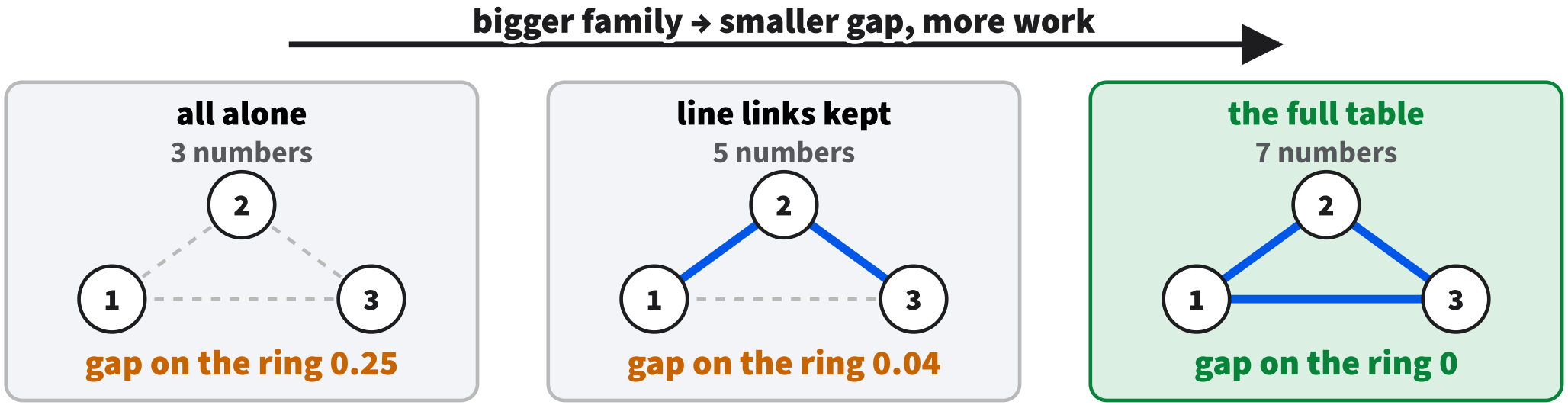
both B almost ruled out: too sure of A

Too confident means **narrower** than the truth: weight on fewer stories, a smaller entropy (Week 3), and a coin agreeing with its neighbours gets **0.866** where the truth is **0.8**.

Fast and scalable, but narrower than the truth: overconfident.

Turn the family dial: keep some links inside the guess

1. **All alone:** one-coin tables multiplied; 3 numbers.
2. **Line links kept:** the guess may say "coin 2 follows coin 1", never the ring link; 5 numbers.
3. **The full table:** 7 numbers, every link; the guess can be the truth.



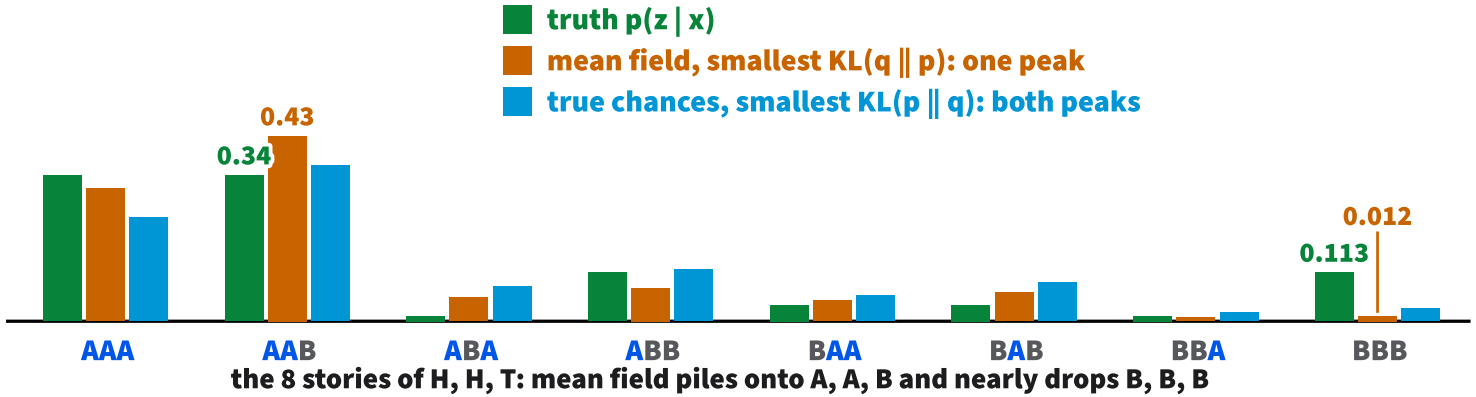
On the line, the line links already **close the gap (Lecture 6, p. 126)**. This is **structured mean field**.

Bigger family: smaller gap, more work.

Theorem C: why mean field is too sure

1. **The truth has two peaks:** all A (A, A, A and A, A, B) and all B (B, B, B). Between them, a **valley** of mixed stories.
2. **A product is one tent:** it can cover both peaks only by covering the valley too.
3. **Mean field** minimizes $\mathrm{KL}(q\|p)$, which **punishes the valley**, not a missed peak. So the tent covers the **big peak only**: B, B, B drops to 0.01. **Too sure.**
4. **The other direction** $\mathrm{KL}(p\|q)$ punishes a **missed peak**: both peaks, the true chances. But it needs the truth.

Same formula, q and p swapped: $\mathrm{KL}(q\|p) = \sum_z q \ln \frac{q}{p}$, weighted by the **guess**, so a peak the guess leaves out costs nothing; $\mathrm{KL}(p\|q) = \sum_z p \ln \frac{p}{q}$, weighted by the **truth**, so a missed peak costs a lot.

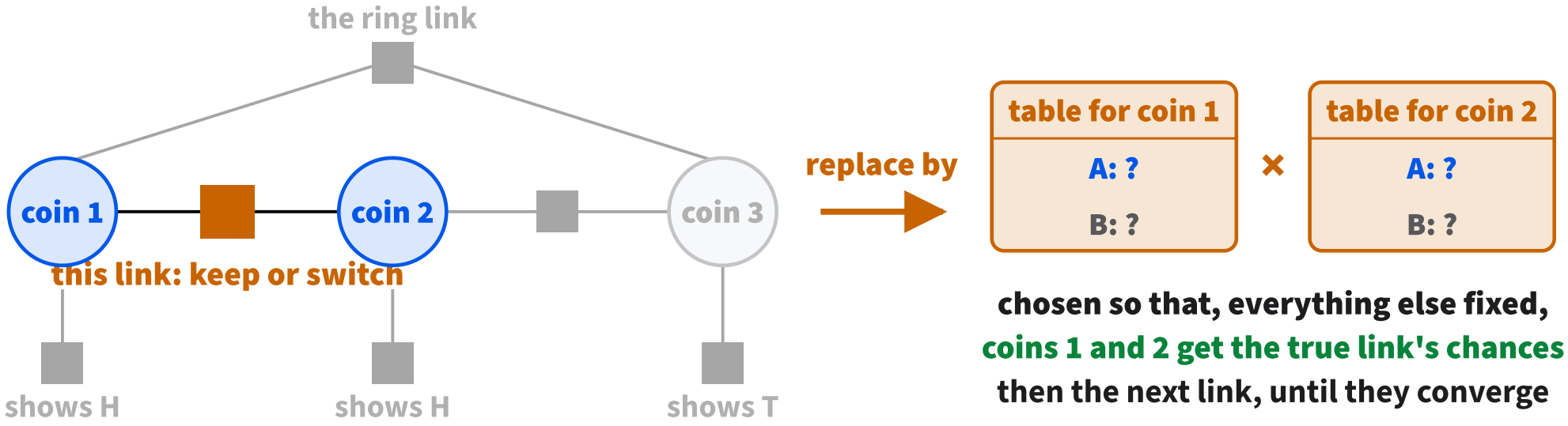


One tent, two peaks: mean field covers the big one and calls it certain.

 KL = Kullback–Leibler

Let the truth judge, one link at a time

1. **Theorem C:** the truth judging, $\mathrm{KL}(p\|q)$, needs the **whole posterior**: impossible.
2. But it can judge **one link at a time**: replace that link by **two one-coin tables**, chosen so that, everything else fixed, the two coins' chances **match the true link's**.
3. Go round the links until the tables **stop changing**. This is **expectation propagation**.



The truth still judges, one link at a time.

 KL = Kullback–Leibler

On our coins, this is exactly Way 4

1. On the line: the **exact chances**.
2. On the ring: **0.787**, the number of messages in a circle **(Lecture 6, p. 55)** (all coins: **(Lecture 6, p. 127)**).
3. Because matching two coins' chances under one true link is **exactly what a message does**.

coin 1's chance of A	line	ring
the truth	**0.8**	**0.75**
one link at a time (expectation propagation)	**0.8**	**0.787**
messages in a circle (Way 4)	**0.8**	**0.787**

Insight. The guess judges all at once (Way 3); the truth judges one link at a time (Way 4): **the two directions of Kullback–Leibler (KL), made local.**

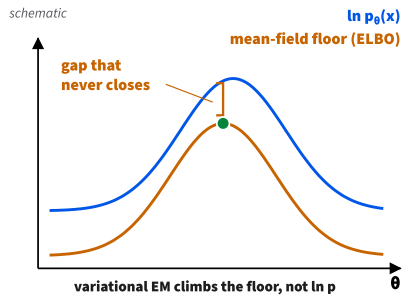
Two directions of KL, two ways out.

Theorem E: variational EM climbs the score, not the likelihood

Theorem E (variational expectation–maximization (EM)). Alternate a mean-field guess step and the count step: each raises B. But $\ln p(x) = B + \text{gap}$ and the gap moves with θ, so $\ln p(x)$ **can fall**. With the exact guess the gap is 0: Theorem 4 **(Lecture 5, p. 49)**.

H, H, T, T: one round **raises** B but **lowers** $\ln p(x)$ from −2.936 to −2.967 **(Lecture 6, p. 123)**.

1. **Guess step:** labels too sure, so the floor sits far below the house.
2. **Count step:** the habits move to where the **floor** is high.
3. The floor rose; the house **dipped**.



Insight. We climb the floor, not the house, so the habits drift to where the **family fits well**.

Remember: exact guess, the likelihood rises; approximate guess, only the score rises.

Way 4: pass messages even when the coins form a circle



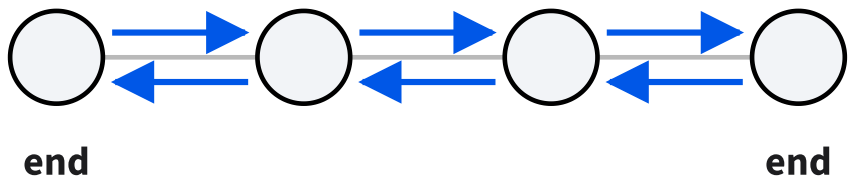
The messages of Way 1 were **exact on a line**. What happens if we pass them around a circle?

> **Way 4 keeps the links between coins that Way 3 drops.**

On a line, messages stop at the ends; on a circle, they return

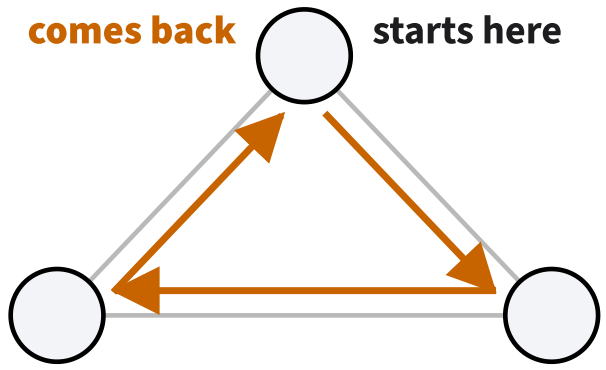
We pass the same messages around the circle **repeatedly** until they **converge**.

a line ✓



a line: messages stop at the ends — exact

a circle



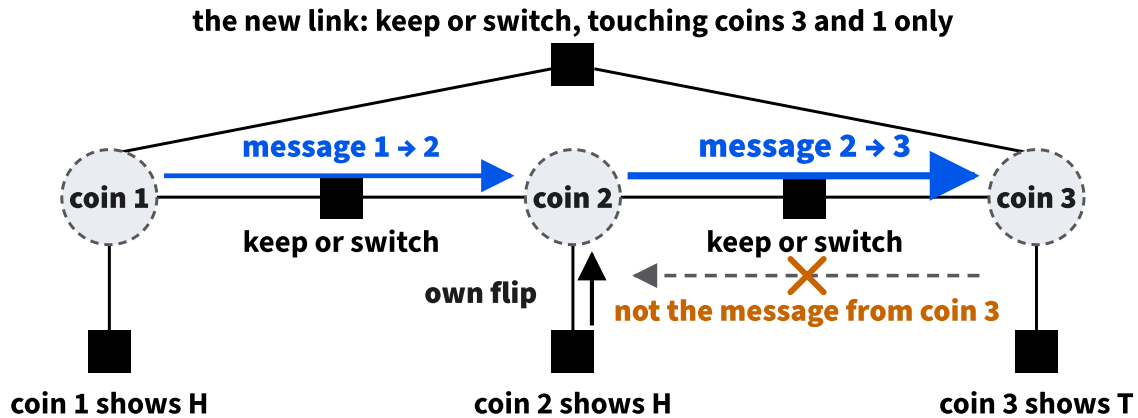
a circle: your own message comes back — counted twice

On a circle, a coin's own evidence returns to it and is counted twice.

A message carries a coin's news, except the receiver's own

Take **coin 2's message to coin 3**. It combines:

1. coin 2's **own flip**;
2. the **messages coin 2 received** from its other neighbours, here coin 1;
3. **not** coin 3's message, because coin 3 already knows that news.



$\mu_{2\to 3}(z_3) = \sum_{z_2} p(z_3 \mid z_2)\, p(x_2 \mid z_2)\, \mu_{1\to 2}(z_2)$ — the sum-product rule

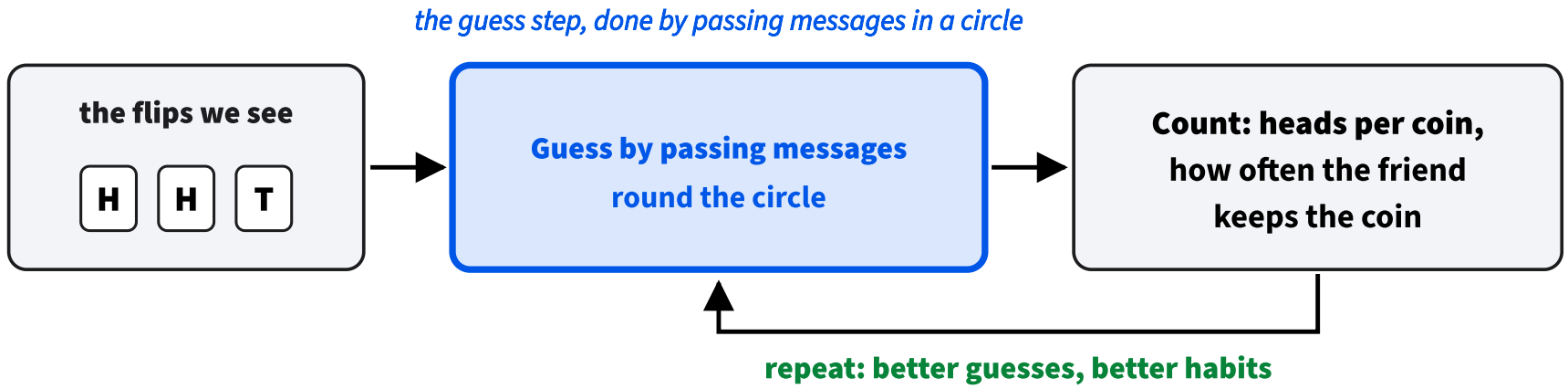
Each coin's **guess** = own flip × all messages received, rescaled to sum to 1.

This is loopy belief propagation: the sum-product rule on a graph with cycles.

On a circle, converged messages give an approximate answer

On a line, the messages are exact **(Lecture 6, p. 26)**. **On a circle,** we use the same rule anyway:

1. **Start** every message at "no information" $(\frac{1}{2}, \frac{1}{2})$.
2. **Update** all messages with the rule, round after round.
3. **Stop** when they have converged, and read off each coin's guess.



The answer is **approximate**: a coin's own evidence returns round the circle and is **counted again**. The messages need not converge.

> **Fast and usually accurate, but without a guarantee.**

Theorem D: on one circle, the messages pick the right coin

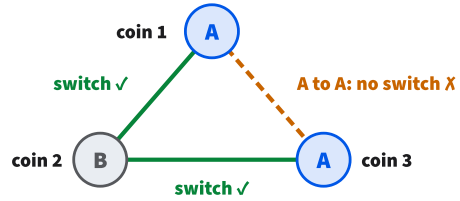
Margin = chance of A − chance of B. **Echo** e: send coin 1's margin once round the circle. Each link keeps only **keep − switch = ½** of it, each flip trims it a little; the fraction that **comes back** is the echo. Our ring: $e = 0.07$.

Theorem D (one circle, two-valued coins). The messages converge, and at every coin

$$\text{true margin} = \frac{1-e}{1+e} \times \text{the messages' margin}.$$

1. The factor is positive: **same winner**.
2. Coins that agree, $e > 0$: **too sure**. Coins that switch, $e < 0$: **not sure enough**.
3. Faint echo, $e \approx 0$: **nearly exact**.

three switches round 3 coins are impossible



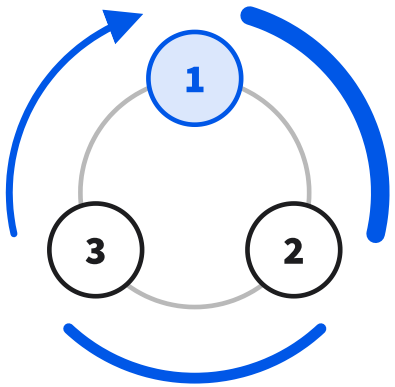
keep 0.05: every link wants a switch

Our ring: factor $\frac{1-0.07}{1+0.07} = 0.87$ **(Lecture 6, p. 125)**.

Remember: on one circle the messages pick the right coin; only their confidence is off, by the echo.

Insight: each link shrinks the echo, or flips its sign

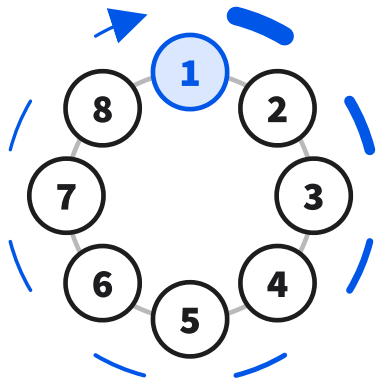
1. Coin 1's margin goes round the circle. **Each link multiplies it by keep − switch.**
2. **Keep ¾:** ×½ per link. After 3 links only ⅛ comes back: a **faint echo**, small error. After 8 links, almost nothing.
3. **Keep 0.05:** ×(−0.9) per link. After 3 links −0.73 comes back: a **strong echo pointing the wrong way**, large error.



circle of 3 coins

returns after 3 links:
still strong, counted twice

error 0.037



circle of 8 coins

returns after 8 links:
almost gone

error 0.0003

The echo and the error for circles of 3 to 8 coins: **(Lecture 6, p. 124)**.

Long circles or weak links: faint echo, accurate messages. Frequent switching: reversed echo, unreliable messages.

What each way costs for n coins with K values

Count the **numbers each way must touch** before it answers once:

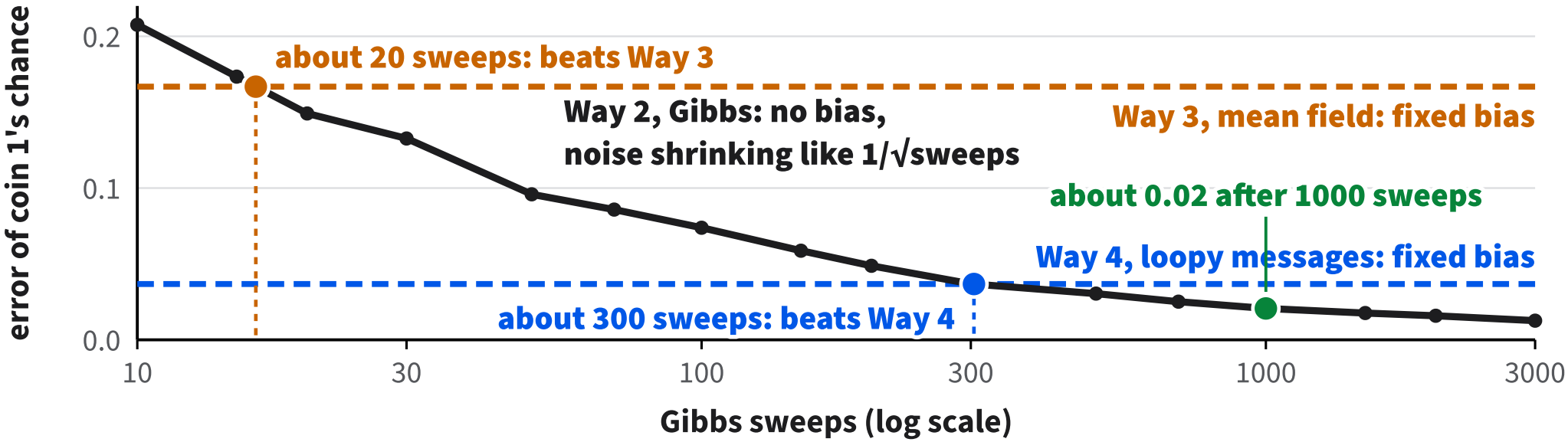
way	work	3 coins, K = 2	300 coins, K = 2
list every story	K^n stories	8	more than atoms in the universe
1 messages along a line	$n\,K^2$ per pass	12	1 200
2 act it out	$n\,K$ per sweep	6	600
3 everyone guesses alone	$n\,K$ per round	6	600
4 messages in a circle	links $\times K^2$ per round	12	1 200

With 300 coins on a ring, **only the last three are possible**: a line needs a line, and 2^{300} stories cannot be listed.

Listing grows with the power; the ways out grow with the count.

Bias you cannot shrink, noise you can

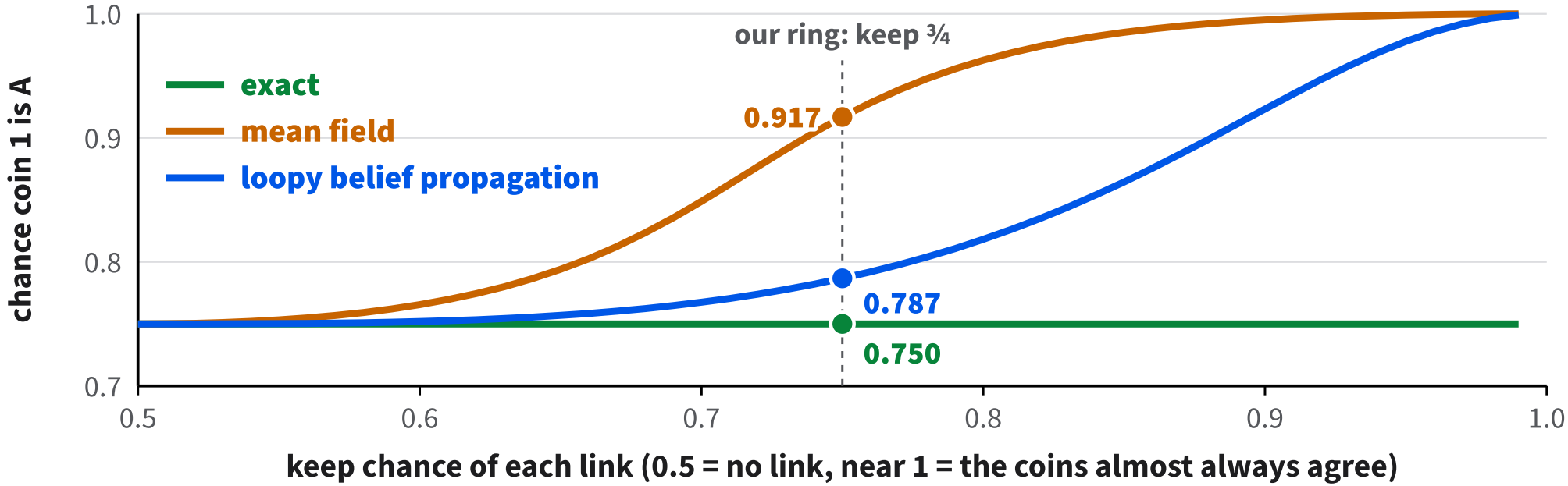
1. **Approximations** (Ways 3, 4) carry a **fixed bias** and no noise: more rounds do not help.
2. **Sampling** (Way 2) carries no bias, but **noise** that shrinks like $1/\sqrt{\text{sweeps}}$.
3. So **sampling wins** once you can afford about **300 sweeps** here; with fewer, the approximation wins.



Little time: approximate. Much time: sample.

The stronger the links, the more both approximations drift

Take the ring over H, H, T and vary only the **keep chance** of every link. Listing the 8 stories gives coin 1 the exact chance **0.75 at every keep**. **Predict:** which approximation drifts first?



Mean field drifts first, 0.917 at keep ¾: separate guesses push each other towards A **(Lecture 6, p. 41)**.

Loopy messages drift later, 0.787 at keep ¾: returning news is counted twice **(Lecture 6, p. 50)**.

Weak links: both are close. Strong links: check against sampling or an exact method.

On the ring, each way trades bias, noise and speed

All four ways on the ring with keep ¾ (the numbers: **(Lecture 6, p. 69)**):

way	error	cost	guarantee
1 messages along a line	**no line here**	**low**	**exact if no cycle**
2 Gibbs sampling	**random** $\propto 1/\sqrt{\text{runs}}$	**slow**	**right on average**
3 mean field	**biased: too sure**	**fast, converges**	$B \leq \ln p(x)$
4 loopy messages	**biased, smaller**	**fast, may not converge**	**exact on trees**

Ways 3 and 4: **bias, no noise**, the same wrong answer every run. Way 2: **noise, no bias**, a new answer every run, right on average.

No way wins everywhere: choose by cycles, link strength and time.

Before believing an answer, make three simple checks

Before believing the learned habits:

1. **Try other starting guesses.** expectation–maximization (EM) can stop at a poor answer, so start from several and keep the best score.
2. **Names do not matter.** A heads $\frac{1}{4}$, B heads $\frac{3}{4}$ fits H, H, T equally well: two mirror answers are one answer.
3. **Some questions have no answer.** With keep $\frac{1}{2}$, the flips look like one fair coin, so **no method** can find heads $\frac{3}{4}$ and $\frac{1}{4}$.

Trust an answer only after these checks.

Four ways to guess when there are too many stories

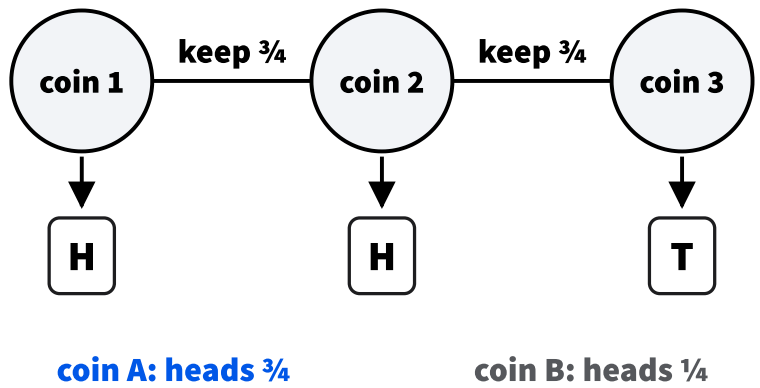
way	picture	good	cost
1	pass messages along a line	exact, low cost	lines only
2	act it out	correct on average, any model	slow
3	everyone guesses alone	fast	overconfident
4	pass messages in a circle	fast, keeps links	approximate

Next: all four ways on one small example.

How close does each way get on the friend's three flips?

We measure one thing: **each coin's chance of being A** given H, H, T, against the **exact answer**.

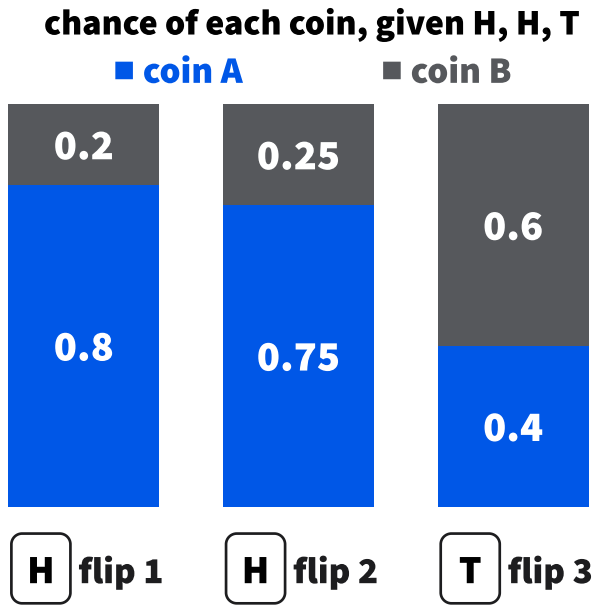
1. **The line:** Way 1 gives the exact answer; we test Ways 2 and 3.
2. **The circle:** link coin 3 to coin 1; the 8 stories give the exact answer; we test Ways 2, 3, 4.
3. **The scoreboard:** all answers on one chart.



Three coins: few enough to know the exact answer and grade every way.

On the line, the exact answer is 0.8, 0.75 and 0.4

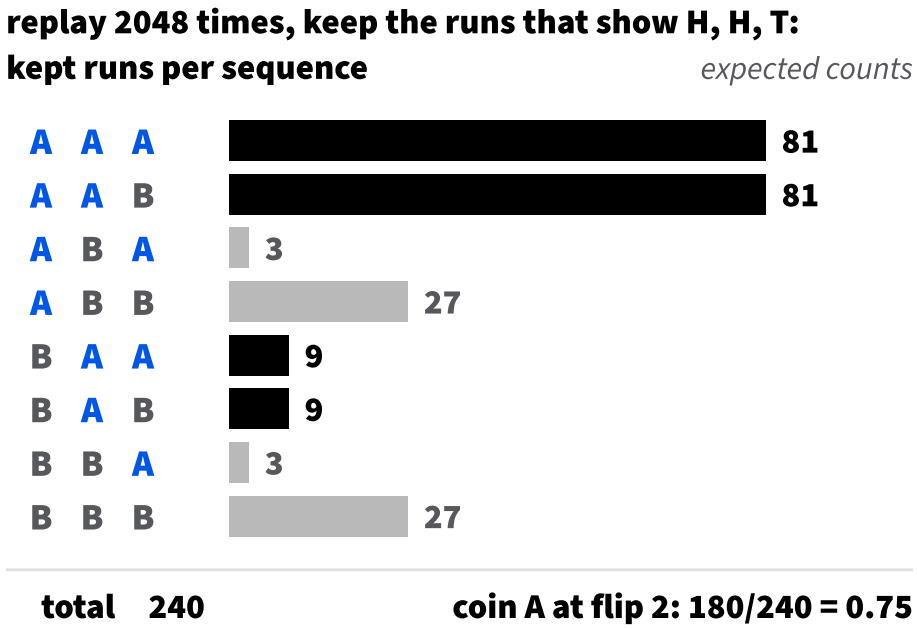
1. **What we compare:** for each coin, the chance that it was **coin A**, given the flips H, H, T.
2. **Where these come from:** Way 1 computed them exactly, step by step **(Lecture 6, p. 24)**. Tails at flip 3 makes coin B likelier there.
3. **What they are for:** they are the **targets**. Ways 2 and 3 are graded by how close they come.



The exact answer on the line: the reference for Ways 2 and 3.

Acting the game out matches the exact answer, but wastes runs

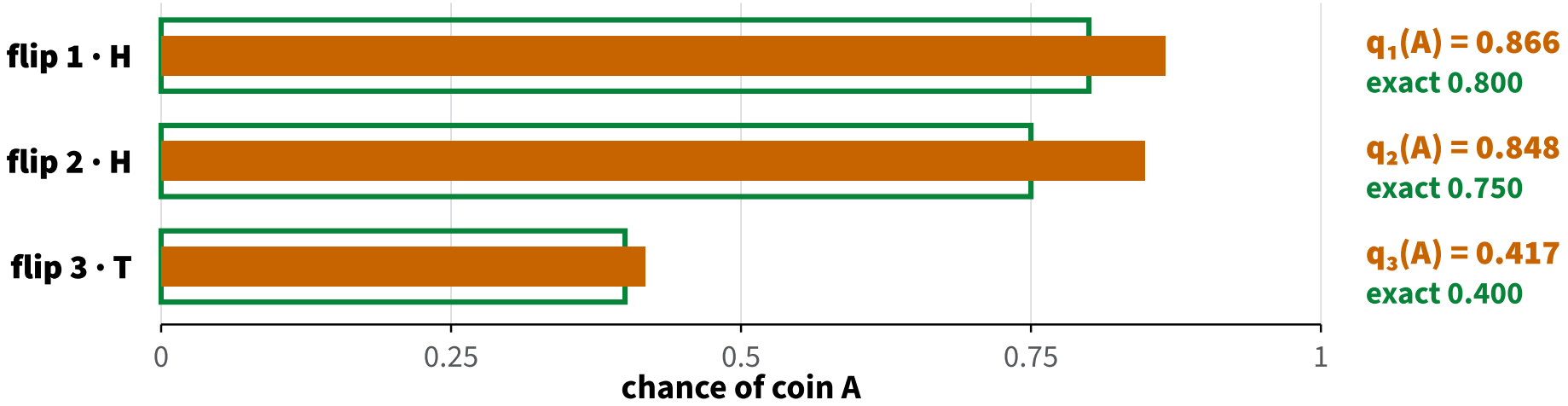
1. **Replay** the game 2048 times. On average only **240 runs** show H, H, T; the other **1808 are thrown away**.
2. Each bar counts the kept runs of one story. **Black bars** are the stories with coin 2 = A.
3. Coin 2 is A in **180 of 240** kept runs: chance **0.75**, the exact value.



Right on average, but most of the work is discarded.

Guessing alone gives 0.866 where the exact answer is 0.8

Press Run: coins 1, 2, 3 update in turn. **Orange bars** are the separate guesses; **green outlines** are the exact chances.



converged: q(A) = **0.866, 0.848, 0.417** · **mean field is too sure: 0.866 vs exact 0.8**

Coins 1 and 2 agree, so each **supports the other**: 0.866 and 0.848 instead of 0.8 and 0.75.

Guessing alone is overconfident where neighbours agree.

On the line, only guessing alone misses the exact answer

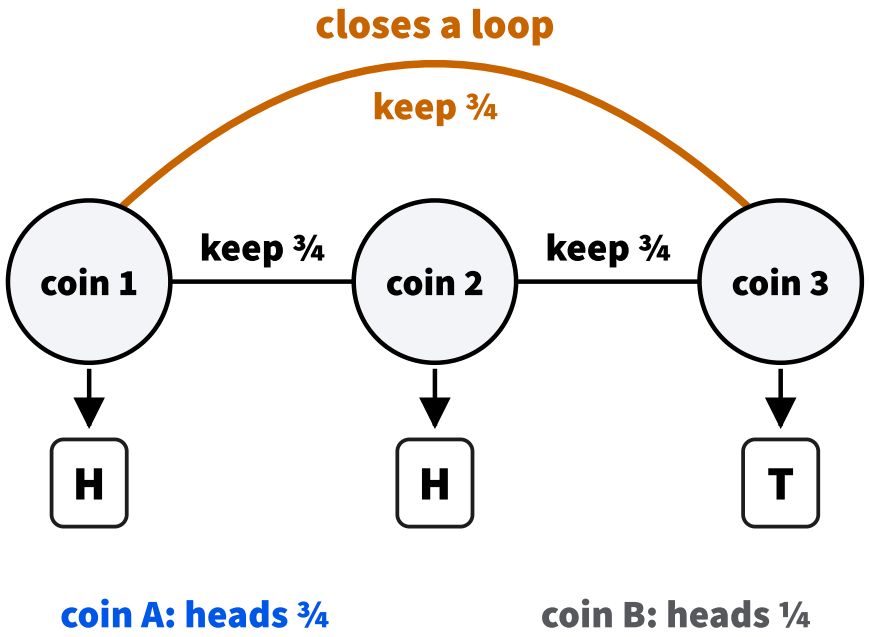
chance of A	coin 1	coin 2	coin 3	verdict
exact (Way 1, messages)	0.8	0.75	0.4	the reference
Way 2, act it out	0.8	0.75	0.4	on average; slow
Way 3, guess alone	0.866	0.848	0.417	too confident

Way 4 is not needed here: on a line, messages in a circle **are** the messages of Way 1.

A line is easy. The real test is a circle.

Linking coin 3 back to coin 1 makes a circle

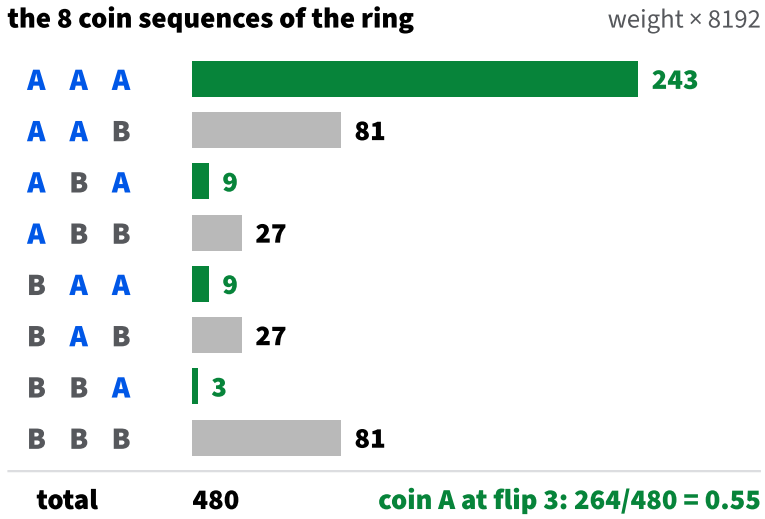
1. The new link uses the same habit: **keep ¾, switch ¼**.
2. Way 1 no longer works, because the messages have **no starting point**.
3. With three coins we can still **list all 8 stories** to get the exact answer.



One extra link turns an easy chain into a hard cycle.

Listing the 8 stories gives the exact chances 0.75, 0.75, 0.55

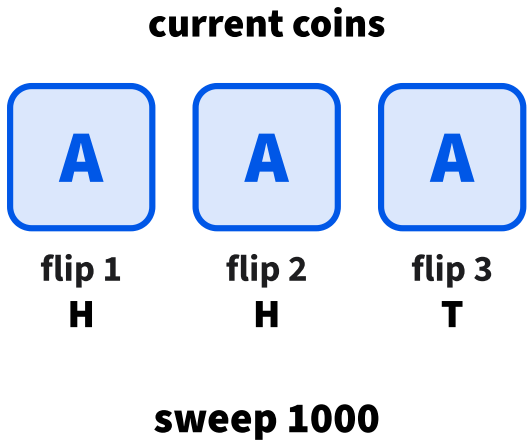
1. A **story** is one choice of coin for each flip, such as A, A, B. Three coins: **8 stories**, one bar each.
2. A story's **weight** is how likely the friend plays it **and** shows H, H, T: multiply the start, every keep or switch (the new link included) and the three flips. (× 8192 only makes whole numbers.) Stories where neighbours agree weigh most, because keep ¾ beats switch ¼.
3. **Chance that coin 3 is A** = weight of the stories with coin 3 = A (green) ÷ weight of all stories = **264 ÷ 480 = 0.55**.

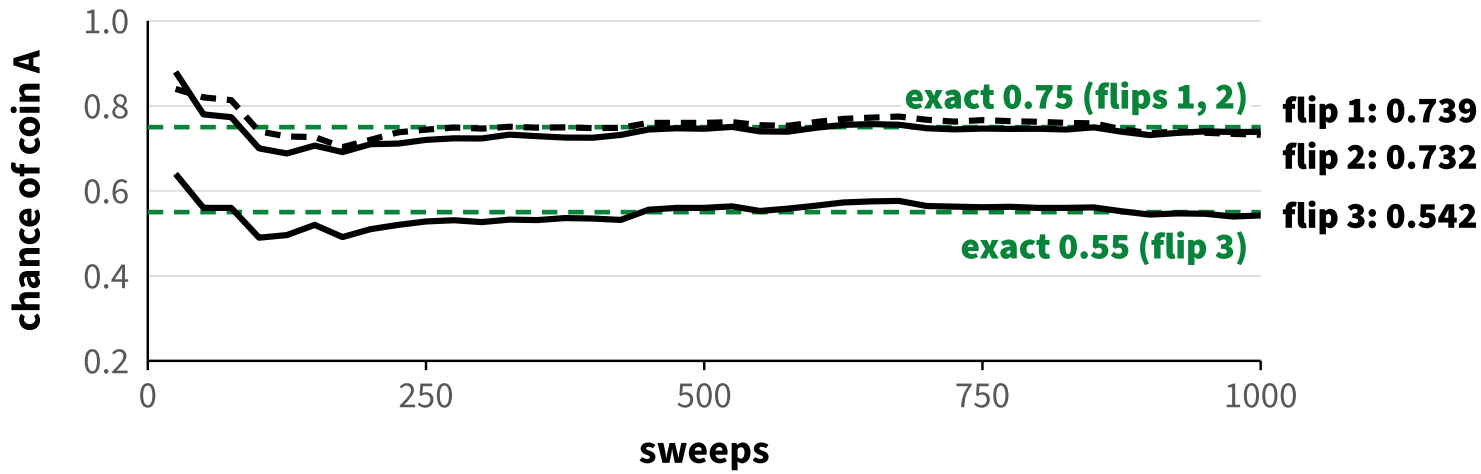


Checking every story is the exact answer; with three coins we can still afford it.

Redrawing one coin at a time reaches 0.739, 0.732, 0.542

Press Run. One **sweep** redraws coins 1, 2 and 3 once. **Black lines** show how often each coin has been A so far; **green dashes** are the exact chances.



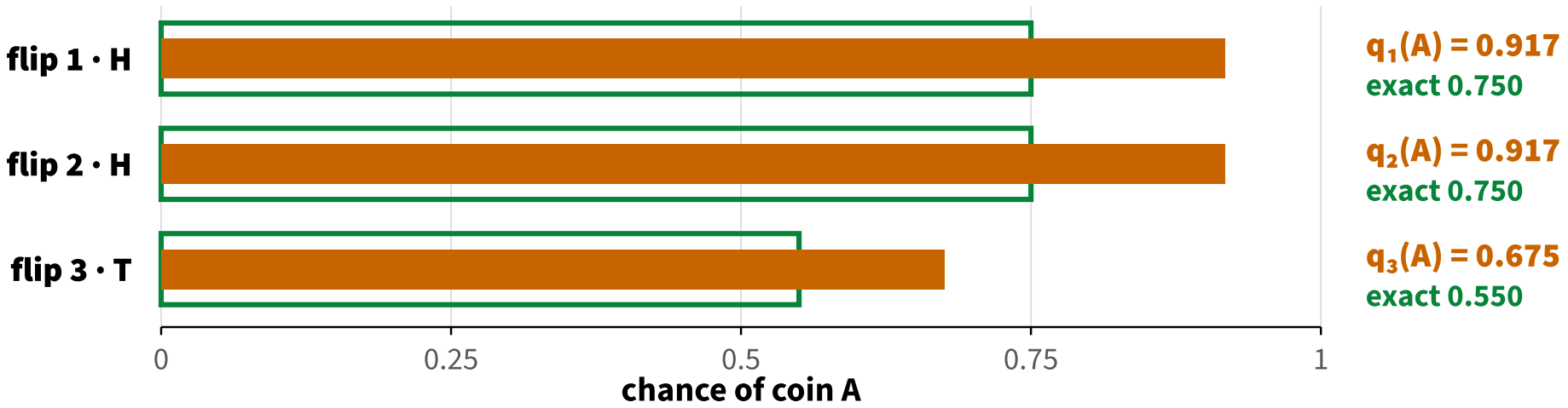


after 1000 sweeps: **0.739, 0.732, 0.542** · exact 0.75, 0.75, 0.55

After 1000 sweeps every estimate is within 0.02 of the exact answer.

On the circle, guessing alone gives 0.917 instead of 0.75

Each coin now has **two neighbours** that agree with it, so each gets more support than on the line.

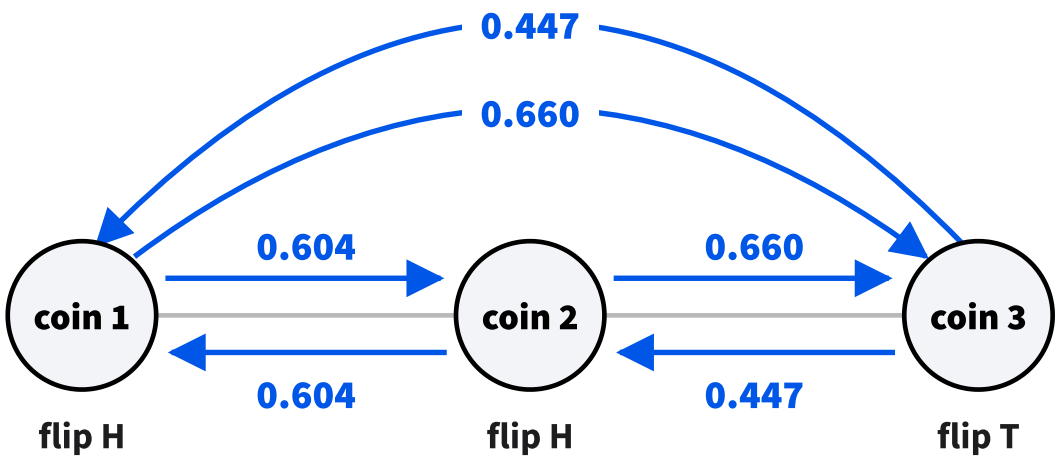


converged: q(A) = **0.917, 0.917, 0.675** · **mean field is too sure: 0.917 vs exact 0.75**

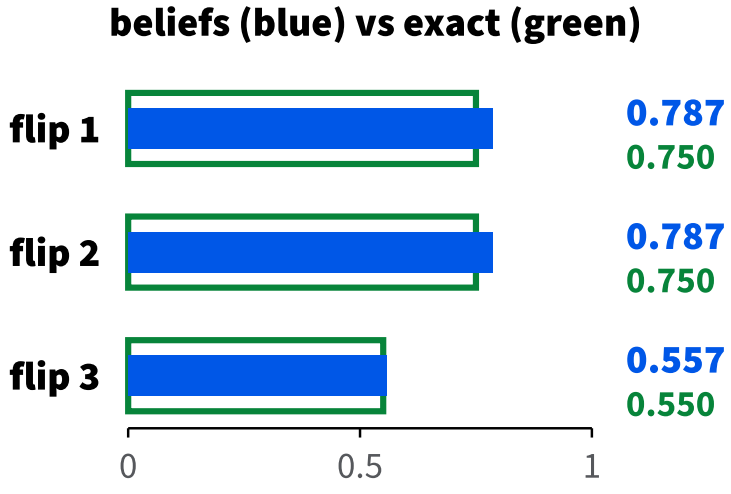
More links between coins make guessing alone more overconfident.

Messages around the circle converge to 0.787 instead of 0.75

Press Run: the messages go round the circle until they converge. **Blue bars** are the messages' guesses; **green outlines** are the exact chances.



arrows: each message's chance of A

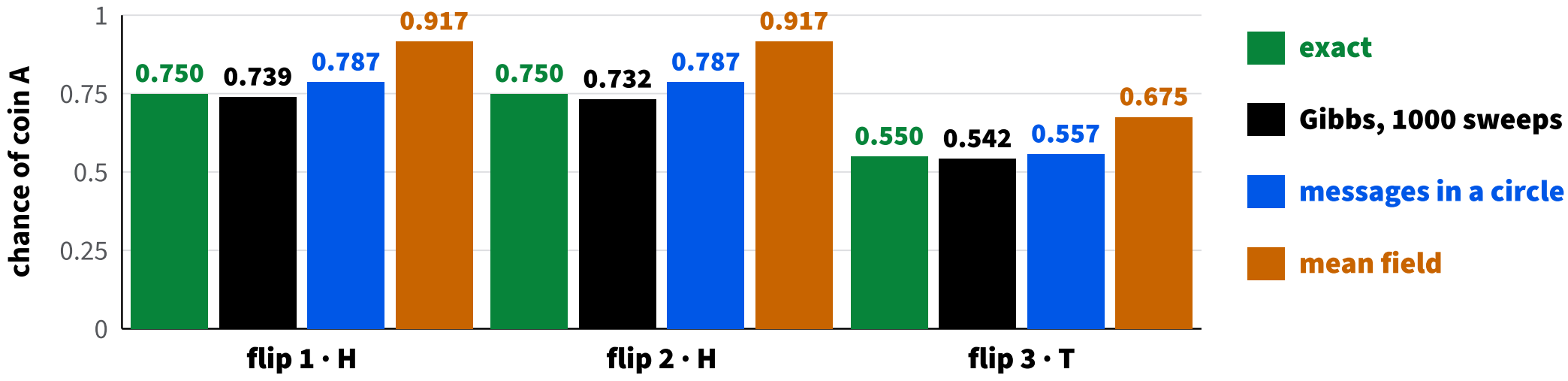


round 10: beliefs **0.787, 0.787, 0.557** · **converges to 0.787, not 0.75: messages go round the loop**

Close but not exact, because some evidence is counted twice.

On the circle, sampling and messages are close; guessing alone is not

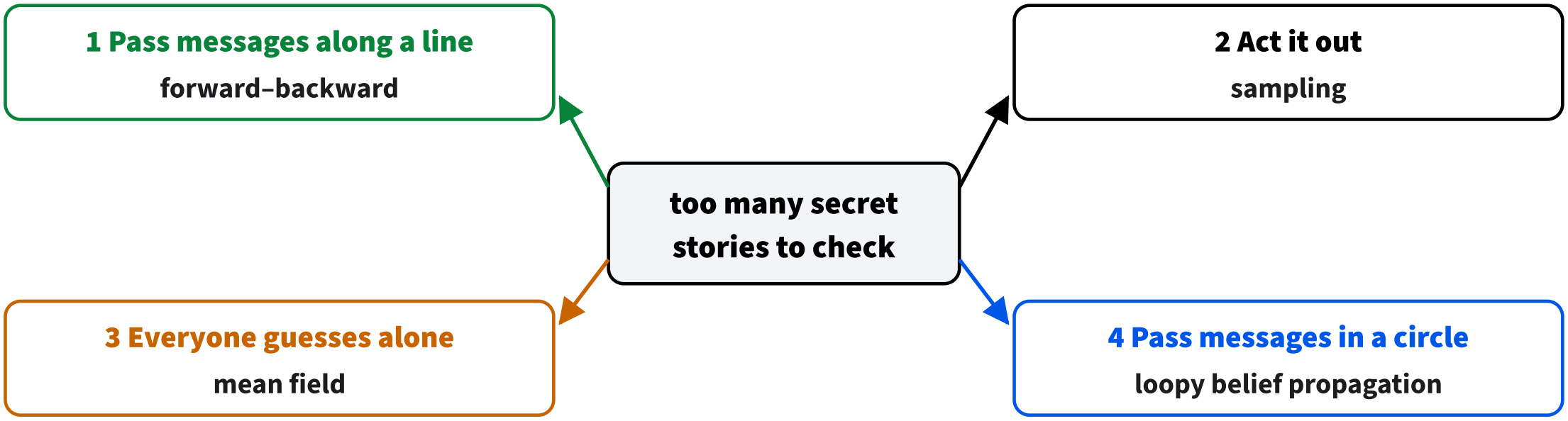
Each group is one coin. Compare every bar with the **green bar**, the exact answer.



Largest error: sampling **0.018**, messages in a circle **0.037**, guessing alone **0.167**.

Choose a way by what you can pay: time, a cycle-free graph, or some accuracy.

The same four ways are used in today's artificial intelligence



Replace the coins by sounds, words, image codes or the bits sent to a phone, and **the same problem** appears.

> **The same problem appears with many more hidden variables.**

Each of the four ways runs inside a system you have used

1 Pass messages along a line — exact

forward–backward (Baum–Welch)

→ speech recognition

messages along the line give the exact answer

2 Act it out — approximate

sampling (Monte Carlo)

→ reasoning models

act out many secret stories and average

3 Everyone guesses alone — approximate

mean field

→ topic models, image autoencoders

each takes the others' guesses as fact

4 Pass messages in a circle — approximate

loopy belief propagation

→ mobile phones decoding data

messages go round the circle: close, not exact

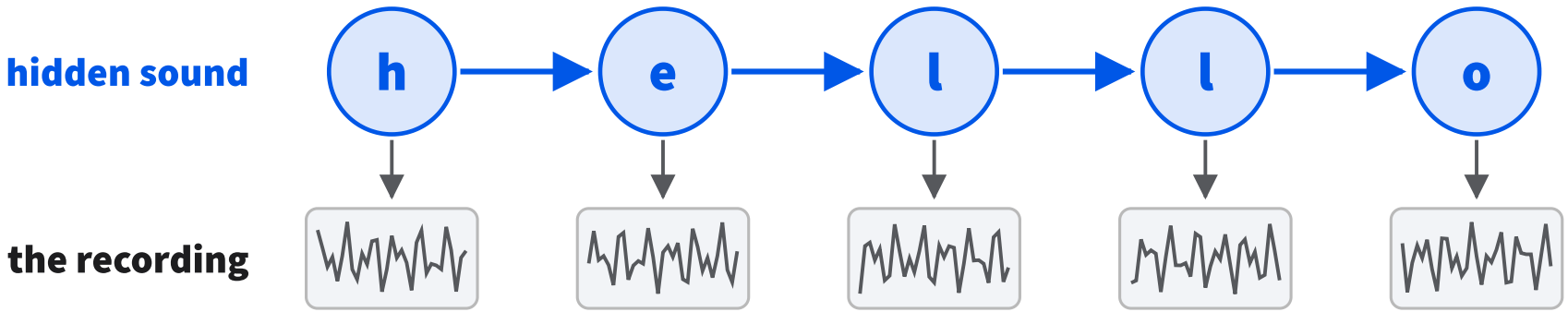
The score and the four ways stay the same; only the hidden variables change.

AI = artificial intelligence

Speech recognition used messages passed along a sentence

Take a recording of the word "hello". It matches our coin game:

1. the hidden coins become the **sounds** h, e, l, l, o;
2. the flips become the **slices of the recording**;
3. keep or switch becomes **stay on a sound or move to the next**, so the sounds form a line.



hidden: which sound is playing · seen: the recording

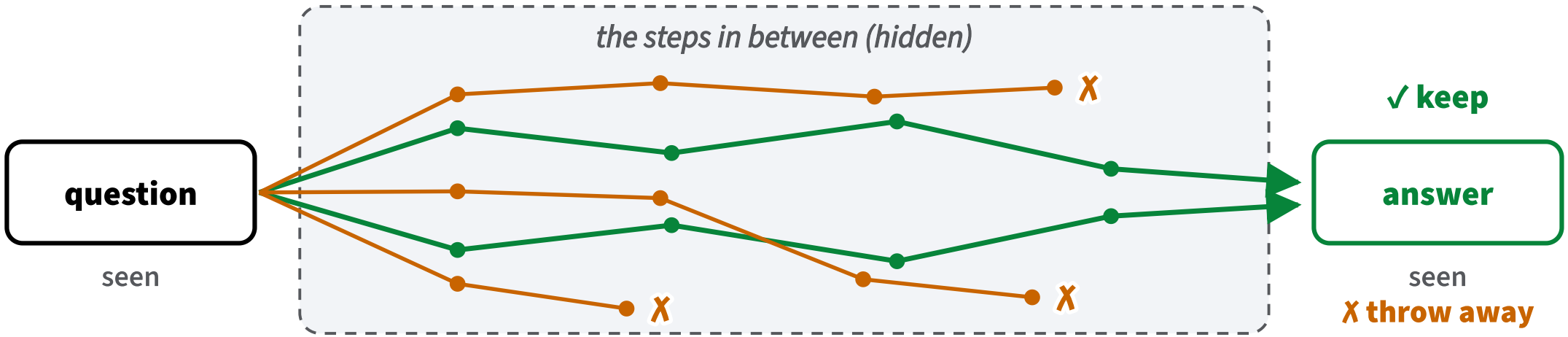
messages along the line learn when each sound starts

For decades, speech recognizers were trained this way.

Reasoning models act out solutions and keep the correct ones

Take a maths question with a known answer. It matches acting the game out:

1. the hidden coins become the **steps of a solution**, which the data never show;
2. replaying the game becomes **writing many attempts**;
3. keeping the runs that match becomes **keeping the attempts with the right answer**, then learning from them.



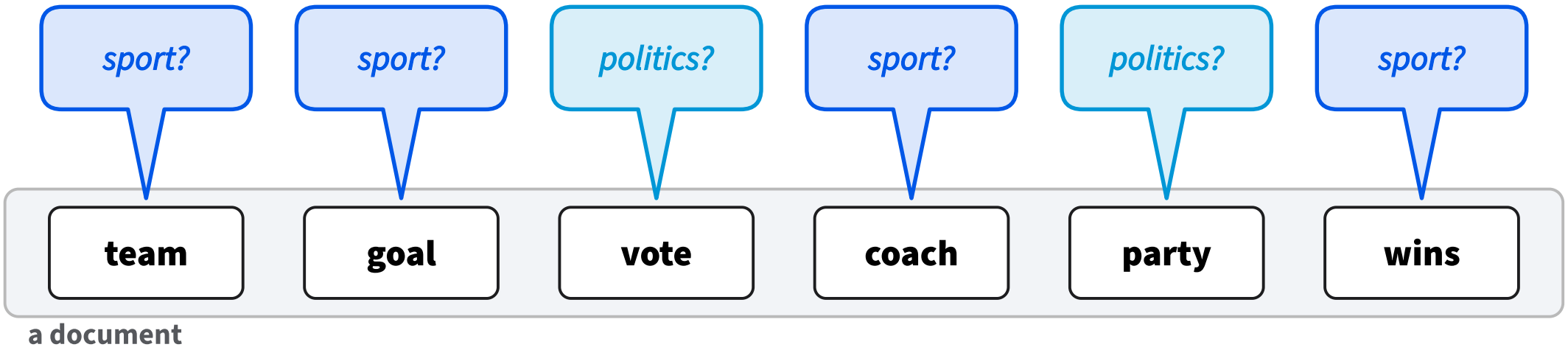
write many attempts, keep the ones that reach the right answer, learn from them

This is Way 2 applied to language models.

Topic models let each word guess its topic alone

Take a short document: team, goal, vote, coach, party, wins. It matches everyone guessing alone:

1. the hidden coins become each word's **topic**, such as sport or politics;
2. the flips become the **words** we read;
3. each word makes **its own guess** of its topic, which is fast enough for whole libraries.



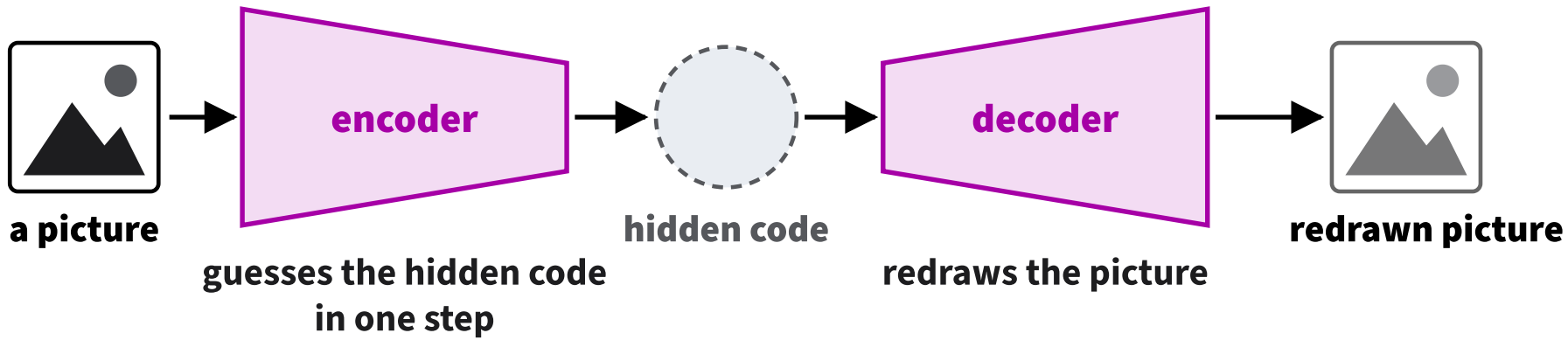
each word guesses its topic alone: fast enough for whole libraries

Fast enough for millions of documents, but somewhat overconfident.

A network can learn to make the guess in one step

Take a picture. It matches our coin game:

1. the hidden coins become the picture's **hidden code**;
2. our guess q is made by a network, the **encoder**, **in one step**;
3. the same score trains it: **redraw the picture well** and **keep the guess simple**.



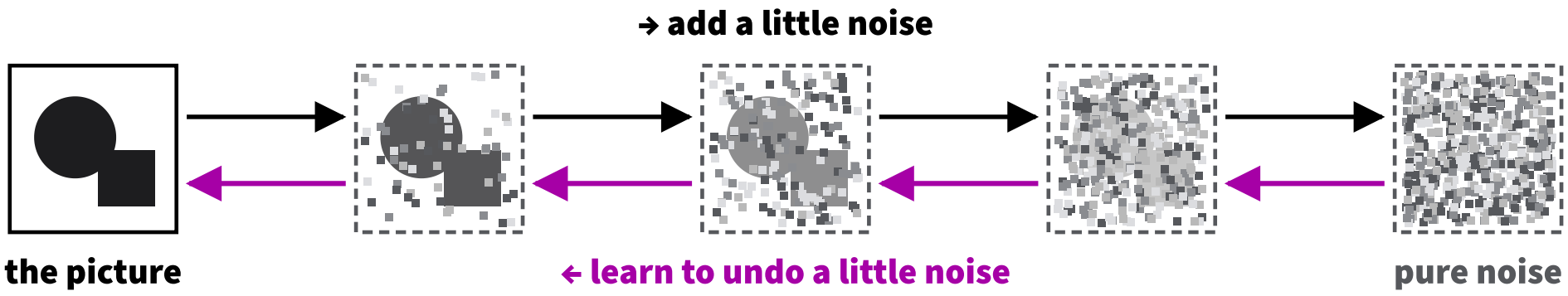
trained with the same score: redraw the picture well, keep the guess simple

Week 10: the variational autoencoder, which learns the guess with a network.

Image generators raise the same score along a chain

Take a picture that is slowly turned into noise. It matches our coin game:

1. the hidden coins become the **noisy pictures** in between;
2. each noisy picture comes from the one before, so they form a **chain**, like our line of coins;
3. learning to **remove a little noise** at each step raises the same score.



the noisy pictures are hidden, in a chain like our coins

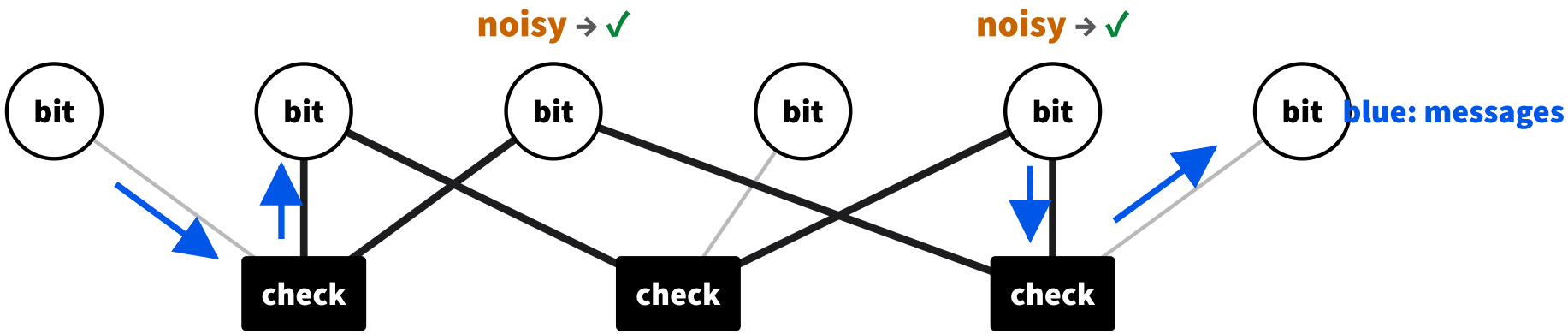
solid: seen · dashed: hidden

Week 11: diffusion models, trained with the same score.

Your phone passes messages in a large network with cycles

Take the bits sent to your phone. They arrive with noise:

1. the hidden coins become the **bits that were sent**;
2. each **check** says its bits add to an even number; the checks link bits into **long cycles**;
3. messages round these cycles **correct the noisy bits** in a fraction of a second.



messages go round long circles and repair the noisy bits

Messages in a circle work best when the cycles are long, as in fifth-generation mobile networks (5G) codes.

With too many stories, we score the guess and choose a way

Too many secret stories to check, so a **score** measures any guess and expectation–maximization (EM) raises it.

1. **Pass messages along a line:** exact on a line.
2. **Act it out:** correct on average, but slow.
3. **Everyone guesses alone:** fast, but overconfident.
4. **Pass messages in a circle:** fast and close, not exact.

Before believing an answer: **try other starting guesses**, **names do not matter**, and **can the data answer at all?**

> **One score, four ways out, three checks.**

Check yourself (1 of 2): the score and the messages

1. Our guess scores far below the truth. What does this tell us?

the habits are wrong | the guess is far from the true answer | nothing

Answer. The guess is far from the true answer: the gap measures exactly this; a perfect guess has no gap.

2. Why can coins in a line pass messages instead of listing every story?

each coin depends only on its neighbour | there are few coins | the flips are all heads

Answer. Each coin depends only on its neighbour, so one message from each side is enough.

Score every guess; use the chain structure when it exists.

Check yourself (2 of 2): acting out and guessing alone

3. Acting out gives an error of about 0.1. How many times as many runs halve it?

twice as many | three times as many | four times as many

Answer. **Four times as many,** because the error shrinks like one over the square root of the runs.

4. Two linked coins both lean to A. Compared with the truth, the separate guesses for A are usually …

higher | lower | the same

Answer. **Higher,** because the separate guesses reinforce each other.

Acting out is slow; guessing alone is overconfident.

References: the bound, the shortcut and sampling

[ELBO, EM as coordinate ascent] R. M. Neal and G. E. Hinton. A view of the EM algorithm that justifies incremental, sparse, and other variants. In M. I. Jordan (ed.), *Learning in Graphical Models*, pp. 355–368. Kluwer, 1998.

[Textbook] C. M. Bishop. *Pattern Recognition and Machine Learning*, Chapters 9–10. Springer, 2006.

[Baum–Welch, Theorem 6] L. E. Baum, T. Petrie, G. Soules and N. Weiss. A maximization technique occurring in the statistical analysis of probabilistic functions of Markov chains. *The Annals of Mathematical Statistics* 41(1):164–171, 1970.

[Speech] L. R. Rabiner. A tutorial on hidden Markov models and selected applications in speech recognition. *Proceedings of the IEEE* 77(2):257–286, 1989.

[Monte Carlo EM] G. C. G. Wei and M. A. Tanner. A Monte Carlo implementation of the EM algorithm and the poor man's data augmentation algorithms. *Journal of the American Statistical Association* 85(411):699–704, 1990.

[Reasoning] E. Zelikman, Y. Wu, J. Mu and N. D. Goodman. STaR: bootstrapping reasoning with reasoning. *Advances in Neural Information Processing Systems* 35, pp. 15476–15488, 2022. · A. Singh et al. Beyond human data: scaling self-training for problem-solving with language models. *Transactions on Machine Learning Research*, 2024.

[Gibbs sampling] S. Geman and D. Geman. Stochastic relaxation, Gibbs distributions, and the Bayesian restoration of images. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 6(6):721–741, 1984.

[Identifiability] H. Teicher. Identifiability of finite mixtures. *The Annals of Mathematical Statistics* 34(4):1265–1269, 1963. · M. Stephens. Dealing with label switching in mixture models. *Journal of the Royal Statistical Society, Series B* 62(4):795–809, 2000. · E. S. Allman, C. Matias and J. A. Rhodes. Identifiability of parameters in latent structure models with many observed variables. *The Annals of Statistics* 37(6A):3099–3132, 2009.

References: mean field, belief propagation and the frontier

[Variational inference, review] D. M. Blei, A. Kucukelbir and J. D. McAuliffe. Variational inference: a review for statisticians. *Journal of the American Statistical Association* 112(518):859–877, 2017.

[Topic models] D. M. Blei, A. Y. Ng and M. I. Jordan. Latent Dirichlet allocation. *Journal of Machine Learning Research* 3:993–1022, 2003. · M. D. Hoffman, D. M. Blei, C. Wang and J. Paisley. Stochastic variational inference. *Journal of Machine Learning Research* 14:1303–1347, 2013.

[Belief propagation] J. Pearl. *Probabilistic Reasoning in Intelligent Systems*. Morgan Kaufmann, 1988.

[Mean field as message passing] J. Winn and C. M. Bishop. Variational message passing. *Journal of Machine Learning Research* 6:661–694, 2005.

[Expectation propagation] T. P. Minka. Expectation propagation for approximate Bayesian inference. *Uncertainty in Artificial Intelligence* (UAI) 17, pp. 362–369, 2001.

[Theorem D, one circle] Y. Weiss. Correctness of local probability propagation in graphical models with loops. *Neural Computation* 12(1):1–41, 2000.

[Bethe free energy] J. S. Yedidia, W. T. Freeman and Y. Weiss. Generalized belief propagation. *Advances in Neural Information Processing Systems* 13, pp. 689–695, 2001; and Constructing free-energy approximations and generalized belief propagation algorithms. *IEEE Transactions on Information Theory* 51(7):2282–2312, 2005. · N. Ruozzi. The Bethe partition function of log-supermodular graphical models. *Advances in Neural Information Processing Systems* 25, pp. 117–125, 2012.

[Linear programs, the big picture] M. J. Wainwright and M. I. Jordan. Graphical models, exponential families, and variational inference. *Foundations and Trends in Machine Learning* 1(1–2):1–305, 2008.

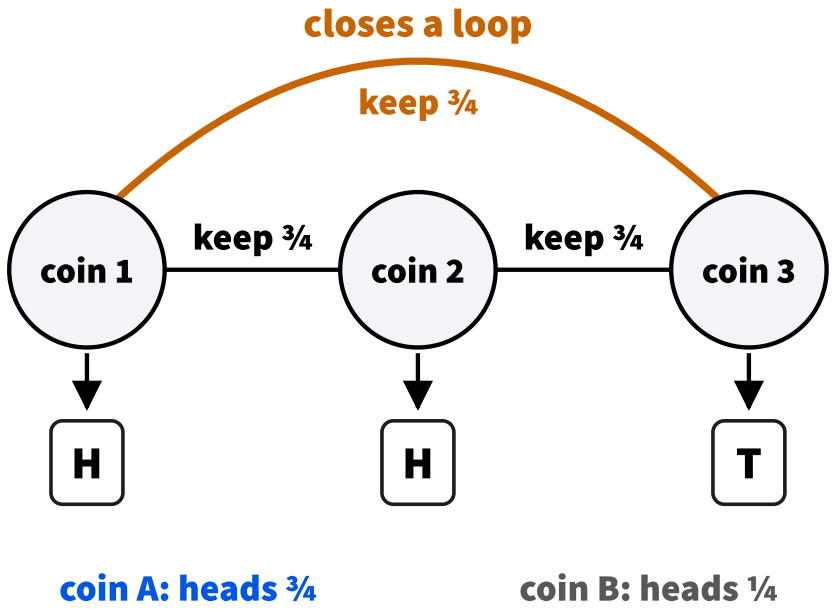
[Codes] R. G. Gallager. Low-density parity-check codes. *IRE Transactions on Information Theory* 8(1):21–28, 1962. · D. J. C. MacKay and R. M. Neal. Near Shannon limit performance of low density parity check codes. *Electronics Letters* 32(18):1645–1646, 1996. · 3GPP TS 38.212, NR; Multiplexing and channel coding (Release 15), 2018.

[Previews] D. P. Kingma and M. Welling. Auto-encoding variational Bayes. ICLR 2014. · A. Kucukelbir et al. Automatic differentiation variational inference. *Journal of Machine Learning Research* 18(14):1–45, 2017. · J. Sohl-Dickstein, E. Weiss, N. Maheswaranathan and S. Ganguli. Deep unsupervised learning using nonequilibrium thermodynamics. ICML 2015, PMLR 37:2256–2265. · J. Ho, A. Jain and P. Abbeel. Denoising diffusion probabilistic models. *Advances in Neural Information Processing Systems* 33, pp. 6840–6851, 2020. · D. Arthur and S. Vassilvitskii. k-means++: the advantages of careful seeding. SODA 2007, pp. 1027–1035.

NEXT WEEK

Next: where do the links between coins come from?

Today, the links were given as a line or a circle. **Adding one link**, coin 3 to coin 1, **changed every answer**.



Week 7, structure learning: learn from data which things should be linked.

> **Today the graph was given. Next week we learn the graph from data.**

The appendix: every proof the lecture pointed to

Way 1 · Baum–Welch

The pair soft label; the M-step's three terms; heads ÷ flips; keeps ÷ moves; forward–backward and one M-step on H, H, T in full.

Way 2 · Monte Carlo expectation–maximization (EM)

The M-step as an average of draws; draws as told labels; replay and keep; Theorem 7; why hard EM can fall; hard EM on the coins.

Way 3 · Mean field

Why the update maximizes the evidence lower bound (ELBO), and the update on our chain and ring.

Way 4 and trust

Loopy belief propagation round by round on the ring; the diagonal is stationary; the saddle on the coins.

Nothing here is needed to follow the lecture; everything here supports a claim made in it.

Proof, E-step: cutting at a move gives the pair soft label

↳ **Why pairs:** keeping links two neighbouring coins, so we need each pair $(z_t, z_{t+1}) = (a, b)$; recall $p(x, z)$ = one factor per arrow **(Lecture 5, p. 70)**.

$$\xi_t(a,b) = p(z_t{=}a,\ z_{t+1}{=}b \mid x) = \frac{p(x,\ z_t{=}a,\ z_{t+1}{=}b)}{p(x)} \qquad \text{(definition; Bayes' rule)}$$

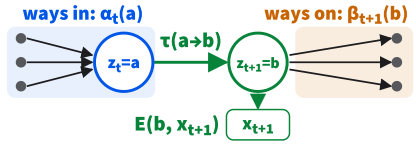
$$= \frac{1}{p(x)} \cdot \sum_{z_{1:t-1}} \sum_{z_{t+2:n}} \text{before}_t(z_{1:t-1}, a) \cdot \tau(a \to b) \cdot E(b, x_{t+1}) \cdot \text{after}_{t+1}(b, z_{t+2:n}) \qquad \text{(add over other coins; split at the move)}$$

$$= \frac{1}{p(x)} \cdot \Big(\sum_{z_{1:t-1}} \text{before}_t(z_{1:t-1}, a) \Big) \cdot \tau(a \to b) \cdot E(b, x_{t+1}) \cdot \Big(\sum_{z_{t+2:n}} \text{after}_{t+1}(b, z_{t+2:n}) \Big) \qquad \text{(distributive law; sums share no coins)}$$

$$= \frac{1}{p(x)} \cdot \underbrace{p(x_{1:t},\ z_t{=}a)}_{\alpha_t(a)} \cdot \tau(a \to b) \cdot E(b, x_{t+1}) \cdot \underbrace{p(x_{t+2:n} \mid z_{t+1}{=}b)}_{\beta_{t+1}(b)} \qquad \text{(brackets are marginals: } \alpha_t(a),\ \beta_{t+1}(b))$$

In words: a pair (a at flip t, b at flip $t{+}1$) is weighted by all the ways to arrive at a, times moving $a \to b$ and showing flip $t{+}1$, times all the ways to go on from b; divide by $p(x)$ to get a chance.

Check: if we do not care which coin comes next (add over b), we get back the one-flip label $\gamma_t(a)$ (since $\sum_b \tau(a \to b) \cdot E(b, x_{t+1}) \cdot \beta_{t+1}(b) = \beta_t(a)$).



Pair label: arrive at a, move to b and show the next flip, go on from b; divide by $p(x)$.

Proof, M-step: start, move and flip terms separate

↳ expectation–maximization (EM)'s M-step maximizes the **expected complete-data log-likelihood** $Q(\theta)$; the posterior $q(z) = p_{\theta'}(z \mid x)$ from the current guess θ' is held fixed. $[\cdot]$ = 1 if true, 0 if false.

$$
\begin{aligned}
Q(\theta) &= \sum_z p_{\theta'}(z \mid x) \cdot \ln p_\theta(x, z) = \sum_z q(z) \cdot \ln p_\theta(x, z) && \text{(the M-step objective)} \\
&= \sum_z q(z) \cdot \Big[\ln \pi(z_1) + \sum_{t=1}^{n-1} \ln \tau(z_t \to z_{t+1}) + \sum_{t=1}^{n} \ln E(z_t, x_t) \Big] && \text{(log of a product = sum of logs)} \\
&= \sum_z q(z) \cdot \ln \pi(z_1) + \sum_{t=1}^{n-1} \sum_z q(z) \cdot \ln \tau(z_t \to z_{t+1}) + \sum_{t=1}^{n} \sum_z q(z) \cdot \ln E(z_t, x_t) && \text{(swap the sums)} \\
&= \sum_a \ln \pi(a) \cdot \sum_z q(z) \cdot [z_1{=}a] + \sum_{t=1}^{n-1} \sum_{a,b} \ln \tau(a \to b) \cdot \sum_z q(z) \cdot [z_t{=}a, z_{t+1}{=}b] && \text{(sort the sequences by the} \\
&\quad + \sum_{t=1}^{n} \sum_c \ln E(c, x_t) \cdot \sum_z q(z) \cdot [z_t{=}c] && \text{coins each log factor uses)} \\
&= \underbrace{\sum_a \gamma_1(a) \cdot \ln \pi(a)}_{\textbf{start}} + \underbrace{\sum_{t=1}^{n-1} \sum_{a,b} \xi_t(a,b) \cdot \ln \tau(a \to b)}_{\textbf{moves}} + \underbrace{\sum_{t=1}^{n} \sum_z \gamma_t(z) \cdot \ln E(z, x_t)}_{\textbf{flips}} && \begin{array}{l}(\sum_z q(z) \cdot [z_t{=}c] \\ \quad = p_{\theta'}(z_t{=}c \mid x) = \gamma_t(c), \\ \sum_z q(z) \cdot [z_t{=}a, z_{t+1}{=}b] = \xi_t(a,b))\end{array}
\end{aligned}
$$

In words: each log factor touches one or two coins, so it needs only γ or ξ, never all K^n sequences.

Why it matters: the three terms share no parameter, so the M-step maximizes each on its own. **Start term:** π is fixed at ½ here, so it needs no update.

Each log factor becomes one term, weighted by the soft label of the coins it touches.

Proof, M-step: the flip terms give heads ÷ flips

↳ Recall Theorem 1 (counting) **(Lecture 5, p. 11)**: $u \cdot \ln\theta + v \cdot \ln(1-\theta)$ is largest at $\theta = \frac{u}{u+v}$, heads ÷ flips. Coin A: $E(A,\mathrm{H}) = \theta_A$, $E(A,\mathrm{T}) = 1-\theta_A$.

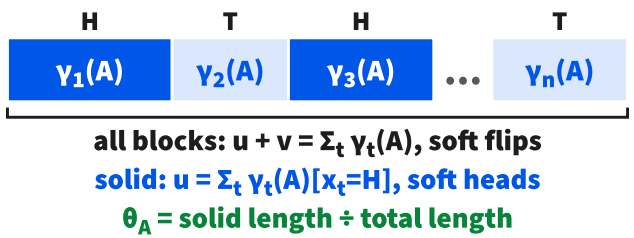
$$\text{flip terms of A} = \sum_{t=1}^{n} \gamma_t(A) \cdot \ln E(A, x_t) \qquad \text{(the } z{=}A \text{ flip terms; } z{=}B \text{ terms hold only } \theta_B\text{)}$$

$$= \underbrace{\Big(\sum_t \gamma_t(A)\cdot[x_t{=}\mathrm{H}]\Big)}_{u:\ \text{soft heads}} \cdot \ln\theta_A + \underbrace{\Big(\sum_t \gamma_t(A)\cdot[x_t{=}\mathrm{T}]\Big)}_{v:\ \text{soft tails}} \cdot \ln(1-\theta_A) \qquad \text{(H uses } \theta_A\text{, T uses } 1-\theta_A\text{)}$$

$$\Rightarrow \quad \theta_A = \frac{u}{u+v} = \frac{\sum_t \gamma_t(A)\cdot[x_t{=}\mathrm{H}]}{\sum_t \gamma_t(A)\cdot[x_t{=}\mathrm{H}] + \sum_t \gamma_t(A)\cdot[x_t{=}\mathrm{T}]} \qquad \text{(Theorem 1)}$$

$$= \frac{\sum_{t=1}^{n} \gamma_t(A)\cdot[x_t{=}\mathrm{H}]}{\sum_{t=1}^{n} \gamma_t(A)} \qquad \text{(each flip is H or T)}$$

In words: credit each flip to coin A by its soft label, then divide heads by flips.



Soft counts in, Theorem 1 out: heads ÷ flips.

Proof, M-step: the move terms give keeps ÷ moves

↳ One keep chance κ for both coins: $\tau(a \to a) = \kappa, \tau(a \to b) = 1 - \kappa$ for $a \neq b$. Theorem 1 again **(Lecture 5, p. 11)**: $u \cdot \ln \kappa + v \cdot \ln(1 - \kappa)$ peaks at $\kappa = \frac{u}{u+v}$.

$$\begin{aligned}
\text{move terms} &= \sum_{t=1}^{n-1} \sum_{a,b} \xi_t(a,b) \cdot \ln \tau(a \to b) && \text{(the moves part)} \\
&= \underbrace{\sum_{t<n} \big(\xi_t(A,A) + \xi_t(B,B)\big)}_{u:\ \text{expected keeps}} \cdot \ln \kappa + \underbrace{\sum_{t<n} \big(\xi_t(A,B) + \xi_t(B,A)\big)}_{v:\ \text{expected switches}} \cdot \ln(1-\kappa) && \text{(kept uses } \kappa\text{, switched } 1-\kappa\text{)} \\
\Rightarrow \quad \kappa &= \frac{u}{u+v} = \frac{\sum_{t<n} \big(\xi_t(A,A) + \xi_t(B,B)\big)}{\sum_{t<n} \sum_{a,b} \xi_t(a,b)} && \text{(Theorem 1)} \\
&= \frac{\sum_{t=1}^{n-1} \big(\xi_t(A,A) + \xi_t(B,B)\big)}{n-1} && \text{(each move's four } \xi \text{ add to 1)}
\end{aligned}$$

In words: new keep chance = expected keeps ÷ number of moves.

If each coin had its own move chances, the same counting gives $\tau(a \to b) = \sum_{t<n} \xi_t(a,b) \big/ \sum_{t<n} \gamma_t(a)$: expected moves $a \to b$ ÷ expected visits to a.

Why it matters: this is Theorem 6's M-step for κ **(Lecture 6, p. 25)**; by Theorem 4 **(Lecture 5, p. 49)**, $p(x)$ never falls. ■

	A at t+1	B at t+1	total
A at t	$\xi_t(A,A)$	$\xi_t(A,B)$	$\gamma_t(A)$
B at t	$\xi_t(B,A)$	$\xi_t(B,B)$	$\gamma_t(B)$
total	$\gamma_{t+1}(A)$	$\gamma_{t+1}(B)$	1

green = kept; the four cells add to 1

Soft counts in, Theorem 1 out: keeps ÷ moves.

Messages from both ends give p(HHT) = 15/128

Quick recap. Flips H, H, T from a hidden chain of coins (Lecture 5, p. 75): keep the coin with ¾, switch with ¼; A shows H with ¾, B with ¼. **Forward** α = chance of the flips so far; **backward** β = chance of the flips still to come.

Step	Number	Why
1. Forward to flip 2	$\alpha_2 = (\frac{15}{64}, \frac{3}{64})$	**Each flip adds up both ways of arriving at a coin.**
2. Backward to flip 2	$\beta_2 = (\frac{3}{8}, \frac{5}{8})$	**The last flip T is likelier from coin B.**
3. Join at flip 2	$p(\mathrm{HHT}) = \frac{15}{128}$	**Past times future, summed over the two coins.**
4. Soft labels for A	**0.8, 0.75, 0.4**	**Each flip's $\alpha \cdot \beta$ for A, divided by $p(\mathrm{HHT})$.**

$$\frac{15}{64} \cdot \frac{3}{8} + \frac{3}{64} \cdot \frac{5}{8} = \frac{15}{128}$$

Two numbers per coin per flip replace the list of all $2^3 = 8$ coin sequences.

The new keep chance is 0.7, and the fit improves to 0.139

Quick recap. The M-step (Lecture 6, p. 118) sets the **keep chance** κ to the average chance that the coin stayed the same between flips (pair labels, Theorem 6 (Lecture 6, p. 25)). Start: $\kappa = \frac{3}{4}, p(\mathrm{HHT}) = \frac{15}{128} = 0.117$ (Lecture 6, p. 89).

Step	Number	Why
1. Kept, flips 1 → 2	0.8	**H then H: the same coin explains both well.**
2. Kept, flips 2 → 3	0.6	**H then T: a switch to B is more plausible.**
3. New keep chance	$\kappa = 0.7$	**The average of the two moves.**
4. New fit	$p(\mathrm{HHT}) = 0.139$	**Forward pass again with $\kappa = 0.7, \theta_A = 0.795, \theta_B = 0.429$.**

$$\kappa = \frac{0.8+0.6}{2} = 0.7, \qquad 0.117 \;\rightarrow\; 0.139$$

One Baum–Welch round raises the fit, as Theorem 4 (Lecture 5, p. 49) promises.

EM's M-step expectation becomes an average over m draws

↳ Recall Section 3 **(Lecture 5, p. 33)** (EM = expectation–maximization): $q_t^{(n)}(z) = p_{\theta_t}(z \mid x^{(n)})$ is record n's posterior (its soft label) at the guess θ_t.

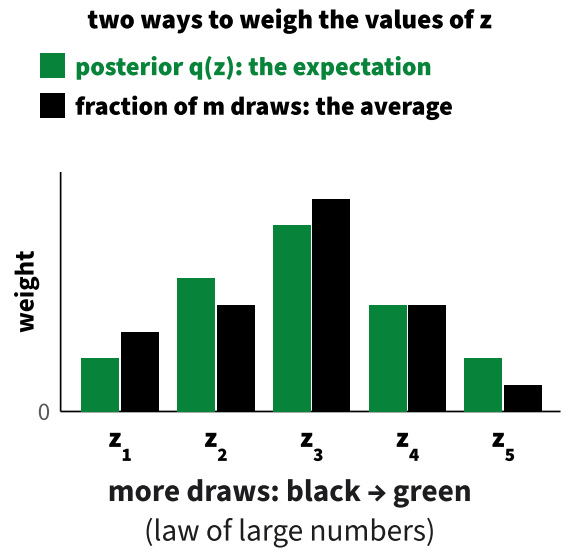
Monte Carlo rule (law of large numbers): for any function g, $\mathbb{E}_{z\sim q}\big[g(z)\big] \approx \dfrac{1}{m}\cdot\sum_{j=1}^{m} g\big(z^{(j)}\big)$, $z^{(j)} \sim q$ independent.

$$\sum_n \sum_z q_t^{(n)}(z)\cdot \ln p_\theta\big(x^{(n)}, z\big) \qquad \text{EM's M-step objective}$$

$$= \sum_n \mathbb{E}_{z\sim q_t^{(n)}}\big[\ln p_\theta\big(x^{(n)}, z\big)\big] \qquad \text{weights } q_t^{(n)}(z) \text{ sum to 1: an expectation}$$

$$\approx \sum_n \frac{1}{m}\cdot\sum_{j=1}^{m} \ln p_\theta\big(x^{(n)}, z^{(n,j)}\big) \qquad \text{Monte Carlo rule, } g(z)=\ln p_\theta(x^{(n)},z),\ z^{(n,j)}\sim q_t^{(n)}$$

$$= \sum_n \sum_z \hat{q}^{(n)}(z)\cdot \ln p_\theta\big(x^{(n)}, z\big) \qquad \text{group equal draws}$$



Line 3 is **Monte Carlo EM's M-step objective**; $\hat{q}^{(n)}(z)$ = **fraction of the m draws** equal to z.

In words: the same log-probability terms, with each posterior share $q_t^{(n)}(z)$ replaced by a fraction of draws $\hat{q}^{(n)}(z)$.

Same log-probability terms, new weights: each posterior share becomes a fraction of draws.

Draws act as told labels: Section 1's maximum likelihood

↳ **Recall Section 1 (Theorem 1) (Lecture 5, p. 11):** when the friend tells every coin $z^{(n)}$, maximum likelihood maximizes $\sum_n \ln p_\theta(x^{(n)}, z^{(n)})$, and the answer is **count and divide**.

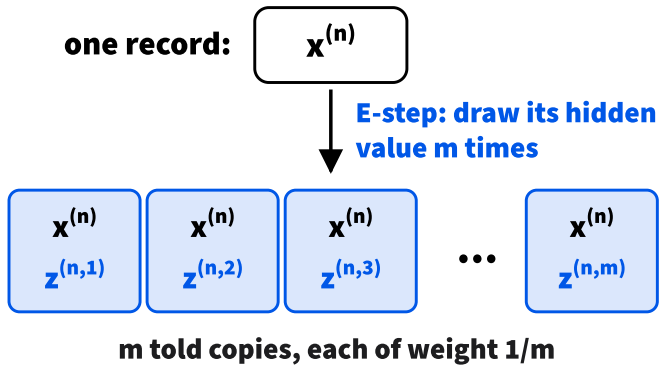
$$\sum_{n=1}^{N} \frac{1}{m} \cdot \sum_{j=1}^{m} \ln p_\theta\left(x^{(n)}, z^{(n,j)}\right) \qquad \text{Monte Carlo EM's M-step objective}$$

$$= \frac{1}{m} \cdot \sum_{\text{pairs } (n,j)} \ln p_\theta\left(x^{(n)}, z^{(n,j)}\right) \qquad \text{one sum over the } N \cdot m \text{ pairs (record, draw)}$$

$$= \frac{1}{m} \cdot \ln \prod_{\text{pairs } (n,j)} p_\theta\left(x^{(n)}, z^{(n,j)}\right) \qquad \text{a sum of logs is the log of a product}$$

$$= \frac{1}{m} \cdot \ln p_\theta(\text{pretend data}) \qquad \text{the } N \cdot m \text{ pairs as independent told records}$$

The factor $\frac{1}{m} > 0$ does not move the arg max: θ_{t+1} is **told-label maximum likelihood**.



In words: **the drawn hidden values are used exactly as if a friend had told them.**

Why it matters: no new optimizer. Any model we can fit with told labels, we can fit with Monte Carlo expectation–maximization (EM).

Each draw is a told label, so the M-step is ordinary maximum likelihood on the pretend data.

Proof: dividing by the keep rate gives the posterior

↳ **Proof of the proposition (replay and keep) (Lecture 6, p. 29).** One run draws $z \sim p_\theta(z)$, then $x' \sim p_\theta(x' \mid z)$; it is **kept** when $x' = x$. For one run:

$$
\begin{aligned}
P(z \mid \text{kept}) &= \frac{P(z,\ \text{kept})}{P(\text{kept})} && \text{conditional probability} \\
&= \frac{p_\theta(z) \cdot p_\theta(x \mid z)}{\sum_{z'} p_\theta(z') \cdot p_\theta(x \mid z')} && \text{draw } z\text{; then } x' \text{ must equal } x\text{; add over } z' \\
&= \frac{p_\theta(x, z)}{\sum_{z'} p_\theta(x, z')} && \text{product rule} \\
&= \frac{p_\theta(x, z)}{\underbrace{p_\theta(x)}_{= P(\text{kept})}} && \text{sum rule} \\
&= p_\theta(z \mid x) && \text{definition } \blacksquare
\end{aligned}
$$

Runs are independent, so the kept values are too.

In words: a value z survives in proportion to how well it explains x, which is exactly the posterior.

The three rules used:

conditional probability
$P(z \mid \text{kept}) = P(z, \text{kept}) / P(\text{kept})$

product rule
$p_\theta(z) \cdot p_\theta(x \mid z) = p_\theta(x, z)$

sum rule
$\sum_{z'} p_\theta(x, z') = p_\theta(x)$

Three textbook rules take "draw, then match" to the posterior, with $p_\theta(x)$ only in the proof.

Proof of Theorem 7: indicators, then means and variances add

↳ **Setup.** Let $I_j = 1$ if draw j is that value, else 0; the fraction is $\frac{1}{m} \cdot \sum_j I_j$.

Mean:

$$\begin{aligned} \mathbb{E}\left[\tfrac{1}{m} \cdot \textstyle\sum_j I_j\right] &= \tfrac{1}{m} \cdot \textstyle\sum_j \mathbb{E}[I_j] && \text{linearity of expectation} \\ &= \tfrac{1}{m} \cdot m \cdot \big(1 \cdot q + 0 \cdot (1-q)\big) = q && \mathbb{E}[I_j] = 1 \cdot q + 0 \cdot (1-q) = q \end{aligned}$$

Standard deviation:

$$\begin{aligned} \mathrm{sd} &= \sqrt{\mathrm{Var}\left[\tfrac{1}{m} \cdot \textstyle\sum_j I_j\right]} && \text{definition} \\ &= \sqrt{\tfrac{1}{m^2} \cdot \textstyle\sum_j \mathrm{Var}\, I_j} && \text{independent: variances add; } \tfrac{1}{m} \text{ out squared} \\ &= \sqrt{\tfrac{1}{m} \cdot \mathrm{Var}\, I_1} && \text{same law for every } j\text{: } m \text{ equal terms} \\ &= \sqrt{\tfrac{1}{m} \cdot q \cdot (1-q)} && \mathrm{Var}\, I_j = \mathbb{E}[I_j^2] - q^2 = q - q^2 = q \cdot (1-q) \;\; (I_j^2 = I_j) \\ &= \sqrt{q \cdot (1-q)/m} && \blacksquare \end{aligned}$$

In words: the fraction is right on average, and its noise falls like $1/\sqrt{m}$.

variance; 0 or 1

$\mathrm{Var}\, X = \mathbb{E}[X^2] - (\mathbb{E}X)^2$; $I_j^2 = I_j$, so $\mathbb{E}[I_j^2] = q$

scaling

$\mathrm{Var}(c \cdot X) = c^2 \cdot \mathrm{Var}\, X$

independent draws

$\mathrm{Var} \sum_j I_j = \sum_j \mathrm{Var}\, I_j$

The two chains give

$$\mathbb{E}\left[\frac{1}{m} \cdot \sum_j I_j\right] = \frac{1}{m} \cdot m \cdot q = q$$

$$\mathrm{Var}\left[\frac{1}{m} \cdot \sum_j I_j\right] = \frac{1}{m^2} \cdot m \cdot q \cdot (1-q) = \frac{q \cdot (1-q)}{m}$$

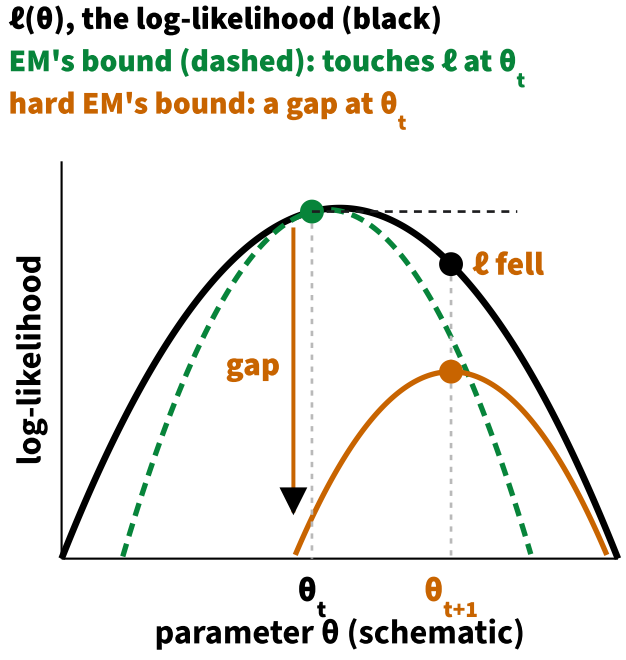
Mean q, standard deviation $\sqrt{q \cdot (1-q)/m}$: right on average, noise $\propto 1/\sqrt{m}$.

Theorem 4 does not cover hard EM: the log-likelihood can fall

↳ **Recall Lemma 1 (Lecture 5, p. 47):** for any label q, $\ell(\theta) \geq \sum_z q(z) \cdot \ln \frac{p_\theta(x,z)}{q(z)}$ per record, with equality only when q is the posterior. Here $\ell(\theta) = \sum_n \ln p_\theta(x^{(n)})$.

$$
\begin{aligned}
\ell(\theta_{t+1}) &\geq \sum_n \sum_z q^{(n)}(z) \cdot \ln \frac{p_{\theta_{t+1}}(x^{(n)}, z)}{q^{(n)}(z)} && \text{Lemma 1, hard label } q^{(n)} \\
&= \sum_n \ln p_{\theta_{t+1}}\left(x^{(n)}, \hat{z}^{(n)}\right) && q^{(n)} = 1 \text{ on } \hat{z}^{(n)}\text{; the 0 terms vanish} \\
&\geq \sum_n \ln p_{\theta_t}\left(x^{(n)}, \hat{z}^{(n)}\right) && \text{the M-step maximizes this sum} \\
&= \sum_n \left[\ln p_{\theta_t}\left(x^{(n)}\right) + \ln p_{\theta_t}\left(\hat{z}^{(n)} \mid x^{(n)}\right)\right] && \text{product rule } p(x,z) = p(x) \cdot p(z \mid x) \\
&= \ell(\theta_t) - \textstyle\sum_n \left(-\ln p_{\theta_t}(\hat{z}^{(n)} \mid x^{(n)})\right) && \text{definition of } \ell\text{; orange} = \text{the gap}
\end{aligned}
$$

The **gap** is > 0 unless every posterior is certain. With expectation–maximization (EM)'s exact posterior it is 0, and the chain ends at $\ell(\theta_t)$: **Theorem 4 (Lecture 5, p. 49)**.



In words: **hard labels are not the posterior, so the bound no longer touches ℓ at the guess.**

Why it matters: only $\ell(\theta_{t+1}) \geq \ell(\theta_t) - \text{gap}$ is guaranteed: hard EM can **end below its start**.

Theorem 4's increase needs the exact posterior as the soft label; hard EM gives that up.

Hard EM gets stuck below where it started

Quick recap. **Hard expectation–maximization (EM)** (Lecture 6, p. 105) gives each record only its likelier coin, not a soft label. Round 1 moved the guess to $(\theta_A, \theta_B) = (\frac{3}{4}, 0)$: coin B now always shows tails. **Score** ℓ = log-likelihood of HH, HT, TT; it started at -4.000.

Step	Number	Why
1. Labels in round 2	**A, A, B**	**B cannot show heads, so HH and HT go to A; TT goes to B.**
2. Next guess	$(\frac{3}{4}, 0)$ **again**	**Same labels give the same counts: EM is stuck.**
3. Score there	$\ell = -4.268$	**Lower than the start's** -4.000.

$$\ell = \ln\left(\tfrac{9}{32} \cdot \tfrac{3}{32} \cdot \tfrac{17}{32}\right) = -4.268 \; < \; -4.000$$

Hard labels are not the posterior, so Theorem 4 does not protect the score.

Proof: as a function of one guess, the ELBO is a constant minus a KL divergence

↳ Claim **(Lecture 6, p. 37)**: with $q = q_i \cdot q_{-i}$ and q_{-i} fixed, the evidence lower bound (ELBO) is largest at $q_i(z_i) \propto \exp\left(\mathbb{E}_{q_{-i}}[\ln p_\theta(x, z)]\right)$. Write $\tilde{q}_i(z_i) = \exp\left(\mathbb{E}_{q_{-i}}[\ln p_\theta(x, z)]\right)/C$, with C making $\tilde{q}_i$ sum to 1.

$$
\begin{aligned}
B(q, \theta) &= \sum_z q(z) \cdot \ln p_\theta(x, z) - \sum_z q(z) \cdot \ln q(z) && \text{(definition of the ELBO)} \\
&= \sum_{z_i} q_i(z_i) \cdot \mathbb{E}_{q_{-i}}\big[\ln p_\theta(x, z)\big] - \sum_{z_i} q_i(z_i) \cdot \ln q_i(z_i) + H(q_{-i}) && \text{(sum over } z_{-i} \text{ first; } \ln q = \ln q_i + \ln q_{-i}\text{)} \\
&= \sum_{z_i} q_i(z_i) \cdot \big(\ln \tilde{q}_i(z_i) + \ln C\big) - \sum_{z_i} q_i(z_i) \cdot \ln q_i(z_i) + H(q_{-i}) && \text{(definition of } \tilde{q}_i\text{)} \\
&= -\sum_{z_i} q_i(z_i) \cdot \ln \frac{q_i(z_i)}{\tilde{q}_i(z_i)} + \ln C + H(q_{-i}) && \text{(}\textstyle\sum_{z_i} q_i(z_i) = 1\text{)} \\
&= -\mathrm{KL}\big(q_i \,\|\, \tilde{q}_i\big) + (\text{terms without } q_i) && \text{(definition of KL)}
\end{aligned}
$$

So the ELBO is largest when the Kullback–Leibler (KL) divergence is 0, i.e. $q_i = \tilde{q}_i$ (Gibbs' inequality, Lecture 5, p. 48). **On our coins:** only the factors containing z_t depend on it, so $\ln \frac{\tilde{q}_t(A)}{\tilde{q}_t(B)} = \ln \frac{E(A,x_t)}{E(B,x_t)} + \sum_s \left[q_s \ln \frac{3}{4} + (1-q_s) \ln \frac{1}{4} - q_s \ln \frac{1}{4} - (1-q_s) \ln \frac{3}{4}\right] = \ln \frac{E(A,x_t)}{E(B,x_t)} + \sum_s (2q_s - 1) \ln 3$. ■

Each mean-field update solves a one-variable problem exactly, so the ELBO never goes down.

Loopy belief propagation on the ring, round by round

↳ Messages **(Lecture 6, p. 49)** give the chance of coin A; all start at ½ and are updated together. **Round 1, coin 1 → coin 2:** coin 1's flip H gives A : B = ¾ : ¼, the message from coin 3 is still ½ : ½, so $m_{1\to 2}(A) \propto \frac{3}{4}\cdot\frac{3}{4}+\frac{1}{4}\cdot\frac{1}{4}=\frac{10}{16}, m_{1\to 2}(B) \propto \frac{3}{4}\cdot\frac{1}{4}+\frac{1}{4}\cdot\frac{3}{4}=\frac{6}{16}$: 0.625.

round	1→2	2→3	3→1	2→1	3→2	1→3	beliefs 1, 2, 3
1	0.625	0.625	0.375	0.625	0.375	0.625	0.750, 0.750, 0.481
2	0.571	0.667	0.429	0.571	0.429	0.667	0.750, 0.750, 0.571
3	0.596	0.650	0.450	0.596	0.450	0.650	0.784, 0.784, 0.535
converged	0.604	0.660	0.447	0.604	0.447	0.660	0.787, 0.787, 0.557

Belief at coin 1 = its flip ¾ : ¼ times the messages from coins 2 and 3, normalized. Exact: 0.75, 0.75, 0.55 **(Lecture 6, p. 65)**.

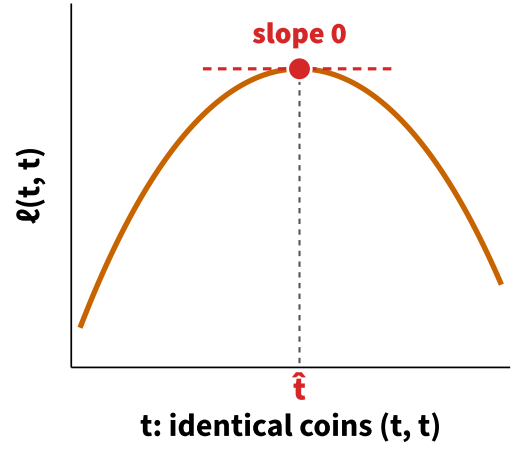
The messages swing for a few rounds, then converge a little away from the exact answer.

On the diagonal, both slopes of ℓ are zero

Recall (Lecture 6, p. 110): $\ell = \sum_x \ln\left[\frac{1}{2}\cdot f(x \mid \theta_A) + \frac{1}{2}\cdot f(x \mid \theta_B)\right]$ with $f(x \mid t) = t^{h_x}\cdot(1-t)^{n_x - h_x}$, one coin's chance of record x; $\hat{t} = \sum_x h_x / \sum_x n_x$.

Diagonal fact, part 2. $(\hat{t}, \hat{t})$ is a **stationary point** of ℓ: $\partial\ell/\partial\theta_A = \partial\ell/\partial\theta_B = 0$ there.

$$\begin{aligned}
\frac{\partial \ell}{\partial \theta_A}\Big|_{(t,t)} &= \sum_x \frac{\frac{1}{2}\cdot f'(x \mid \theta_A)}{\frac{1}{2}\cdot f(x \mid \theta_A) + \frac{1}{2}\cdot f(x \mid \theta_B)}\Big|_{(t,t)} && \text{(chain rule for } \ln\text{; only A's term has } \theta_A\text{)} \\
&= \tfrac{1}{2}\cdot \sum_x \frac{f'(x \mid t)}{f(x \mid t)} && \text{(at } (t,t) \text{ the two halves add to } f\text{)} \\
&= \tfrac{1}{2}\cdot \frac{d}{dt}\sum_x \ln f(x \mid t) && (f'/f = (\ln f)') \\
&= 0 \ \text{ at } t = \hat{t} && \text{(Theorem 1: best fit = heads/flips)}
\end{aligned}$$



one coin's ℓ,
top at t̂ (Theorem 1)

both slopes vanish there

In words: on the diagonal each slope is half the slope of one coin's log-likelihood, which is flat at $\hat{t}$. The same steps give $\partial\ell/\partial\theta_B = 0$.

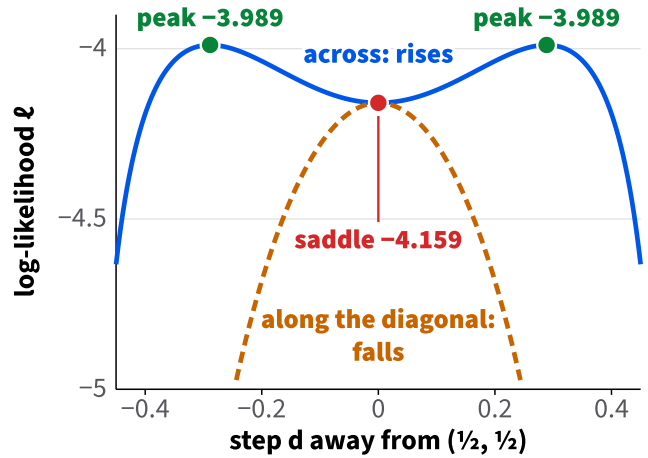
Why it matters: stationary is not the same as best: whether $(\hat{t}, \hat{t})$ is a peak or a trap depends on how ℓ curves across the diagonal.

The resting point on the diagonal is flat in every direction, but not necessarily a peak.

Two identical coins sit on a saddle

Quick recap. Two mirror peaks (Lecture 6, p. 116); between them sit two identical fair coins, $(\theta_A, \theta_B) = (\frac{1}{2}, \frac{1}{2})$. **Saddle** (Lecture 6, p. 112) = highest in one direction, lowest in another. **Along**: move both coins the same way; **across**: move them apart.

Step	Number	Why
1. Middle	$\ell = -4.159$	**Each record has chance $\frac{1}{4}$: $3 \cdot \ln \frac{1}{4}$.**
2. Along	$3 \cdot \ln(\frac{1}{4} - d^2)$	**Any move lowers ℓ: highest at the middle.**
3. Across	$\approx -4.159 + 4 \cdot d^2$	**Any move raises ℓ: lowest at the middle.**



the same point is lowest across and highest along

Highest along, lowest across: a saddle, a worse fit than the mirror peaks.

More details: Lecture 5's two coins, Bethe, mean-field sweeps

Way 2 on the two coins

Hard expectation–maximization (EM); replay and keep on HH; one Monte Carlo EM round; Theorem 7 on HT and hard EM stuck.

Ways 3 and 4

The first mean-field sweeps by hand; the Bethe free energy formula; the most likely state as a linear program.

Can we trust it?

The three answers; mirror peaks; identical coins; the saddle; restarts; the counting test; Teicher; two settings, same data.

Everything here is the full version of a claim the lecture makes in one line.

Hard EM replaces each soft label by the single most likely hidden value

Definition (hard expectation–maximization (EM)). At the guess θ_t, for each record n:

E-step: take the most likely hidden value (ties by a fixed rule)

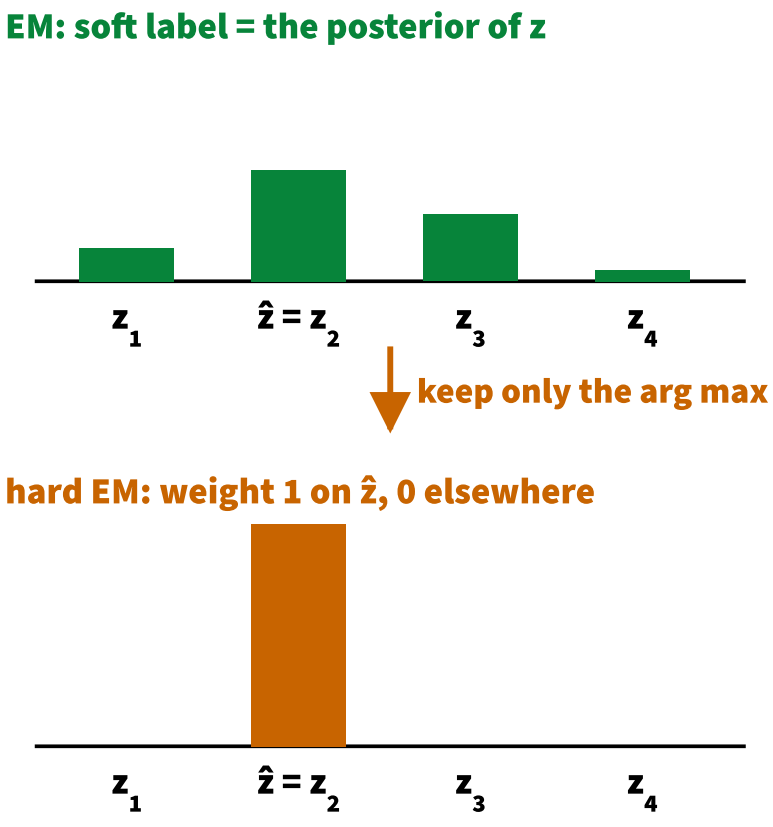
$$\hat{z}^{(n)} = \arg\max_z p_{\theta_t}\left(z \mid x^{(n)}\right) = \arg\max_z p_{\theta_t}\left(x^{(n)}, z\right)$$

(same denominator $p_{\theta_t}(x^{(n)})$), and use the label 1 on $\hat{z}^{(n)}$, 0 elsewhere.

M-step: $\theta_{t+1} = \arg\max_\theta \sum_n \ln p_\theta\left(x^{(n)}, \hat{z}^{(n)}\right)$, maximum likelihood with these told labels.

It is Monte Carlo EM with **one "draw" per record that is always the most likely value**, as in k-means (Lecture 5, p. 58).

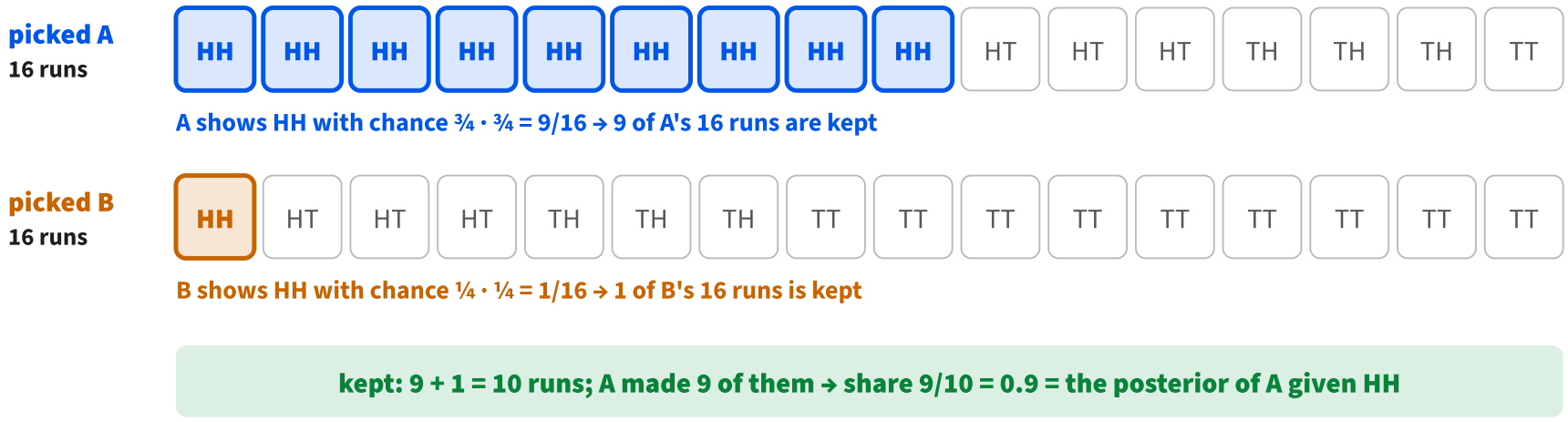
Why it matters: no normalizing sum and no random draws are needed, but the label is **no longer the posterior**, so Theorem 4's increase is lost: the log-likelihood can fall (Appendix, Way 2 (Lecture 6, p. 95)).



Hard EM rounds each posterior to 0 or 1; it is cheap, but it throws the uncertainty away.

Replay the game and keep the runs that match: the kept runs follow the posterior

↳ **Proposition (replay and keep) (Lecture 6, p. 29) applied to HH**, one record of Lecture 5's two-coin game (pick A or B with chance ½, flip it twice). Replay the game at the guess $\theta_A = \frac{3}{4}, \theta_B = \frac{1}{4}$ and **keep** a run only if it shows HH. Shown: 32 replays at their expected counts; a fair toss sends $32 \cdot \frac{1}{2} = 16$ runs to each coin.



No sum was computed: we only ran the game forward and compared. **Its cost:** $32 - 10 = 22$ of the 32 runs are thrown away. The keep rate $10/32 = 9/32 + 1/32$ is exactly $p(\mathrm{HH})$, which we never computed. So **rare records waste most runs**.

Kept runs show A : B = 9 : 1, so each kept run's coin is a posterior draw: A with chance 0.9.

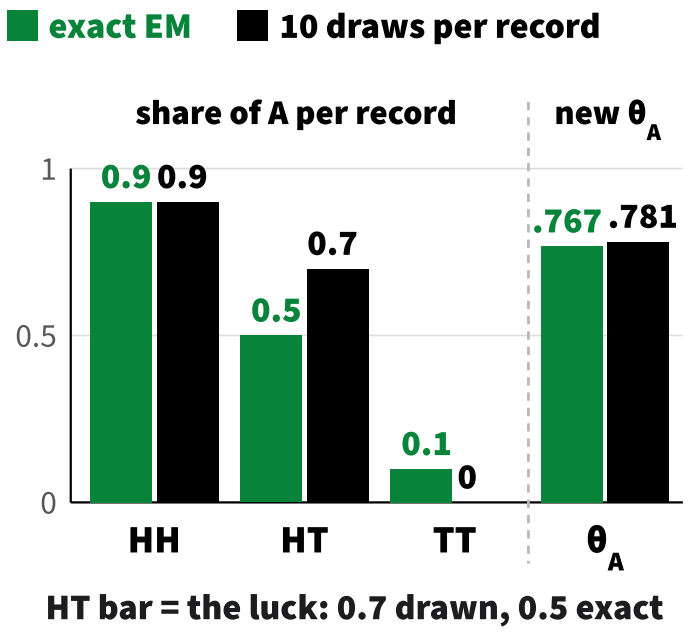
One Monte Carlo EM round with ten draws gives $\theta_A = 0.781$, close to EM's 0.767

↳ **Definition (Monte Carlo expectation–maximization (EM)) (Lecture 6, p. 30) applied to the coins,** $m = 10$. At $\theta_A = \frac{3}{4}, \theta_B = \frac{1}{4}$, we drew each record's coin 10 times with a computer's random numbers (A with chance 0.9, 0.5, 0.1, the posteriors) **(Lecture 5, p. 41)**. **In this one run A came up 9, 7, 0 times**; another run gives other counts. M-step: count and divide (Theorem 1) **(Lecture 5, p. 11)** with these fractions.

record	fraction of A's	heads to A	flips to A
HH	9/10 = 0.9	0.9 · 2 = 1.8	0.9 · 2 = 1.8
HT	7/10 = 0.7	0.7 · 1 = 0.7	0.7 · 2 = 1.4
TT	0/10 = 0	0 · 0 = 0	0 · 2 = 0
total		**2.5**	**3.2**

$$\theta_A = \frac{\text{heads to A}}{\text{flips to A}} = \frac{2.5}{3.2} = 0.781 \quad \text{EM (recall): } \frac{2.3}{3.0} = 0.767$$

Coin B, from the remaining fractions 0.1, 0.3, 1: $\theta_B = 0.5/2.8 = 0.179$.



Then draw afresh at $(0.781,\ 0.179)$ **and repeat: EM with "share" replaced by "fraction of draws".**

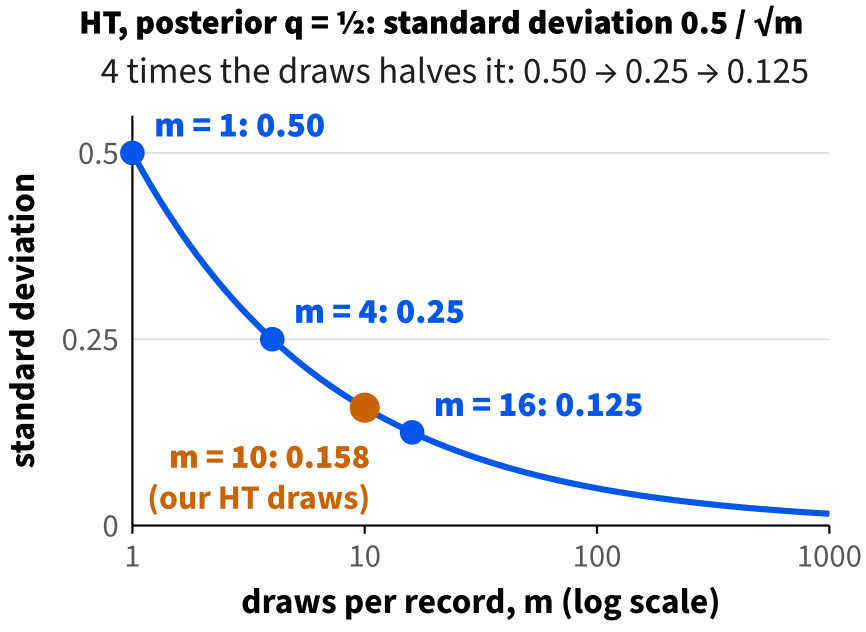
Theorem 7 on HT: ten draws give noise 0.158, and four times the draws halve it

↳ **Theorem 7 (Lecture 6, p. 31) applied to HT.** At $\theta_A = \frac{3}{4}, \theta_B = \frac{1}{4}$ the posterior of A given HT is $q = \frac{1}{2}$ (recall) **(Lecture 5, p. 41)**; we drew $m = 10$ times.

$$\text{standard deviation} = \sqrt{q \cdot (1-q)/m} \qquad \text{Theorem 7}$$

$$= \sqrt{\tfrac{1}{2} \cdot \tfrac{1}{2}/10} = 0.158 \qquad \tfrac{1}{4}/10 = 0.025,\ \sqrt{0.025} = 0.158$$

Ten draws typically miss by about 0.16, so HT's 0.7 (posterior 0.5) is **ordinary luck**, which the new guess inherits: $\theta_A = 0.781$, not expectation–maximization (EM)'s 0.767 **(Lecture 5, p. 42)**.



Hard EM (Lecture 6, p. 102) instead: round, do not draw

Give each record wholly to its likelier coin: HH → A, HT → A (tie), TT → B. Counting then gives $\theta_A = \frac{3}{4}, \theta_B = 0$, and the labels stay A, A, B: hard EM is **stuck**. Its fit $\ell = -4.268$ is below the start's -4.000 **(Lecture 5, p. 28)**: rounding throws away information (all steps: Appendix **(Lecture 6, p. 96)**).

Draws add noise $\propto 1/\sqrt{m}$ that more draws remove; rounding to one coin can get stuck.

The first mean-field sweeps, by hand

↳ Update **(Lecture 6, p. 37)**: $\ln \frac{q_t(A)}{q_t(B)} = \pm \ln 3 + \sum_s (2q_s - 1) \ln 3$, with $+\ln 3$ for H and $-\ln 3$ for T; start from all $q = \frac{1}{2}$, order 1, 2, 3.

update	chain	ring
flip 1 (H)	$\ln 3 + 0 \;\Rightarrow\; q_1 = \frac{3}{4} = 0.75$	$\ln 3 + 0 + 0 \;\Rightarrow\; 0.75$
flip 2 (H)	$\ln 3 + 0.5 \ln 3 = 1.5 \ln 3 \;\Rightarrow\; \frac{3^{1.5}}{1+3^{1.5}} = 0.839$	**the same: 0.839**
flip 3 (T)	$-\ln 3 + 0.677 \ln 3 \;\Rightarrow\; 0.412$	$-\ln 3 + (0.677 + 0.5) \ln 3 \;\Rightarrow\; 0.549$
converged	**0.866, 0.848, 0.417**	**0.917, 0.917, 0.675**

$2 \cdot 0.839 - 1 = 0.677$; converting back: $q = 3^u/(1 + 3^u)$ when the log-odds is $u \ln 3$.

Each update is one line of arithmetic; repeating the sweeps converges the guesses.

The Bethe free energy, and the most likely state

Bethe free energy (pairwise model; b_t, b_{st} = single and pairwise beliefs that agree on their overlaps):

$$F_{\text{Bethe}} = \underbrace{-\sum_{\text{factors } a} \mathbb{E}_{b_a}[\ln f_a]}_{\text{energy}} \; \underbrace{-\left(\sum_{\text{links } st} H(b_{st}) - \sum_{t} (d_t - 1)\, H(b_t)\right)}_{\text{entropy from pieces}}$$

H = entropy, d_t = number of links at t. On a tree this equals $-\ln p_\theta(x)$ at the exact marginals.

Most likely state. Asking for the single most likely hidden sequence instead relaxes to a **linear program** over the same locally consistent beliefs; it is exact on trees, and max-product messages are closely related (Wainwright and Jordan, 2008, ch. 8).

belief propagation (BP)'s beliefs are where this free energy is flat; on a tree that is the exact answer.

Next: an answer can be stuck, or best yet not unique

run on HH, HT, TT (Lecture 5's two coins)	start (θ_A, θ_B)	ends at	ℓ
expectation–maximization (EM) (soft labels) (Lecture 5, p. 43)	(¾, ¼)	(0.789, 0.211)	−3.989, **best**
hard EM (Lecture 6, p. 102)	(¾, ¼)	(¾, 0), stuck	−4.268
EM, identical start	(½, ½)	(½, ½), never moves	3 ln ¼ = −4.159

ℓ = log-likelihood of the three records. Identical start: each record has chance $\frac{1}{2} \cdot \frac{1}{4} + \frac{1}{2} \cdot \frac{1}{4} = \frac{1}{4}$, so $\ell = 3 \cdot \ln \frac{1}{4}$; equal coins split every record ½ : ½, so the guess never moves.

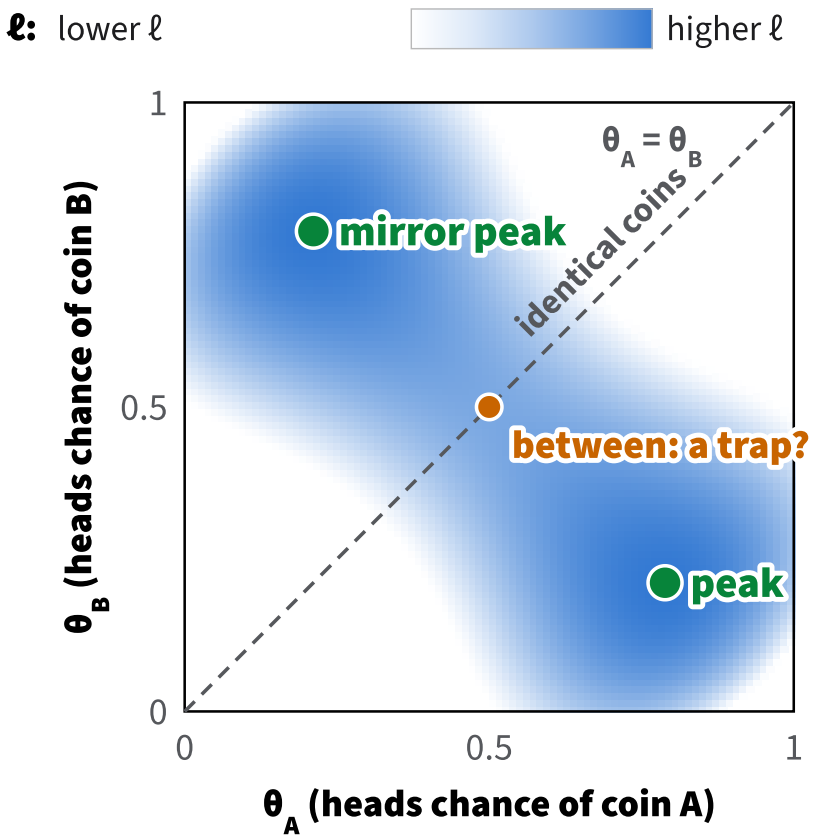
Swap the names: $\theta_A = 0.211,\ \theta_B = 0.789$. Is the fit better, worse, or the same?

the same | better | worse

Answer. The same. $p(\mathrm{HH}) = \frac{1}{2} \cdot 0.789^2 + \frac{1}{2} \cdot 0.211^2$ either way: the two paths swap places, since each coin is picked with chance $\frac{1}{2}$.

Every way climbs a floor or converges somewhere: which answer to trust is the last question.

The coin fit has two mirror-image peaks



two equally high peaks, mirror images across the diagonal

Can a learned $\theta = (\theta_A, \theta_B)$ be trusted? Colour the **log-likelihood** ℓ of Lecture 5's records HH, HT, TT over every pair of heads chances.

Two equally high peaks appear, mirror images across the diagonal $\theta_A = \theta_B$ (identical coins), with an orange point between them.

Three questions:

1. Why are the peaks mirror images?
2. Can expectation–maximization (EM) get stuck between them?
3. Can the data always pick one answer?

Before trusting a fit, ask whether a different setting fits just as well.

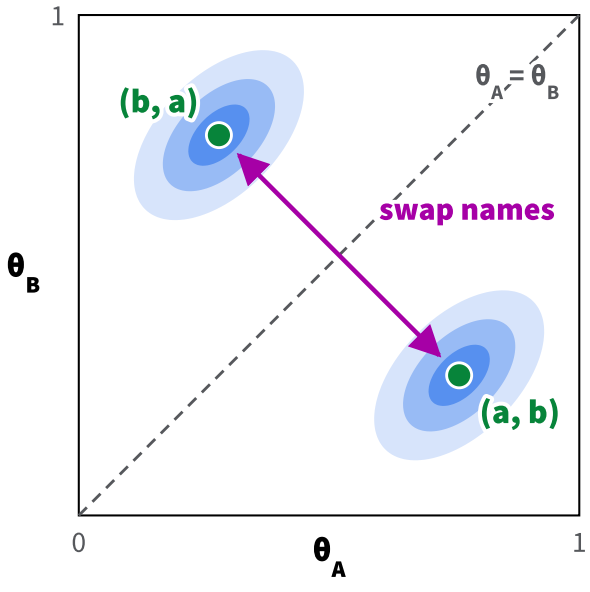
Renaming the coins never changes the likelihood

Label-switching lemma. If the friend picks each coin with chance $\frac{1}{2}$, then $\ell(\theta_A, \theta_B) = \ell(\theta_B, \theta_A)$ for every pair of heads chances.

Write $f(x \mid t)$ for the chance that one coin with heads chance t produces record x.

$$
\begin{aligned}
\ell(\theta_A, \theta_B) &= \sum_x \ln p_\theta(x) && \text{(independent records)} \\
&= \sum_x \ln\left[\tfrac{1}{2} \cdot f(x \mid \theta_A) + \tfrac{1}{2} \cdot f(x \mid \theta_B)\right] && \text{(sum over hidden coin } z\text{)} \\
&= \sum_x \ln\left[\tfrac{1}{2} \cdot f(x \mid \theta_B) + \tfrac{1}{2} \cdot f(x \mid \theta_A)\right] && \text{(addition commutes)} \\
&= \ell(\theta_B, \theta_A) && \text{(line 2, names swapped)}
\end{aligned}
$$

Why it matters: every peak (a, b) has an equally high **mirror peak** (b, a). With an unknown pick chance π, the same steps give $\ell(\pi, \theta_A, \theta_B) = \ell(1 - \pi, \theta_B, \theta_A)$.



same ℓ at both points:
the hills are mirror images across the diagonal

Names are labels, not facts: the data see only the unordered pair of coins.

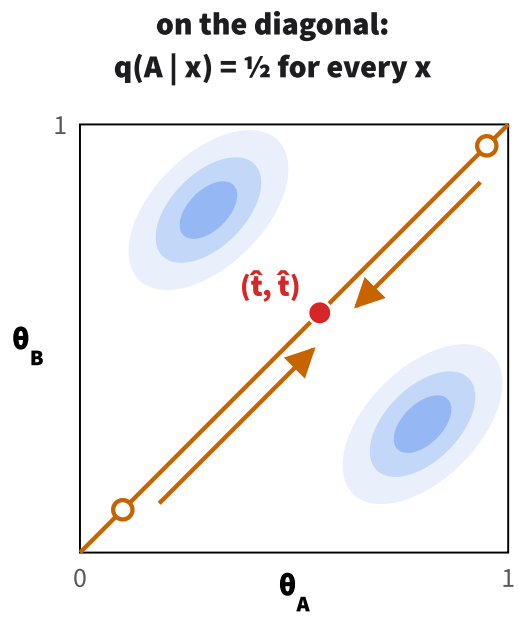
Identical coins stay identical under EM

Setting. Pick chance $\frac{1}{2}$. Record x has h_x heads in n_x flips, so one coin with heads chance t gives it $f(x \mid t) = t^{h_x} \cdot (1-t)^{n_x - h_x}$. Let $\hat{t} = \sum_x h_x / \sum_x n_x$: all heads over all flips.

Diagonal fact. From identical coins (t, t), one expectation–maximization (EM) step lands on $(\hat{t}, \hat{t})$; later steps stay there.

$$
\begin{aligned}
q(A \mid x) &= \frac{\frac{1}{2} \cdot f(x \mid t)}{\frac{1}{2} \cdot f(x \mid t) + \frac{1}{2} \cdot f(x \mid t)} = \frac{1}{2} && \text{(Bayes' rule, identical coins)} \\
\theta_A^{\text{new}} &= \frac{\sum_x q(A \mid x) \cdot h_x}{\sum_x q(A \mid x) \cdot n_x} && \text{(M-step: credited heads ÷ credited flips)} \\
&= \frac{\frac{1}{2} \cdot \sum_x h_x}{\frac{1}{2} \cdot \sum_x n_x} = \hat{t} && (q(A \mid x) = \tfrac{1}{2} \text{ for every } x) \\
&= \theta_B^{\text{new}} && (q(B \mid x) = \tfrac{1}{2} \text{ too: the same sums})
\end{aligned}
$$

Why it matters: a start with $\theta_A = \theta_B$ never leaves the diagonal: EM cannot break the symmetry. There both slopes of ℓ are zero (Appendix, Trust (Lecture 6, p. 99)).



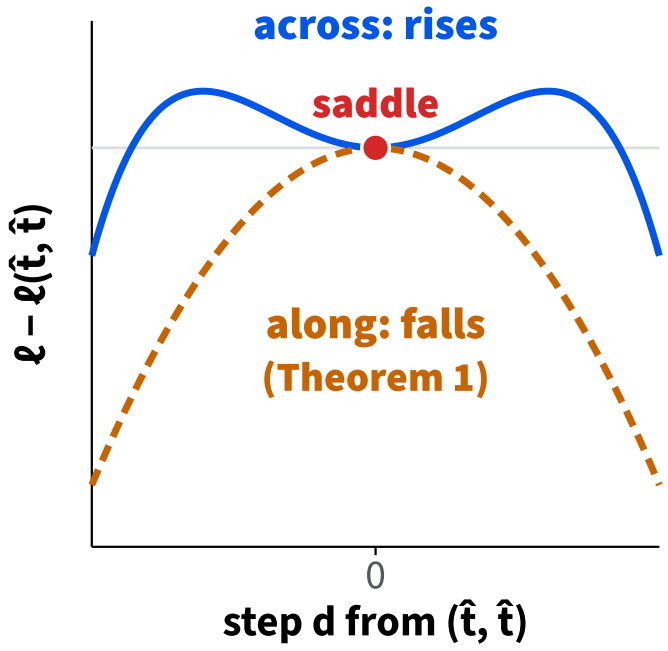
Identical starts stay identical: EM moves only along the diagonal.

Across the diagonal, ℓ can curve upward: a saddle

Saddle test. **Along** the diagonal, $\ell(t,t)$ is one coin's log-likelihood, largest at $t = \hat{t}$ (Theorem 1) (Lecture 5, p. 11). **Across** it, ℓ rises if $\sum_x f''(x \mid \hat{t})/f(x \mid \hat{t}) > 0$; then $(\hat{t},\hat{t})$ is a **saddle**.

$$\ell(\hat{t}+d,\,\hat{t}-d) \approx \ell(\hat{t},\hat{t}) + \frac{d^2}{2}\cdot\sum_x \frac{f''(x \mid \hat{t})}{f(x \mid \hat{t})}$$ (averaging $\pm d$ cancels the d terms)

In words: along the diagonal it is a **top**, across it a **dip**. By the diagonal fact (Lecture 6, p. 111), expectation–maximization (EM) started there stops there anyway.



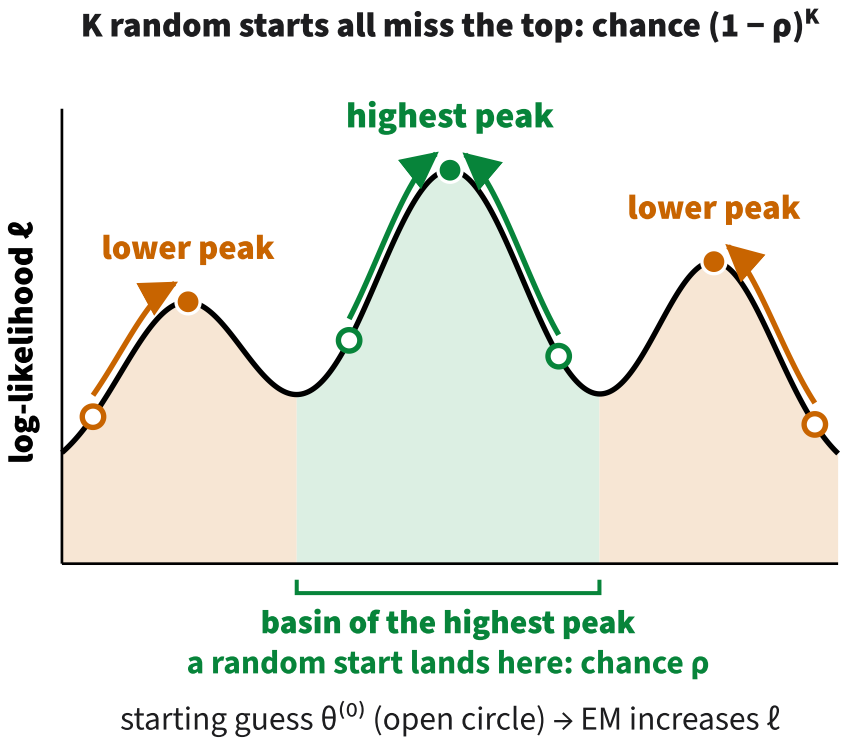
A saddle is a worse fit, and a symmetric start makes EM stop exactly there.

Random restarts reach the top with high probability

Theorem 4 **(Lecture 5, p. 49)**: ℓ never goes down, but expectation–maximization (EM) may stop at a lower peak or a saddle. The same holds for the floor of variational EM and for where loopy belief propagation (BP) converges.

Restart principle. Run from K independent random starts; keep the run with the largest ℓ (or the largest evidence lower bound (ELBO)).

If a random start reaches the top with chance $\rho > 0$, all K runs miss with chance $(1-\rho)^K \to 0$.



Several random starts escape saddles and lower peaks; keep the best score.

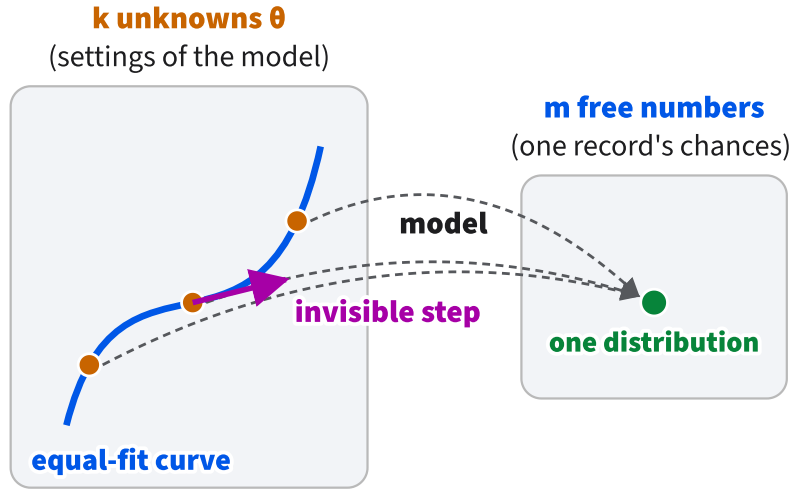
More unknowns than free numbers means no unique fit

↳ Question 3: can the data always pick one answer?

Theorem 8 (counting test). If a model has **more unknowns** than one record's distribution has **free numbers** (its chances, minus one because they sum to 1), then a **whole curve** of settings fits equally well.

Coins: 3 unknowns $\pi = p(A),\ \theta_A,\ \theta_B$; two flips give $p(0), p(1), p(2)$, summing to 1: **2 free numbers** < 3.

Why: 3 knobs, but the data read only 2 dials. One direction of turning the knobs leaves both dials unchanged.



k > m: a whole curve of settings
lands on the same distribution of one record

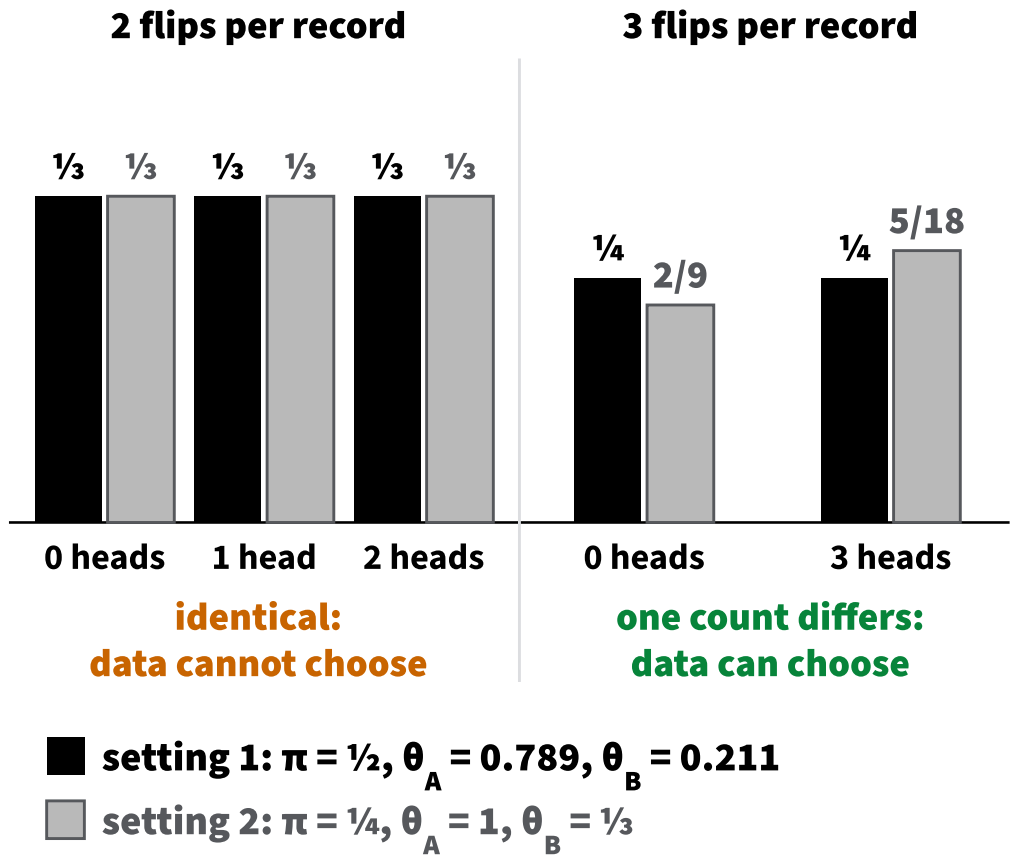
Count before you fit: no algorithm can choose between settings the data cannot tell apart.

With three flips per record, the two coins are pinned down

Theorem (Teicher). With **three or more flips** per record (and two really different coins), the data pin down π, θ_A, θ_B: only renaming A ↔ B is left.

In words: two flips leave a curve of equal fits; three flips leave only a setting and its mirror.

Teicher, 1963; Allman, Matias & Rhodes, 2009.



Observe more per record, and the curve of equal fits shrinks back to two mirror peaks.

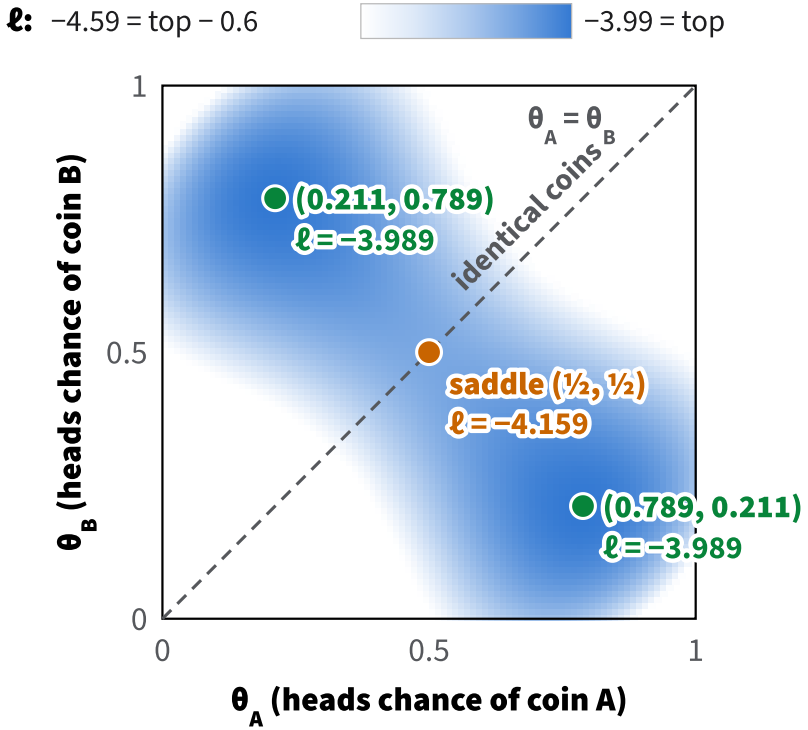
Both mirror peaks have ℓ = −3.989; the saddle only −4.159

↳ expectation–maximization (EM)'s limit on HH, HT, TT **(Lecture 5, p. 43)**: $\theta_A = 0.789,\ \theta_B = 1 - \theta_A = 0.211$, so $p(\mathrm{HT}) = \theta_A \cdot \theta_B = \frac{1}{6}$ and $p(\mathrm{HH}) = p(\mathrm{TT}) = \frac{1}{2} \cdot (1 - \frac{2}{6}) = \frac{1}{3}$.

point (θ_A, θ_B)	$p(\mathrm{HH})$	$p(\mathrm{HT})$	$p(\mathrm{TT})$	ℓ
peak $(0.789,\ 0.211)$	$\frac{1}{3}$	$\frac{1}{6}$	$\frac{1}{3}$	-3.989
mirror $(0.211,\ 0.789)$	$\frac{1}{3}$	$\frac{1}{6}$	$\frac{1}{3}$	-3.989
saddle $(\frac{1}{2}, \frac{1}{2})$	$\frac{1}{4}$	$\frac{1}{4}$	$\frac{1}{4}$	-4.159

Peaks: $\ell = 2 \cdot \ln \frac{1}{3} + \ln \frac{1}{6} = -3.989$. Saddle: $\ell = 3 \cdot \ln \frac{1}{4} = -4.159$.

Why $(\frac{1}{2}, \frac{1}{2})$ is a saddle, step by step: Appendix **(Lecture 6, p. 100)**.



equal peaks, mirror images; the saddle between is lower

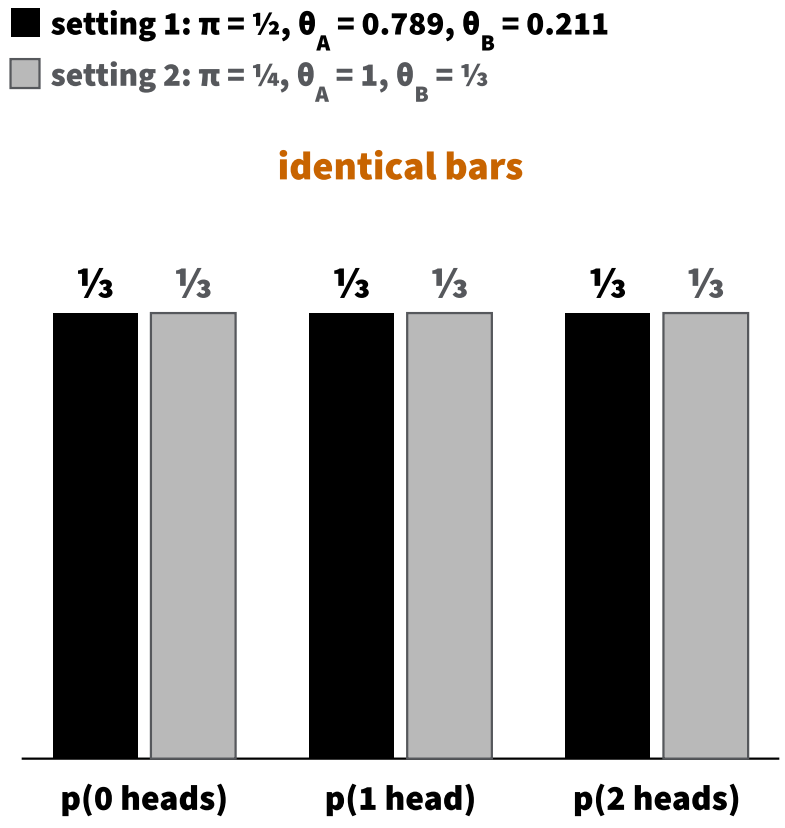
A start with $\theta_A = \theta_B$ stays at the worse saddle; restart off the diagonal.

Two different coin settings give exactly the same data

↳ Theorem 8 **(Lecture 6, p. 114)** on the coins, two flips per record, $p(h)$ = chance of h heads.

$\pi,\ \theta_A,\ \theta_B$	**setting 1 (the fit found):** $\frac{1}{2},\ 0.789,\ 0.211$	**setting 2:** $\frac{1}{4},\ 1,\ \frac{1}{3}$
$p(2)$	$\frac{1}{2}\cdot(\theta_A^2+\theta_B^2)=\frac{1}{3}$	$\frac{1}{4}\cdot 1+\frac{3}{4}\cdot\frac{1}{9}=\frac{1}{3}$
$p(0)$	$=p(2)\ \ (1-\theta_A=\theta_B)$	$\frac{1}{4}\cdot 0+\frac{3}{4}\cdot\frac{4}{9}=\frac{1}{3}$
$p(1)$	$1-\frac{1}{3}-\frac{1}{3}=\frac{1}{3}$	$1-\frac{1}{3}-\frac{1}{3}=\frac{1}{3}$

Same data, same $\ell=-3.989$: a **ridge** of equal fits. A **third flip** separates them: $p(3\text{ heads})=\frac{1}{4}$ vs $\frac{5}{18}$ (Teicher **(Lecture 6, p. 115)**).



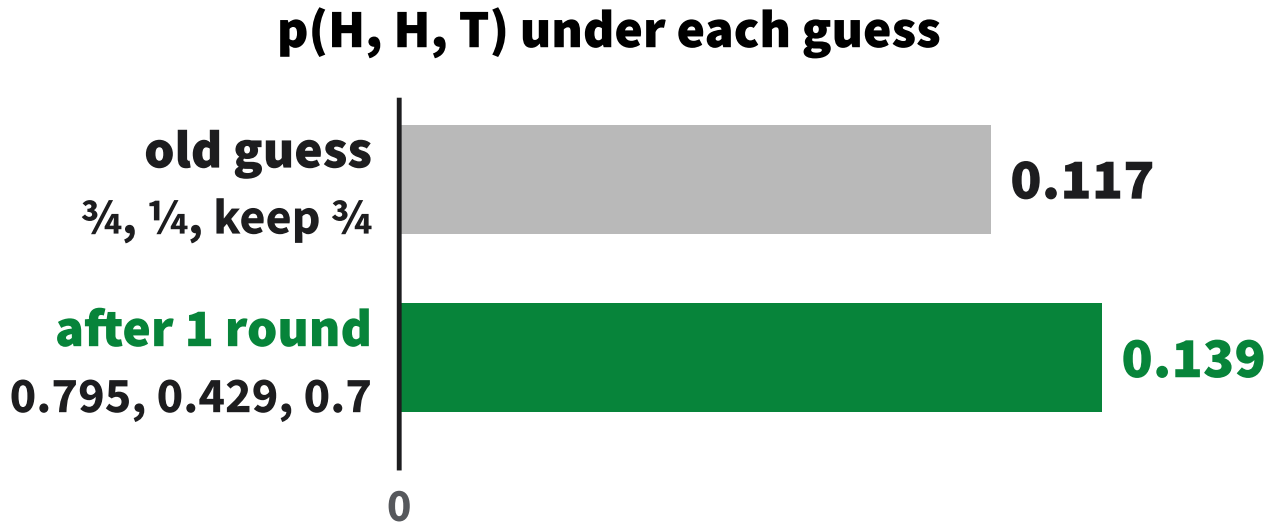
each setting: ⅓ + ⅓ + ⅓ = 1

Different knobs, identical readings: the data cannot choose, so no algorithm can.

Counting with those guesses makes the flips more likely

One count step **(Lecture 6, p. 25)** replaces the first guess of the habits:

1. coin A's heads chance: $\frac{3}{4}$ becomes **0.795**;
2. coin B's heads chance: $\frac{1}{4}$ becomes **0.429**;
3. the chance to keep the coin: $\frac{3}{4}$ becomes **0.7**.



One round of Baum–Welch improves the fit, as every expectation–maximization (EM) round does.

Coin 2's update, worked: from the two weights to 0.839

Coin 2 showed H. Its neighbours guess $q_1(A) = 0.75$ and $q_3(A) = 0.5$. Using the rule (Lecture 6, p. 37), what is the new $q_2(A)$?

0.750 | 0.839 | 0.900

Answer. The rule (Lecture 6, p. 37) gives coin 2 **one weight per value**, $w(A)$ and $w(B)$; then $q_2(A) = \dfrac{w(A)}{w(A)+w(B)}$. In logs, with keep $\frac{3}{4}$, switch $\frac{1}{4}$, A shows H $\frac{3}{4}$, B shows H $\frac{1}{4}$, $q_1(B) = 0.25$, $q_3(B) = 0.5$:

$$\begin{aligned}\ln w(A) &= \ln(\text{A shows H}) + \big[q_1(A)\ln\text{keep} + q_1(B)\ln\text{switch}\big] + \big[q_3(A)\ln\text{keep} + q_3(B)\ln\text{switch}\big]\\ &= \ln\tfrac{3}{4} + \big[0.75\ln\tfrac{3}{4} + 0.25\ln\tfrac{1}{4}\big] + \big[0.5\ln\tfrac{3}{4} + 0.5\ln\tfrac{1}{4}\big] = -0.288 - 0.562 - 0.837 = \mathbf{-1.687}\\ \ln w(B) &= \ln(\text{B shows H}) + \big[q_1(A)\ln\text{switch} + q_1(B)\ln\text{keep}\big] + \big[q_3(A)\ln\text{switch} + q_3(B)\ln\text{keep}\big]\\ &= \ln\tfrac{1}{4} + \big[0.75\ln\tfrac{1}{4} + 0.25\ln\tfrac{3}{4}\big] + \big[0.5\ln\tfrac{1}{4} + 0.5\ln\tfrac{3}{4}\big] = -1.386 - 1.112 - 0.837 = \mathbf{-3.335}\end{aligned}$$

A's share of the two weights: $q_2(A) = \dfrac{w(A)}{w(A)+w(B)} = \dfrac{e^{-1.687}}{e^{-1.687}+e^{-3.335}} = \mathbf{0.839}$

Coin 1's lean towards A pulls coin 2 from 0.75 up to 0.839.

The hand computation: every update raises B

↳ Slide **(Lecture 6, p. 40)**: start from $q = (\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$, score -2.799 **(Lecture 6, p. 36)**, and update coin 1, coin 2, coin 3, and repeat, each time with the rule in log form **(Lecture 6, p. 119)**, which on our coins simplifies to:

$$\ln \frac{q_t(A)}{q_t(B)} = \pm \ln 3 \ (\text{flip H: } +,\ \text{T: } -) \ + \sum_{\text{neighbours } j} (2\, q_j(A) - 1) \ln 3$$

update	$q_1(A)$	$q_2(A)$	$q_3(A)$	score B
start	0.5	0.5	0.5	−2.799
round 1: coin 1	0.750	0.5	0.5	−2.655
round 1: coin 2	0.750	0.839	0.5	−2.348
round 1: coin 3	0.750	0.839	0.412	−2.332
round 2: coin 1	0.863	0.839	0.412	−2.2870
round 2: coin 2	0.863	0.846	0.412	−2.2868
round 2: coin 3	0.863	0.846	0.416	−2.2867
converged	0.866	0.848	0.417	−2.287
exact **(Lecture 6, p. 24)**	0.8	0.75	0.4	$\ln p(x) = -2.144$

The score never falls and converges fast, but at 0.143 below $\ln p(x)$, with chances too sure.

One redraw, worked: coin 2 becomes A with chance 0.75

↳ Slide **(Lecture 6, p. 32)**: the current story is A, A, B over H, H, T on the ring with keep ¾. Coin 2 is redrawn from the three things that touch it; each value of coin 2 gets a weight, the product of the three.

if coin 2 is …	coin 1 is A, so the move 1 → 2 is	coin 3 is B, so the move 2 → 3 is	coin 2 shows H	weight
A	**keep: ¾**	**switch: ¼**	**A shows H: ¾**	**¾ · ¼ · ¾ = 9/64**
B	**switch: ¼**	**keep: ¾**	**B shows H: ¼**	**¼ · ¾ · ¼ = 3/64**

Coin 2 is redrawn as **A with chance 9 ÷ (9 + 3) = 0.75**, and as B with chance 0.25: the same three squares as the formula $p(z_2 \mid z_1, z_3, x_2) \propto p(z_2 \mid z_1)\, p(z_3 \mid z_2)\, p(x_2 \mid z_2)$.

Then redraw coin 3 the same way from coin 2, coin 1 and its flip T; then coin 1; then coin 2 again, and so on. Each redraw uses only the **current values** of the two neighbours, so a redraw costs three factors however long the circle is.

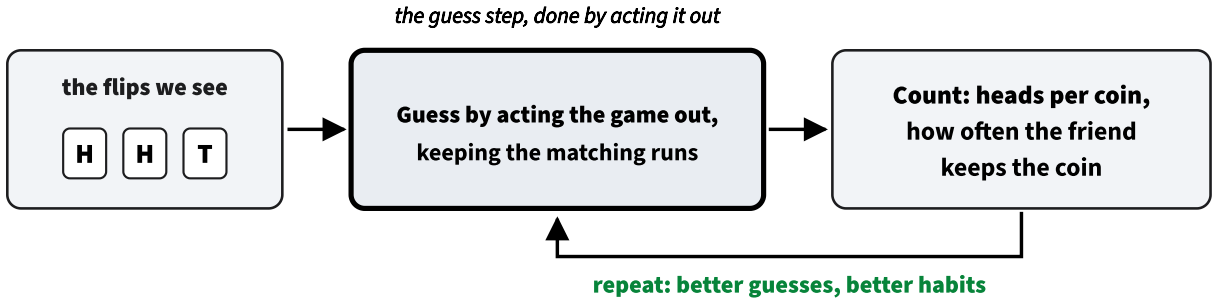
Only the squares that touch coin 2 enter the redraw: three factors, two weights, one chance.

Counting the kept runs gives Baum–Welch's habits

↳ Slide **(Lecture 6, p. 30)**: the 240 kept runs of H, H, T **(Lecture 6, p. 29)**, each counted like kept run 1 (coin A: 2 flips, 2 heads; coin B: 1 flip, 0 heads; kept once, switched once), then added up.

habit	count	of which	new habit
coin A's heads chance	**468 flips**	**372 heads**	**372 ÷ 468 = 0.795**
coin B's heads chance	**252 flips**	**108 heads**	**108 ÷ 252 = 0.429**
chance to keep the coin	**480 moves**	**336 keeps**	**336 ÷ 480 = 0.7**

These are the habits of Baum–Welch's count step **(Lecture 6, p. 118)**: the kept runs come in the **right proportions**, so counting them is the same as weighting every story by its chance.



Count the kept runs and you have the count step, no sum over stories needed.

Variational EM on H, H, T, T: the score rises, ln p(x) falls

↳ Theorem E **(Lecture 6, p. 46)** on four flips H, H, T, T: one mean-field guess step with the habits before, then one count step with those soft labels; the score B is taken with the same guess before and after.

four flips H, H, T, T	start A	keep	A heads	B heads	score B	$\ln p(x)$
before	**0.363**	**0.820**	**0.744**	**0.365**	**−3.416**	**−2.936**
after one round	**0.354**	**0.719**	**0.805**	**0.429**	**−3.288**	**−2.967**

1. **Guess step:** mean field with the habits before gives each flip a soft label, the chance of A: 0.354, 0.252, 0.053, 0.094.
2. **Count step:** those labels, counted as Baum–Welch counts **(Lecture 6, p. 118)**, give the habits after one round.
3. **Compare:** the score rises by 0.128, as Theorem E promises, but only because the gap $\ln p(x) - B$ shrank from 0.480 to 0.321; the likelihood itself **falls by 0.031**. With the exact guess the gap would be 0 and $\ln p(x)$ could not fall **(Lecture 5, p. 49)**.

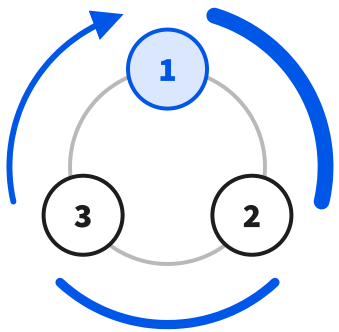
A higher score is not a higher likelihood when the gap moves with the habits.

Longer circles: the echo and the error, by circle length

↳ Slide **(Lecture 6, p. 52)**: circles of 3 to 8 coins with keep ¾ and the flips H, H, T repeated; λ_2/λ_1 from Theorem D's matrix C **(Lecture 6, p. 51)**, the error from the converged messages against the exact sum.

coins in the circle	3	4	5	6	8
λ_2/λ_1, the echo	0.069	0.019	0.005	0.005	0.0004
largest error of the messages	0.037	0.014	0.004	0.003	0.0003

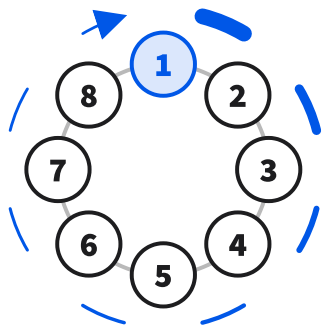
Each link multiplies a lean by keep − switch = ½, so the echo fades link by link, and the error fades with it. The circle of 3 is the **worst case**; by 8 coins the messages are **exact to three decimals**.



circle of 3 coins

returns after 3 links:
still strong, counted twice

error 0.037



circle of 8 coins

returns after 8 links:
almost gone

error 0.0003

The longer the circle, the fainter the echo, the smaller the error.

Theorem D on our ring: the echo as an eigenvalue ratio

↳ Recall **(Lecture 6, p. 51)**. Two kinds of 2×2 table: a **link** $M = \begin{pmatrix} \text{keep} & \text{switch} \\ \text{switch} & \text{keep} \end{pmatrix} = \begin{pmatrix} \frac{3}{4} & \frac{1}{4} \\ \frac{1}{4} & \frac{3}{4} \end{pmatrix}$ and a **flip** $D_t = \begin{pmatrix} \text{A shows } x_t & 0 \\ 0 & \text{B shows } x_t \end{pmatrix}$, e.g. $D_1 = \begin{pmatrix} \frac{3}{4} & 0 \\ 0 & \frac{1}{4} \end{pmatrix}$ for H.

Once round the circle from coin 1: $C = M\,D_3\,M\,D_2\,M\,D_1$ (link, flip, link, flip, link, flip). Its eigenvalues $\lambda_1 > |\lambda_2|$ give the echo $e = \lambda_2/\lambda_1$, and the theorem reads $p(A) - p(B) = \dfrac{\lambda_1 - \lambda_2}{\lambda_1 + \lambda_2}\big(\tilde{p}(A) - \tilde{p}(B)\big)$, $\tilde{p}$ = the messages' answer.

coin 1 on the ring	$e = \lambda_2/\lambda_1$	factor $\frac{1-e}{1+e}$	messages' margin	true margin	exact	messages
keep ¾ (coins like to agree)	**+0.069**	**0.87**	**0.787 − 0.213 = 0.574**	**0.50**	**0.75**	**0.787**
keep 0.05 (coins like to switch)	**−0.59**	**3.9**	**0.564 − 0.436 = 0.128**	**0.50**	**0.75**	**0.564**

Both rows give the true margin 0.50, that is, the exact chance 0.75: the messages name the right coin, and the factor repairs the margin.

The margin is off by exactly the echo; the winner never is.

The family dial: the best guess in each family, line and ring

↳ Slide **(Lecture 6, p. 42)**: keep ¾, flips H, H, T. Each row is the member of its family with the highest score B; the gap is $\ln p(x) - B$, with $\ln p(x) = -2.144$ on the line and -2.837 on the ring. **All alone:** the mean-field fixed point. **Line links kept:** $q(z_1)\, q(z_2 \mid z_1)\, q(z_3 \mid z_2)$, its 5 numbers chosen by coordinate ascent on B. **Full table:** the posterior itself.

family	numbers	line: coins 1, 2, 3	line gap	ring: coins 1, 2, 3	ring gap
all alone	**3**	**0.866, 0.848, 0.417**	**0.143**	**0.917, 0.917, 0.675**	**0.249**
line links kept	**5**	**0.800, 0.750, 0.400**	**0**	**0.771, 0.750, 0.558**	**0.041**
the full table	**7**	**0.800, 0.750, 0.400**	**0**	**0.750, 0.750, 0.550**	**0**

The scores: on the line $B = -2.287,\ -2.144,\ -2.144$; on the ring $B = -3.086,\ -2.879,\ -2.837$. On the line the chain family already contains the posterior, so its gap is 0. On the ring, leaving out the one ring link costs 0.041; leaving out every link costs 0.249.

Every link kept inside the guess buys back part of the gap.

One link at a time: the update in symbols, and the results

↳ Slide **(Lecture 6, p. 44)**. Every link factor $\psi(z_i, z_j)$ is replaced by two one-coin tables $s_{ij}(z_i)\, s_{ji}(z_j)$, so the guess is a product, $q(z) \propto \prod_t \phi_t(z_t) \prod_{\text{links}} s_{ij}(z_i)\, s_{ji}(z_j)$, with ϕ_t the flip factor of coin t. For one link (i, j), three steps:

1. **Cavity:** the product of all the other tables, $q^{\setminus ij}(z_i)$ and $q^{\setminus ij}(z_j)$.
2. **Tilt by the true link:** $\tilde{p}(z_i, z_j) \propto q^{\setminus ij}(z_i)\, q^{\setminus ij}(z_j)\, \psi(z_i, z_j)$; take its one-coin chances $\tilde{p}(z_i), \tilde{p}(z_j)$.
3. **New tables:** $s_{ij}(z_i) = \tilde{p}(z_i)/q^{\setminus ij}(z_i)$, $s_{ji}(z_j) = \tilde{p}(z_j)/q^{\setminus ij}(z_j)$, so cavity × tables gives the matched chances.

chance of A, coins 1, 2, 3	line	ring
the truth	**0.800, 0.750, 0.400**	**0.750, 0.750, 0.550**
one link at a time (expectation propagation)	**0.800, 0.750, 0.400**	**0.787, 0.787, 0.557**
messages in a circle (Way 4)	**0.800, 0.750, 0.400**	**0.787, 0.787, 0.557**
everyone guesses alone (Way 3)	**0.866, 0.848, 0.417**	**0.917, 0.917, 0.675**

Step 2 for coin j is $\tilde{p}(z_j) \propto q^{\setminus ij}(z_j) \sum_{z_i} q^{\setminus ij}(z_i)\, \psi(z_i, z_j)$: the sum is the note coin i sends to coin j in Way 4, so step 3 makes s_{ij} that note. With one-coin tables the converged tables are the converged messages: the same answer on the line and on the ring.

Expectation propagation with one-coin tables is loopy belief propagation.